Online Resource 1

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Validating a synthetic segmentation benchmark by rank correlation: how the methods compared, the clinical reference and training noise change the answer in wound imaging

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Notation

General. Numbers are rounded to 3 decimals (p values to 2 significant digits, percentages to 1 decimal); negative numbers use the minus sign. Intervals are 95% percentile bootstrap intervals; each caption names the bootstrap and its number of draws B.

Datasets. Medetec, CO2Wounds-V2, FUSeg, AZH, Kvasir, Kvasir-sessile, CVC-ClinicDB, CVC-ColonDB, ETIS-Larib, PH2, TN3K. FUSeg means the FUSeg challenge validation images used as the reference; AZH means the AZH test images. Control: the AZH training slice, a second sample from the AZH source. Table S9a gives the images kept in each set.

Method sets (the codes used in the tables; number of methods in brackets): U (18), untrained classical methods; F (14), methods trained from scratch on FUSeg; K (14), the same 14 methods trained from scratch on Kvasir-SEG; M (11), the 10 original ImageNet-pretrained architectures, each at training seed 0, plus nnU-Net; M11s (11), the same 10 architectures as seed averages, plus nnU-Net; M19 (19), all 18 ImageNet-pretrained architectures (10 original, 8 new) as seed averages, plus nnU-Net (the modern models of the main text); N8s (8), the 8 new architectures as seed averages; U+F (32), U and F pooled; U+K (32), U and K pooled; U+M (29), U and M pooled; U+M11s (29), U and M11s pooled; U+M19 (37), U and M19 pooled; ALL57 (57), U, F, K and M

pooled; ALL (65), U, F, K and M19 pooled. A seed average is one column per architecture: per image, the mean of the scores of the training seeds that exist, then the mean over images (equal to the mean of the seed models' means). Further sets: P10s (10), the 10 original architectures as seed averages; K11 (11), K without its 3 pixel classifiers (logreg_all_kv, gbm_all_kv, logreg_colour_kv), used where those were not rescored (Table S6).

Single-seed sets (suffix _sk for training seed k = 0, 1 or 2): P10_sk, the original architectures at training seed k (the models that exist); N8_sk, the new architectures at seed k; M_sk, P10_sk plus nnU-Net (M_s0 is the set M); M19_sk, P10_sk, N8_sk and nnU-Net.

Seed-analysis sets. The architectures with three usable training seeds: P9, the 9 original architectures with three seeds (unet_r18, unetpp_r18, deeplabv3p_r18, fpn_r18, pspnet_r18, linknet_r18, segformer_mitb0, unet_mbv2, unet_effb0); N7, the 7 new architectures with three seeds (unet_r50, deeplabv3p_r50, upernet_convnext_t, unet_swin_t, segformer_mitb2, unet_effb3, fpn_regnety016); C16, the 16 architectures of P9 and N7 together. Not in these sets: manet_r18, manet_r50, nnunet_2d (a diverged training or a single training).

Agreement. Agreement is the Spearman correlation of the per-method mean scores of two image sets over the methods of a method set. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice.

Method keys (for example unet_r18, segformer_mitb2) are listed with their architecture in Table S1. Online Resource 2 gives every per-method mean and every pairwise agreement.

Table S1. Methods, training and held-out Dice. Every method in the study. Group: U, untrained classical methods (family in the third column); F, trained from scratch on FUSeg; K, the same methods trained from scratch on Kvasir-SEG; modern, original 10 and new 8: ImageNet-pretrained architectures trained on FUSeg at training seeds 0, 1 and 2; nnU-Net; SAM/MedSAM: box-prompted foundation models, used only as oracle references and never pooled. Architecture: decoder, encoder. Modern models: the training split is the FUSeg training images minus duplicates; 40 epochs; the decision threshold of each model was chosen on 148 held-out FUSeg training images, and 'Held-out Dice' is the mean Dice on those images at that threshold (optimistic, because the threshold was tuned on the same images). nnU-Net: one training, fixed threshold, held-out Dice on the same 148 images, which it never saw. Seed 0 of the original 10 was trained in a first GPU session, seeds 1 and 2 and all seeds of the new 8 in a second session (41 of 44 second-session trainings saved; manet_r18_in_s2, manet_r50_s0, manet_r50_s2 diverged with non-finite values under mixed precision and were not imputed; Table S9c). The modern rows name the seed-0 key of an original architecture or the seed-average key of a new one; individual models carry the suffix _s1 or _s2 (original 10) or _s0, _s1, _s2 (new 8) in Online Resource 2. n/a: not applicable. nr: not stored in a results file. div.: diverged.

Method	Group	Architecture or algorithm	Training data	Training seeds available	Threshold (seed 0 / 1 / 2)	Held-out Dice, seed 0	Seed 1	Seed 2
adaptive_gaussian	U	classical, background-referenced	none	n/a	n/a	n/a	n/a	n/a
adaptive_mean	U	classical, background-referenced	none	n/a	n/a	n/a	n/a	n/a
otsu_gray	U	classical, background-referenced	none	n/a	n/a	n/a	n/a	n/a
bgr_corner_distance	U	classical, background-referenced	none	n/a	n/a	n/a	n/a	n/a
hsv_corner_distance	U	classical, background-referenced	none	n/a	n/a	n/a	n/a	n/a
otsu_a	U	classical, colour-referenced	none	n/a	n/a	n/a	n/a	n/a
pct92_a	U	classical, colour-referenced	none	n/a	n/a	n/a	n/a	n/a
redness_ratio	U	classical, colour-referenced	none	n/a	n/a	n/a	n/a	n/a
otsu_saturation	U	classical, colour-referenced	none	n/a	n/a	n/a	n/a	n/a
kmeans_lab_k3	U	classical, clustering	none	n/a	n/a	n/a	n/a	n/a
kmeans_lab_k5	U	classical, clustering	none	n/a	n/a	n/a	n/a	n/a
kmeans_lab_k4_higha	U	classical, clustering	none	n/a	n/a	n/a	n/a	n/a
meanshift_otsu_a	U	classical, clustering	none	n/a	n/a	n/a	n/a	n/a
watershed_a	U	classical, region/contour	none	n/a	n/a	n/a	n/a	n/a
grabcut_center	U	classical, region/contour	none	n/a	n/a	n/a	n/a	n/a
floodfill_a	U	classical, region/contour	none	n/a	n/a	n/a	n/a	n/a
chanvese_a	U	classical, region/contour	none	n/a	n/a	n/a	n/a	n/a
canny_fill	U	classical, region/contour	none	n/a	n/a	n/a	n/a	n/a
logreg_all	F	pixel classifier, logistic regression (all pixel features)	FUSeg training images, from scratch	one training	nr	nr	n/a	n/a
gbm_all	F	pixel classifier, gradient boosting (all pixel features)	FUSeg training images, from scratch	one training	nr	nr	n/a	n/a
logreg_colour	F	pixel classifier, logistic regression (colour features)	FUSeg training images, from scratch	one training	nr	nr	n/a	n/a
unet	F	small U-Net	FUSeg training images, from scratch	one training	nr	nr	n/a	n/a
unet_r18	F	U-Net, ResNet-18	FUSeg training images, from scratch	one training	0.10	nr	n/a	n/a
unetpp_r18	F	U-Net++, ResNet-18	FUSeg training images, from scratch	one training	0.20	nr	n/a	n/a
deeplabv3p_r18	F	DeepLabV3+, ResNet-18	FUSeg training images, from scratch	one training	0.40	nr	n/a	n/a
fpn_r18	F	FPN, ResNet-18	FUSeg training images, from scratch	one training	0.45	nr	n/a	n/a
pspnet_r18	F	PSPNet, ResNet-18	FUSeg training images, from scratch	one training	0.10	nr	n/a	n/a
linknet_r18	F	LinkNet, ResNet-18	FUSeg training images, from scratch	one training	0.20	nr	n/a	n/a
manet_r18	F	MA-Net, ResNet-18	FUSeg training images, from scratch	one training	0.10	nr	n/a	n/a

Method	Group	Architecture or algorithm	Training data	Training seeds available	Threshold (seed 0 / 1 / 2)	Held-out Dice, seed 0	Seed 1	Seed 2
segformer_mitb0	F	SegFormer, MiT-B0	FUSeg training images, from scratch	one training	0.35	nr	n/a	n/a
unet_mbv2	F	U-Net, MobileNetV2	FUSeg training images, from scratch	one training	0.10	nr	n/a	n/a
unet_effb0	F	U-Net, EfficientNet-B0	FUSeg training images, from scratch	one training	0.35	nr	n/a	n/a
logreg_all_kv	K	pixel classifier, logistic regression (all pixel features)	Kvasir-SEG training images, from scratch	one training	nr	nr	n/a	n/a
gbm_all_kv	K	pixel classifier, gradient boosting (all pixel features)	Kvasir-SEG training images, from scratch	one training	nr	nr	n/a	n/a
logreg_colour_kv	K	pixel classifier, logistic regression (colour features)	Kvasir-SEG training images, from scratch	one training	nr	nr	n/a	n/a
unet_kv	K	small U-Net	Kvasir-SEG training images, from scratch	one training	nr	nr	n/a	n/a
unet_r18_kv	K	U-Net, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.55	nr	n/a	n/a
unetpp_r18_kv	K	U-Net++, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.35	nr	n/a	n/a
deeplabv3p_r18_kv	K	DeepLabV3+, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.45	nr	n/a	n/a
fpn_r18_kv	K	FPN, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.10	nr	n/a	n/a
pspnet_r18_kv	K	PSPNet, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.15	nr	n/a	n/a
linknet_r18_kv	K	LinkNet, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.55	nr	n/a	n/a
manet_r18_kv	K	MA-Net, ResNet-18	Kvasir-SEG training images, from scratch	one training	0.40	nr	n/a	n/a
segformer_mitb0_kv	K	SegFormer, MiT-B0	Kvasir-SEG training images, from scratch	one training	0.55	nr	n/a	n/a
unet_mbv2_kv	K	U-Net, MobileNetV2	Kvasir-SEG training images, from scratch	one training	0.10	nr	n/a	n/a
unet_effb0_kv	K	U-Net, EfficientNet-B0	Kvasir-SEG training images, from scratch	one training	0.35	nr	n/a	n/a
unet_r18_in	modern, original 10	U-Net, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.35 / 0.30 / 0.35	0.827	0.831	0.835
unetpp_r18_in	modern, original 10	U-Net++, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.40 / 0.30 / 0.35	0.835	0.838	0.837
deeplabv3p_r18_in	modern, original 10	DeepLabV3+, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.35 / 0.30 / 0.30	0.808	0.800	0.794
fpn_r18_in	modern, original 10	FPN, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.30 / 0.35 / 0.30	0.808	0.798	0.819
pspnet_r18_in	modern, original 10	PSPNet, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.15 / 0.20 / 0.30	0.748	0.748	0.754
linknet_r18_in	modern, original 10	LinkNet, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.65 / 0.45 / 0.55	0.806	0.809	0.818
manet_r18_in	modern, original 10	MA-Net, ResNet-18	FUSeg training split (392 images), ImageNet initialisation	0, 1 (seed 2 diverged)	0.40 / 0.25 / div.	0.843	0.812	diverged

Method	Group	Architecture or algorithm	Training data	Training seeds available	Threshold (seed 0 / 1 / 2)	Held-out Dice, seed 0	Seed 1	Seed 2
segformer_mitb0_in	modern, original 10	SegFormer, MiT-B0	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.35 / 0.40 / 0.35	0.778	0.781	0.783
unet_mbv2_in	modern, original 10	U-Net, MobileNetV2	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.40 / 0.35 / 0.40	0.816	0.819	0.817
unet_effb0_in	modern, original 10	U-Net, EfficientNet-B0	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.35 / 0.35 / 0.35	0.806	0.812	0.818
unet_r50@seeds	modern, new 8	U-Net, ResNet-50	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.45 / 0.50 / 0.35	0.811	0.815	0.818
deeplabv3p_r50@seeds	modern, new 8	DeepLabV3+, ResNet-50	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.30 / 0.30 / 0.25	0.800	0.804	0.799
upernet_convnext_t@seeds	modern, new 8	UPerNet, ConvNeXt-T	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.50 / 0.40 / 0.40	0.809	0.814	0.826
unet_swin_t@seeds	modern, new 8	U-Net, Swin-T	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.40 / 0.25 / 0.50	0.838	0.827	0.830
segformer_mitb2@seeds	modern, new 8	SegFormer, MiT-B2	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.40 / 0.30 / 0.40	0.820	0.803	0.808
unet_effb3@seeds	modern, new 8	U-Net, EfficientNet-B3	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.20 / 0.40 / 0.25	0.805	0.807	0.807
fpn_regnety016@seeds	modern, new 8	FPN, RegNetY-016	FUSeg training split (392 images), ImageNet initialisation	0, 1, 2	0.35 / 0.35 / 0.20	0.789	0.811	0.811
manet_r50@seeds	modern, new 8	MA-Net, ResNet-50	FUSeg training split (392 images), ImageNet initialisation	1 (seed 0, 2 diverged)	div. / 0.40 / div.	diverged	0.826	diverged
nnunet_2d	nnU-Net	nnU-Net, 2D configuration	FUSeg training split	one training	0.50 (fixed)	0.833	n/a	n/a
sam_vitb_box	SAM/MedSAM	SAM ViT-B, reference box	none (published weights)	n/a	n/a	n/a	n/a	n/a
medsam_vitb_box	SAM/MedSAM	MedSAM ViT-B, reference box	none (published weights)	n/a	n/a	n/a	n/a	n/a
medsam_fixed_gtbox	SAM/MedSAM	MedSAM ViT-B, official preprocessing, reference box	none (published weights)	n/a	n/a	n/a	n/a	n/a
sam_nnbox	SAM/MedSAM	SAM ViT-B, nnU-Net boxes	none (published weights)	n/a	n/a	n/a	n/a	n/a
medsam_fixed_nnbox	SAM/MedSAM	MedSAM ViT-B, official preprocessing, nnU-Net boxes	none (published weights)	n/a	n/a	n/a	n/a	n/a

Table S2a. Agreement levels by pair category (Dice). Point estimates on the full data. Each column is the mean agreement over the pairs of one category (number of pairs in brackets). Same source: two datasets from one source (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice. Agreement is the Spearman correlation of the per-method mean scores of two image sets over the methods of a method set. Method sets and their sizes as defined in the Notation.

Set	n	Control	Same source (3)	Different source, same modality (13)	Across modality (39)	Same modality (16)	Wound pairs (6)	Wound, different source (5)	FUSeg with AZH	Colonoscopy pairs (10)	Modality difference (55)	Modality difference (52)
U	18	0.948	0.672	0.668	0.072	0.668	0.351	0.401	0.096	0.859	0.596	0.595
F	14	0.912	0.659	0.600	0.144	0.611	0.164	0.184	0.064	0.879	0.467	0.456
K	14	0.877	0.543	0.680	0.156	0.654	0.363	0.443	−0.037	0.829	0.498	0.523
M	11	0.809	0.561	0.474	0.025	0.490	0.105	0.100	0.127	0.722	0.465	0.449
M11s	11	0.882	0.409	0.390	0.064	0.394	0.274	0.296	0.164	0.465	0.329	0.326
M19	19	0.916	0.549	0.278	0.104	0.329	0.206	0.205	0.211	0.403	0.225	0.174
N8s	8	1.000	0.619	0.007	0.067	0.122	0.258	0.238	0.357	0.040	0.055	−0.060
U+F	32	0.699	0.854	0.736	−0.075	0.758	0.523	0.498	0.650	0.899	0.834	0.812
U+K	32	0.951	0.578	0.701	0.000	0.678	0.270	0.362	−0.187	0.922	0.677	0.700
U+M	29	0.973	0.857	0.841	−0.245	0.844	0.796	0.808	0.737	0.873	1.089	1.086
U+M11s	29	0.975	0.851	0.844	−0.256	0.845	0.806	0.819	0.739	0.869	1.102	1.100
U+M19	37	0.978	0.858	0.810	−0.281	0.819	0.818	0.824	0.789	0.820	1.101	1.092
ALL57	57	0.916	0.879	0.826	−0.182	0.836	0.686	0.677	0.733	0.926	1.018	1.008
ALL	65	0.940	0.895	0.839	−0.212	0.850	0.770	0.764	0.801	0.897	1.062	1.051

Table S2b. Agreement levels by pair category (NSD). Point estimates on the full data. Each column is the mean agreement over the pairs of one category (number of pairs in brackets). Same source: two datasets from one source (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice. Agreement is the Spearman correlation of the per-method mean scores of two image sets over the methods of a method set. Method sets and their sizes as defined in the Notation.

Set	n	Control	Same source (3)	Different source, same modality (13)	Across modality (39)	Same modality (16)	Wound pairs (6)	Wound, different source (5)	FUSeg with AZH	Colonoscopy pairs (10)	Modality difference (55)	Modality difference (52)
U	18	0.957	0.855	0.687	0.298	0.719	0.622	0.601	0.725	0.777	0.421	0.389
F	14	0.868	0.937	0.607	0.106	0.669	0.314	0.205	0.855	0.882	0.563	0.501
K	14	0.464	0.654	0.644	−0.031	0.646	0.260	0.262	0.248	0.877	0.677	0.675
M	11	0.845	0.761	0.520	0.121	0.565	0.308	0.262	0.536	0.719	0.444	0.399
M11s	11	0.855	0.518	0.634	0.317	0.612	0.561	0.615	0.291	0.643	0.295	0.316
M19	19	0.865	0.647	0.516	0.180	0.540	0.479	0.495	0.396	0.577	0.360	0.336
N8s	8	0.786	0.810	−0.082	0.081	0.085	0.393	0.329	0.714	−0.100	0.004	−0.164
U+F	32	0.930	0.898	0.710	0.033	0.746	0.555	0.509	0.783	0.860	0.713	0.678
U+K	32	0.827	0.823	0.728	0.028	0.746	0.554	0.535	0.647	0.862	0.718	0.700
U+M	29	0.977	0.910	0.776	−0.121	0.801	0.864	0.857	0.901	0.763	0.922	0.897
U+M11s	29	0.975	0.906	0.800	−0.130	0.820	0.884	0.882	0.894	0.782	0.950	0.930
U+M19	37	0.973	0.909	0.788	−0.199	0.811	0.885	0.885	0.886	0.766	1.010	0.987
ALL57	57	0.950	0.928	0.831	−0.173	0.849	0.780	0.758	0.893	0.890	1.022	1.004
ALL	65	0.963	0.930	0.847	−0.211	0.862	0.843	0.829	0.916	0.874	1.073	1.057

Table S2c. Agreement with method and hierarchical bootstrap intervals (Dice). Each cell: estimate, then the m and h intervals. m: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000; h: hierarchical bootstrap (methods within group and resampling units within dataset), B = 1,000; both 95% percentile intervals. Resampling units are photographs for Medetec, AZH and the control (Medetec 38 units for 152 images, AZH 118 units for 146 images, Control 71 units for 78 images) and single images otherwise. Resampling the units adds noise to every method mean and so pulls agreement down in each draw; h intervals are therefore shifted towards zero and understate how strong agreement is. Wound with non-wound: mean over the 28 pairs of a wound dataset with a non-wound dataset. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice.

Set	n	Wound pairs (6)	Wound with non-wound (28)	Colonoscopy pairs (10)	Modality difference (55)	Modality difference (52)	Control
U	18	0.351 m [0.122, 0.594] h [0.107, 0.619]	0.087 m [−0.155, 0.295] h [−0.169, 0.291]	0.859 m [0.676, 0.938] h [0.640, 0.925]	0.596 m [0.349, 0.830] h [0.350, 0.817]	0.595 m [0.359, 0.822] h [0.355, 0.812]	0.948 m [0.806, 0.988] h [0.572, 0.994]
F	14	0.164 m [−0.152, 0.531] h [−0.132, 0.513]	0.145 m [−0.094, 0.366] h [−0.119, 0.352]	0.879 m [0.627, 0.964] h [0.640, 0.960]	0.467 m [0.209, 0.722] h [0.210, 0.704]	0.456 m [0.157, 0.731] h [0.168, 0.701]	0.912 m [0.659, 1.000] h [0.431, 0.987]
K	14	0.363 m [0.048, 0.636] h [−0.032, 0.634]	0.188 m [−0.122, 0.473] h [−0.135, 0.478]	0.829 m [0.596, 0.922] h [0.489, 0.917]	0.498 m [0.240, 0.730] h [0.177, 0.716]	0.523 m [0.256, 0.756] h [0.203, 0.736]	0.877 m [0.545, 0.996] h [0.185, 0.969]
M	11	0.105 m [−0.121, 0.377] h [−0.188, 0.430]	−0.030 m [−0.296, 0.245] h [−0.288, 0.268]	0.722 m [0.305, 0.911] h [0.188, 0.882]	0.465 m [0.170, 0.770] h [0.071, 0.722]	0.449 m [0.131, 0.775] h [0.030, 0.697]	0.809 m [0.346, 0.972] h [0.014, 0.991]
M11s	11	0.274 m [−0.111, 0.650] h [−0.142, 0.687]	0.074 m [−0.266, 0.477] h [−0.229, 0.480]	0.465 m [0.062, 0.757] h [−0.018, 0.716]	0.329 m [−0.027, 0.766] h [−0.048, 0.682]	0.326 m [−0.032, 0.763] h [−0.076, 0.660]	0.882 m [0.486, 1.000] h [0.093, 1.000]
M19	19	0.206 m [−0.070, 0.491] h [−0.098, 0.484]	0.143 m [−0.055, 0.373] h [−0.050, 0.355]	0.403 m [0.154, 0.630] h [0.063, 0.593]	0.225 m [0.048, 0.457] h [−0.007, 0.401]	0.174 m [−0.009, 0.424] h [−0.072, 0.370]	0.916 m [0.688, 0.989] h [0.456, 0.973]
N8s	8	0.258 m [−0.149, 0.722] h [−0.241, 0.658]	0.065 m [−0.078, 0.192] h [−0.072, 0.274]	0.040 m [−0.200, 0.432] h [−0.205, 0.459]	0.055 m [−0.148, 0.316] h [−0.204, 0.270]	−0.060 m [−0.288, 0.251] h [−0.344, 0.221]	1.000 m [1.000, 1.000] h [0.344, 1.000]
U+F	32	0.523 m [0.294, 0.694] h [0.254, 0.691]	−0.265 m [−0.441, −0.093] h [−0.420, −0.064]	0.899 m [0.814, 0.935] h [0.794, 0.931]	0.834 m [0.638, 0.999] h [0.586, 0.977]	0.812 m [0.606, 0.981] h [0.556, 0.964]	0.699 m [0.383, 0.884] h [0.262, 0.944]
U+K	32	0.270 m [0.105, 0.432] h [0.066, 0.446]	0.099 m [−0.027, 0.209] h [−0.022, 0.216]	0.922 m [0.835, 0.958] h [0.824, 0.951]	0.677 m [0.583, 0.755] h [0.559, 0.748]	0.700 m [0.610, 0.773] h [0.588, 0.764]	0.951 m [0.874, 0.979] h [0.667, 0.970]
U+M	29	0.796 m [0.742, 0.858] h [0.724, 0.866]	−0.499 m [−0.621, −0.357] h [−0.630, −0.347]	0.873 m [0.760, 0.922] h [0.685, 0.909]	1.089 m [0.953, 1.196] h [0.913, 1.181]	1.086 m [0.947, 1.195] h [0.900, 1.180]	0.973 m [0.929, 0.987] h [0.803, 0.988]
U+M11s	29	0.806 m [0.748, 0.868] h [0.731, 0.870]	−0.513 m [−0.633, −0.374] h [−0.639, −0.349]	0.869 m [0.761, 0.921] h [0.680, 0.911]	1.102 m [0.975, 1.209] h [0.916, 1.198]	1.100 m [0.970, 1.211] h [0.899, 1.197]	0.975 m [0.930, 0.990] h [0.797, 0.991]
U+M19	37	0.818 m [0.773, 0.866] h [0.763, 0.871]	−0.561 m [−0.664, −0.446] h [−0.661, −0.428]	0.820 m [0.714, 0.890] h [0.636, 0.873]	1.101 m [0.983, 1.197] h [0.921, 1.174]	1.092 m [0.974, 1.187] h [0.908, 1.170]	0.978 m [0.937, 0.992] h [0.865, 0.989]
ALL57	57	0.686 m [0.594, 0.764] h [0.556, 0.762]	−0.393 m [−0.480, −0.304] h [−0.478, −0.286]	0.926 m [0.888, 0.948] h [0.872, 0.944]	1.018 m [0.924, 1.101] h [0.895, 1.090]	1.008 m [0.909, 1.095] h [0.879, 1.084]	0.916 m [0.840, 0.959] h [0.756, 0.968]

Set	n	Wound pairs (6)	Wound with non-wound (28)	Colonoscopy pairs (10)	Modality difference (55)	Modality difference (52)	Control
ALL	65	0.770 m [0.711, 0.825] h [0.682, 0.826]	−0.452 m [−0.533, −0.374] h [−0.530, −0.348]	0.897 m [0.851, 0.926] h [0.832, 0.924]	1.062 m [0.977, 1.136] h [0.938, 1.129]	1.051 m [0.966, 1.127] h [0.930, 1.123]	0.940 m [0.885, 0.968] h [0.813, 0.975]

Table S2d. Agreement with method and hierarchical bootstrap intervals (NSD). Each cell: estimate, then the m and h intervals. m: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000; h: hierarchical bootstrap (methods within group and resampling units within dataset), B = 1,000; both 95% percentile intervals. Resampling units are photographs for Medetec, AZH and the control (Medetec 38 units for 152 images, AZH 118 units for 146 images, Control 71 units for 78 images) and single images otherwise. Resampling the units adds noise to every method mean and so pulls agreement down in each draw; h intervals are therefore shifted towards zero and understate how strong agreement is. Wound with non-wound: mean over the 28 pairs of a wound dataset with a non-wound dataset. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice.

Set	n	Wound pairs (6)	Wound with non-wound (28)	Colonoscopy pairs (10)	Modality difference (55)	Modality difference (52)	Control
U	18	0.622 m [0.337, 0.834] h [0.279, 0.825]	0.387 m [0.127, 0.585] h [0.100, 0.567]	0.777 m [0.565, 0.874] h [0.531, 0.884]	0.421 m [0.270, 0.543] h [0.269, 0.558]	0.389 m [0.240, 0.514] h [0.231, 0.528]	0.957 m [0.818, 1.000] h [0.769, 1.000]
F	14	0.314 m [−0.033, 0.626] h [−0.041, 0.641]	0.101 m [−0.228, 0.400] h [−0.208, 0.375]	0.882 m [0.677, 0.956] h [0.588, 0.942]	0.563 m [0.302, 0.822] h [0.253, 0.781]	0.501 m [0.221, 0.770] h [0.185, 0.744]	0.868 m [0.580, 0.986] h [0.233, 0.964]
K	14	0.260 m [−0.161, 0.655] h [−0.136, 0.667]	0.014 m [−0.249, 0.213] h [−0.257, 0.243]	0.877 m [0.705, 0.944] h [0.662, 0.932]	0.677 m [0.398, 0.957] h [0.363, 0.941]	0.675 m [0.374, 0.971] h [0.359, 0.949]	0.464 m [−0.169, 0.824] h [−0.472, 0.880]
M	11	0.308 m [−0.070, 0.643] h [−0.099, 0.673]	0.167 m [−0.085, 0.390] h [−0.112, 0.398]	0.719 m [0.327, 0.909] h [0.231, 0.869]	0.444 m [0.157, 0.693] h [0.103, 0.701]	0.399 m [0.069, 0.698] h [0.015, 0.682]	0.845 m [0.483, 0.972] h [0.177, 0.991]
M11s	11	0.561 m [0.120, 0.845] h [0.109, 0.851]	0.386 m [0.049, 0.696] h [0.049, 0.701]	0.643 m [0.268, 0.834] h [0.235, 0.837]	0.295 m [0.003, 0.560] h [−0.011, 0.559]	0.316 m [0.019, 0.600] h [−0.013, 0.582]	0.855 m [0.365, 1.000] h [0.213, 0.991]
M19	19	0.479 m [0.169, 0.725] h [0.122, 0.699]	0.251 m [0.005, 0.486] h [−0.006, 0.486]	0.577 m [0.302, 0.752] h [0.241, 0.745]	0.360 m [0.179, 0.522] h [0.133, 0.517]	0.336 m [0.133, 0.520] h [0.082, 0.508]	0.865 m [0.575, 0.977] h [0.379, 0.966]
N8s	8	0.393 m [−0.137, 0.800] h [−0.177, 0.785]	0.090 m [−0.038, 0.242] h [−0.128, 0.290]	−0.100 m [−0.230, 0.170] h [−0.227, 0.313]	0.004 m [−0.178, 0.195] h [−0.202, 0.253]	−0.164 m [−0.362, 0.087] h [−0.356, 0.200]	0.786 m [0.038, 1.000] h [−0.074, 1.000]
U+F	32	0.555 m [0.340, 0.729] h [0.302, 0.728]	−0.105 m [−0.273, 0.055] h [−0.278, 0.084]	0.860 m [0.754, 0.907] h [0.726, 0.901]	0.713 m [0.587, 0.829] h [0.536, 0.832]	0.678 m [0.548, 0.802] h [0.485, 0.802]	0.930 m [0.829, 0.966] h [0.696, 0.971]
U+K	32	0.554 m [0.352, 0.726] h [0.336, 0.726]	0.133 m [−0.066, 0.314] h [−0.055, 0.302]	0.862 m [0.738, 0.921] h [0.711, 0.923]	0.718 m [0.562, 0.872] h [0.547, 0.866]	0.700 m [0.541, 0.856] h [0.521, 0.843]	0.827 m [0.616, 0.947] h [0.510, 0.945]
U+M	29	0.864 m [0.787, 0.922] h [0.773, 0.920]	−0.303 m [−0.487, −0.141] h [−0.512, −0.144]	0.763 m [0.581, 0.860] h [0.575, 0.864]	0.922 m [0.760, 1.061] h [0.754, 1.074]	0.897 m [0.726, 1.043] h [0.722, 1.061]	0.977 m [0.936, 0.993] h [0.864, 0.991]
U+M11s	29	0.884 m [0.812, 0.937] h [0.790, 0.931]	−0.327 m [−0.504, −0.177] h [−0.506, −0.183]	0.782 m [0.611, 0.868] h [0.590, 0.868]	0.950 m [0.792, 1.088] h [0.770, 1.083]	0.930 m [0.764, 1.075] h [0.740, 1.075]	0.975 m [0.933, 0.992] h [0.863, 0.988]
U+M19	37	0.885 m [0.834, 0.928] h [0.807, 0.921]	−0.436 m [−0.576, −0.312] h [−0.574, −0.288]	0.766 m [0.627, 0.855] h [0.560, 0.859]	1.010 m [0.875, 1.123] h [0.823, 1.128]	0.987 m [0.843, 1.111] h [0.783, 1.119]	0.973 m [0.931, 0.989] h [0.850, 0.987]
ALL57	57	0.780 m [0.698, 0.850] h [0.674, 0.849]	−0.378 m [−0.481, −0.285] h [−0.486, −0.268]	0.890 m [0.837, 0.921] h [0.812, 0.913]	1.022 m [0.948, 1.099] h [0.914, 1.092]	1.004 m [0.925, 1.084] h [0.891, 1.077]	0.950 m [0.911, 0.971] h [0.836, 0.969]

Set	n	Wound pairs (6)	Wound with non-wound (28)	Colonoscopy pairs (10)	Modality difference (55)	Modality difference (52)	Control
ALL	65	0.843 m [0.788, 0.888] h [0.770, 0.888]	−0.450 m [−0.546, −0.362] h [−0.551, −0.352]	0.874 m [0.817, 0.908] h [0.801, 0.906]	1.073 m [0.998, 1.149] h [0.972, 1.148]	1.057 m [0.979, 1.137] h [0.951, 1.138]	0.963 m [0.937, 0.977] h [0.876, 0.974]

Table S2e. Exact tests of the modality difference (55 and 52 pairs). Exact relabelling tests of the modality difference. Exact p: share of the 13,860 distinct relabellings of the 11 modality labels whose modality difference is at least the observed one; (min) marks the smallest attainable p (2/13,860 = 0.00014). The 52-pair test drops the 3 same-source pairs under every relabelling. The last three columns give the category means that enter the 52-pair difference and the mean of the dropped same-source pairs. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice. Method sets and their sizes as defined in the Notation.

Metric	Set	n	Difference (55)	Exact p (55)	Difference (52)	Exact p (52)	Different source, same modality (13)	Across modality (39)	Same source (3, dropped)
Dice	U	18	0.596	0.00029	0.595	0.00014 (min)	0.668	0.072	0.672
Dice	F	14	0.467	0.0046	0.456	0.0091	0.600	0.144	0.659
Dice	K	14	0.498	0.00058	0.523	0.00014 (min)	0.680	0.156	0.543
Dice	M	11	0.465	0.00087	0.449	0.00043	0.474	0.025	0.561
Dice	M11s	11	0.329	0.0020	0.326	0.0019	0.390	0.064	0.409
Dice	M19	19	0.225	0.0061	0.174	0.018	0.278	0.104	0.549
Dice	N8s	8	0.055	0.33	−0.060	0.61	0.007	0.067	0.619
Dice	U+F	32	0.834	0.00014 (min)	0.812	0.00014 (min)	0.736	−0.075	0.854
Dice	U+K	32	0.677	0.00014 (min)	0.700	0.00014 (min)	0.701	0.000	0.578
Dice	U+M	29	1.089	0.00014 (min)	1.086	0.00014 (min)	0.841	−0.245	0.857
Dice	U+M11s	29	1.102	0.00014 (min)	1.100	0.00014 (min)	0.844	−0.256	0.851
Dice	U+M19	37	1.101	0.00014 (min)	1.092	0.00014 (min)	0.810	−0.281	0.858
Dice	ALL57	57	1.018	0.00014 (min)	1.008	0.00014 (min)	0.826	−0.182	0.879
Dice	ALL	65	1.062	0.00014 (min)	1.051	0.00014 (min)	0.839	−0.212	0.895
NSD	U	18	0.421	0.00014 (min)	0.389	0.00014 (min)	0.687	0.298	0.855
NSD	F	14	0.563	0.00014 (min)	0.501	0.00029	0.607	0.106	0.937
NSD	K	14	0.677	0.00043	0.675	0.00029	0.644	−0.031	0.654
NSD	M	11	0.444	0.00072	0.399	0.00087	0.520	0.121	0.761
NSD	M11s	11	0.295	0.0030	0.316	0.0016	0.634	0.317	0.518
NSD	M19	19	0.360	0.00029	0.336	0.00029	0.516	0.180	0.647
NSD	N8s	8	0.004	0.41	−0.164	0.88	−0.082	0.081	0.810
NSD	U+F	32	0.713	0.00014 (min)	0.678	0.00014 (min)	0.710	0.033	0.898
NSD	U+K	32	0.718	0.00014 (min)	0.700	0.00014 (min)	0.728	0.028	0.823
NSD	U+M	29	0.922	0.00014 (min)	0.897	0.00014 (min)	0.776	−0.121	0.910
NSD	U+M11s	29	0.950	0.00014 (min)	0.930	0.00014 (min)	0.800	−0.130	0.906
NSD	U+M19	37	1.010	0.00014 (min)	0.987	0.00014 (min)	0.788	−0.199	0.909
NSD	ALL57	57	1.022	0.00014 (min)	1.004	0.00014 (min)	0.831	−0.173	0.928
NSD	ALL	65	1.073	0.00014 (min)	1.057	0.00014 (min)	0.847	−0.211	0.930

Table S2f. Empty-mask-excluded Dice. Dice with images whose reference mask is empty excluded (per-method means recomputed without them; empty reference masks: FUSeg 4, AZH 10, Control 6; all other datasets have none). With empties: the main-analysis value. Exact p: share of the 13,860 distinct relabellings of the 11 modality labels whose modality difference is at least the observed one; (min) marks the smallest attainable p (2/13,860 = 0.00014). Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice.

Set	n	Modality difference (55), no empty	Exact p	Modality difference (52), no empty	Exact p	Modality difference (55), with empties	Wound pairs, no empty	Wound pairs, with empties
U	18	0.597	0.00029	0.595	0.00014 (min)	0.596	0.343	0.351
F	14	0.461	0.0068	0.448	0.014	0.467	0.187	0.164
K	14	0.501	0.00058	0.528	0.00029	0.498	0.349	0.363
M	11	0.464	0.0010	0.443	0.00087	0.465	0.120	0.105
M11s	11	0.343	0.0016	0.337	0.0014	0.329	0.315	0.274
M19	19	0.230	0.0049	0.182	0.012	0.225	0.237	0.206
N8s	8	0.055	0.33	−0.060	0.61	0.055	0.258	0.258
U+F	32	0.760	0.00014 (min)	0.753	0.00014 (min)	0.834	0.439	0.523
U+K	32	0.675	0.00014 (min)	0.699	0.00014 (min)	0.677	0.248	0.270
U+M	29	1.086	0.00014 (min)	1.083	0.00014 (min)	1.089	0.789	0.796
U+M11s	29	1.098	0.00014 (min)	1.097	0.00014 (min)	1.102	0.798	0.806
U+M19	37	1.097	0.00014 (min)	1.088	0.00014 (min)	1.101	0.815	0.818
ALL57	57	0.991	0.00014 (min)	0.986	0.00014 (min)	1.018	0.653	0.686
ALL	65	1.039	0.00014 (min)	1.033	0.00014 (min)	1.062	0.746	0.770

Table S3a. Leave one dataset out. Modality difference over the 45 pairs of the remaining 10 clinical references, with its exact relabelling p in brackets. The number of distinct relabellings depends on the dropped dataset (1,260, 5,040, 6,300); (min) marks the smallest attainable p for that design. Sets: U (18), M (11), M11s (11), M19 (19), N8s (8); definitions in the Notation. Agreement is the Spearman correlation of the per-method mean scores of two image sets over the methods of a method set.

Metric	Dropped dataset	U: difference (p)	M: difference (p)	M11s: difference (p)	M19: difference (p)	N8s: difference (p)
Dice	Medetec	0.648 (0.0012)	0.515 (0.0024)	0.332 (0.0044)	0.262 (0.0063)	0.060 (0.29)
Dice	CO2Wounds-V2	0.748 (0.00040 (min))	0.511 (0.0024)	0.362 (0.0016)	0.252 (0.0067)	−0.033 (0.54)
Dice	FUSeg	0.637 (0.00079)	0.621 (0.0012)	0.425 (0.00079)	0.310 (0.0016)	0.081 (0.24)
Dice	AZH	0.657 (0.00040 (min))	0.520 (0.0016)	0.316 (0.0032)	0.222 (0.0067)	−0.016 (0.48)
Dice	Kvasir	0.524 (0.0013)	0.364 (0.0038)	0.291 (0.0076)	0.200 (0.0070)	0.115 (0.26)
Dice	Kvasir-sessile	0.535 (0.0013)	0.379 (0.0038)	0.240 (0.0076)	0.168 (0.024)	0.066 (0.38)
Dice	CVC-ClinicDB	0.543 (0.0013)	0.350 (0.0038)	0.243 (0.011)	0.196 (0.039)	0.109 (0.26)
Dice	CVC-ColonDB	0.551 (0.0013)	0.461 (0.0038)	0.343 (0.0044)	0.191 (0.040)	0.064 (0.32)
Dice	ETIS-Larib	0.557 (0.0013)	0.375 (0.0038)	0.359 (0.0044)	0.243 (0.0083)	0.106 (0.26)
Dice	PH2	0.588 (0.00079 (min))	0.522 (0.0016)	0.351 (0.0063)	0.217 (0.0079)	−0.036 (0.53)
Dice	TN3K	0.535 (0.0016)	0.472 (0.0016)	0.357 (0.0016)	0.199 (0.020)	0.093 (0.25)
NSD	Medetec	0.448 (0.00040 (min))	0.487 (0.0016)	0.307 (0.0048)	0.390 (0.00040 (min))	−0.078 (0.67)
NSD	CO2Wounds-V2	0.500 (0.00040 (min))	0.466 (0.0016)	0.313 (0.0060)	0.372 (0.0012)	−0.098 (0.71)
NSD	FUSeg	0.456 (0.00040 (min))	0.565 (0.00040 (min))	0.336 (0.0028)	0.424 (0.00040 (min))	−0.007 (0.44)
NSD	AZH	0.447 (0.00040 (min))	0.538 (0.0016)	0.330 (0.0040)	0.375 (0.00040 (min))	−0.079 (0.66)
NSD	Kvasir	0.407 (0.00063 (min))	0.376 (0.0032)	0.259 (0.0095)	0.322 (0.00063 (min))	0.084 (0.30)
NSD	Kvasir-sessile	0.414 (0.00063 (min))	0.368 (0.0025)	0.252 (0.014)	0.344 (0.00063 (min))	0.067 (0.34)
NSD	CVC-ClinicDB	0.410 (0.00063 (min))	0.351 (0.0038)	0.252 (0.013)	0.350 (0.0019)	0.062 (0.32)
NSD	CVC-ColonDB	0.417 (0.00063 (min))	0.397 (0.0038)	0.291 (0.0044)	0.306 (0.0032)	0.034 (0.37)
NSD	ETIS-Larib	0.451 (0.00063 (min))	0.460 (0.0025)	0.326 (0.0070)	0.399 (0.00063 (min))	0.100 (0.26)
NSD	PH2	0.353 (0.00079 (min))	0.505 (0.00079 (min))	0.338 (0.00079 (min))	0.373 (0.0016)	−0.081 (0.64)
NSD	TN3K	0.299 (0.00079 (min))	0.351 (0.0040)	0.232 (0.011)	0.300 (0.0016)	0.054 (0.28)

Table S3b. Leave one method out of M19. Agreement summaries over the remaining 18 methods. Method keys as in Table S1, seed averages shown by architecture key (for example unet_r18 is the seed average of the three unet_r18 models). Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice. Agreement is the Spearman correlation of the per-method mean scores of two image sets over the methods of a method set.

Dropped method	Dice: modality difference (55)	Dice: modality difference (52)	Dice: wound pairs	NSD: modality difference (55)	NSD: modality difference (52)	NSD: wound pairs
none (all 19)	**0.225**	**0.174**	**0.206**	**0.360**	**0.336**	**0.479**
unet_r18	0.233	0.179	0.202	0.363	0.330	0.449
unetpp_r18	0.201	0.153	0.239	0.355	0.327	0.462
deeplabv3p_r18	0.227	0.183	0.246	0.341	0.320	0.508
fpn_r18	0.202	0.152	0.216	0.351	0.328	0.465
pspnet_r18	0.266	0.202	0.104	0.402	0.374	0.387
linknet_r18	0.213	0.163	0.197	0.341	0.317	0.459
manet_r18	0.194	0.141	0.169	0.356	0.326	0.444
segformer_mitb0	0.248	0.191	0.222	0.385	0.353	0.488
unet_mbv2	0.216	0.158	0.258	0.327	0.296	0.520
unet_effb0	0.246	0.189	0.221	0.381	0.355	0.493
unet_r50	0.234	0.189	0.228	0.357	0.340	0.480
deeplabv3p_r50	0.234	0.186	0.194	0.389	0.375	0.535
upernet_convnext_t	0.231	0.181	0.214	0.357	0.335	0.472
unet_swin_t	0.242	0.200	0.226	0.352	0.334	0.470
segformer_mitb2	0.258	0.217	0.174	0.353	0.329	0.453
unet_effb3	0.218	0.170	0.154	0.358	0.340	0.471
fpn_regnety016	0.227	0.171	0.210	0.366	0.340	0.457
manet_r50	0.222	0.175	0.256	0.379	0.363	0.530
nnunet_2d	0.149	0.095	0.168	0.316	0.281	0.530

Table S4a. Ranking reliability across training seeds (Dice). Same dataset, different training seed. r_st: Spearman agreement between the per-architecture mean scores of seed s and seed t on the same images. r-bar: mean of r01, r02 and r12. SB3: Spearman-Brown reliability of the 3-seed average, 3 r-bar / (1 + 2 r-bar), undefined (undef.) when r-bar is at most −0.5; negative values mean no reliability. Sets P9, N7 and C16 as defined in the Notation. Seed 0 of the original architectures comes from the first GPU session and seeds 1 and 2 from the second, so for P9 seed and session effects cannot be separated.

Dataset	P9 r01	P9 r02	P9 r12	P9 r-bar	P9 SB3	N7 r01	N7 r02	N7 r12	N7 r-bar	N7 SB3	C16 r01	C16 r02	C16 r12	C16 r-bar	C16 SB3
Medetec	0.450	0.733	0.750	0.644	0.845	0.536	−0.714	−0.393	−0.190	−0.923	0.456	0.300	0.244	0.333	0.600
CO2Wounds-V2	0.517	0.083	0.400	0.333	0.600	0.179	0.179	0.143	0.167	0.375	0.338	0.112	0.341	0.264	0.518
FUSeg	0.333	0.633	0.783	0.583	0.808	0.500	0.786	0.357	0.548	0.784	0.432	0.703	0.632	0.589	0.811
AZH	0.700	0.783	0.483	0.656	0.851	0.607	0.571	0.464	0.548	0.784	0.547	0.618	0.559	0.575	0.802
Kvasir	0.233	0.367	0.483	0.361	0.629	0.643	0.143	−0.179	0.202	0.432	0.509	0.591	0.503	0.534	0.775
Kvasir-sessile	0.433	0.650	0.633	0.572	0.801	0.643	0.429	0.643	0.571	0.800	0.456	0.726	0.647	0.610	0.824
CVC-ClinicDB	−0.417	−0.033	0.300	−0.050	−0.167	0.036	0.286	0.786	0.369	0.637	0.018	0.232	0.618	0.289	0.550
CVC-ColonDB	0.233	−0.250	0.250	0.078	0.202	−0.179	−0.036	0.214	0.000	0.000	−0.026	−0.138	0.268	0.034	0.096
ETIS-Larib	0.400	0.400	0.617	0.472	0.729	0.571	0.679	0.893	0.714	0.882	0.526	0.606	0.788	0.640	0.842
PH2	0.667	0.183	0.700	0.517	0.762	0.929	0.571	0.571	0.690	0.870	0.694	0.344	0.576	0.538	0.778
TN3K	0.289	0.251	0.559	0.367	0.635	−0.071	−0.536	0.000	−0.202	−1.020	0.388	0.056	0.484	0.309	0.573
Control	0.833	0.883	0.833	0.850	0.944	0.821	0.821	0.714	0.786	0.917	0.806	0.879	0.812	0.832	0.937

Table S4b. Ranking reliability across training seeds (NSD). Same dataset, different training seed. r_st: Spearman agreement between the per-architecture mean scores of seed s and seed t on the same images. r-bar: mean of r01, r02 and r12. SB3: Spearman-Brown reliability of the 3-seed average, 3 r-bar / (1 + 2 r-bar), undefined (undef.) when r-bar is at most −0.5; negative values mean no reliability. Sets P9, N7 and C16 as defined in the Notation. Seed 0 of the original architectures comes from the first GPU session and seeds 1 and 2 from the second, so for P9 seed and session effects cannot be separated.

Dataset	P9 r01	P9 r02	P9 r12	P9 r-bar	P9 SB3	N7 r01	N7 r02	N7 r12	N7 r-bar	N7 SB3	C16 r01	C16 r02	C16 r12	C16 r-bar	C16 SB3
Medetec	0.383	0.800	0.650	0.611	0.825	0.464	−0.143	−0.750	−0.143	−0.600	0.497	0.338	−0.003	0.277	0.535
CO2Wounds-V2	0.433	−0.100	0.700	0.344	0.612	0.143	0.179	−0.071	0.083	0.214	0.309	0.206	0.218	0.244	0.492
FUSeg	0.850	0.833	0.983	0.889	0.960	0.321	0.821	0.464	0.536	0.776	0.656	0.824	0.768	0.749	0.900
AZH	0.850	0.833	0.783	0.822	0.933	0.321	0.107	0.464	0.298	0.560	0.482	0.535	0.662	0.560	0.792
Kvasir	0.433	0.567	0.500	0.500	0.750	0.821	0.464	0.143	0.476	0.732	0.641	0.741	0.509	0.630	0.837
Kvasir-sessile	0.717	0.817	0.817	0.783	0.916	0.857	0.786	0.571	0.738	0.894	0.621	0.744	0.771	0.712	0.881
CVC-ClinicDB	−0.067	0.233	0.000	0.056	0.150	−0.714	−0.286	0.357	−0.214	−1.125	−0.259	0.321	0.182	0.081	0.210
CVC-ColonDB	0.217	0.150	−0.167	0.067	0.176	−0.714	0.500	−0.464	−0.226	−1.239	−0.050	0.550	−0.159	0.114	0.278
ETIS-Larib	0.600	0.317	0.717	0.544	0.782	0.500	−0.071	0.429	0.286	0.545	0.409	0.121	0.594	0.375	0.642
PH2	0.383	0.500	0.250	0.378	0.646	0.857	0.714	0.393	0.655	0.851	0.459	0.409	0.359	0.409	0.675
TN3K	−0.120	0.093	0.093	0.022	0.063	0.198	−0.559	−0.179	−0.180	−0.841	0.203	−0.012	0.234	0.141	0.330
Control	0.983	0.917	0.867	0.922	0.973	0.321	0.643	0.714	0.560	0.792	0.735	0.809	0.832	0.792	0.920

Table S4c. Seed-to-seed agreement on the same dataset, by dataset group. Mean r-bar (Table S4a, S4b) over dataset groups: the 4 wound datasets, the 5 colonoscopy datasets, all 11 clinical references, and the control. Each cell: estimate, then m and h intervals. m: architecture bootstrap (architectures resampled with replacement; the same draw for all seeds and for the seed average), B = 2,000; h: hierarchical bootstrap (architectures and resampling units), B = 1,000; both 95% percentile intervals.

Metric	Set	n	Wound (4)	Colonoscopy (5)	All 11 references	Control
Dice	P9	9	0.554 m [0.154, 0.802] h [0.199, 0.766]	0.287 m [−0.150, 0.600] h [−0.125, 0.617]	0.412 m [0.039, 0.662] h [0.067, 0.648]	0.850 m [0.478, 0.988] h [0.214, 1.000]
Dice	N7	7	0.268 m [−0.095, 0.590] h [−0.134, 0.617]	0.371 m [0.101, 0.488] h [0.054, 0.567]	0.311 m [0.141, 0.445] h [0.106, 0.457]	0.786 m [0.216, 0.974] h [0.137, 1.000]
Dice	C16	16	0.440 m [0.174, 0.643] h [0.186, 0.646]	0.422 m [0.198, 0.586] h [0.171, 0.582]	0.429 m [0.240, 0.562] h [0.236, 0.559]	0.832 m [0.547, 0.942] h [0.480, 0.935]
NSD	P9	9	0.667 m [0.333, 0.859] h [0.267, 0.865]	0.390 m [−0.023, 0.656] h [−0.010, 0.671]	0.456 m [0.185, 0.633] h [0.195, 0.617]	0.922 m [0.667, 1.000] h [0.417, 1.000]
NSD	N7	7	0.193 m [−0.186, 0.552] h [−0.167, 0.585]	0.212 m [−0.030, 0.414] h [−0.067, 0.535]	0.210 m [0.013, 0.364] h [0.039, 0.417]	0.560 m [−0.333, 0.869] h [−0.333, 0.926]
NSD	C16	16	0.458 m [0.178, 0.682] h [0.180, 0.678]	0.382 m [0.177, 0.544] h [0.168, 0.555]	0.390 m [0.213, 0.530] h [0.217, 0.523]	0.792 m [0.469, 0.946] h [0.387, 0.934]

Table S4d. Single seeds, seed averages and the dataset gap (Dice). Cross-dataset agreement with single seeds and with seed averages. Seed s: every architecture represented by its seed-s model. Mean of single seeds: mean over seeds 0, 1 and 2. Seed average: the seed-averaged columns of the same architectures. Different wound dataset, different seed: seed s on one wound dataset against seed t on another (s different from t; 6 ordered seed pairs times 6 wound pairs), the counterpart of the same-dataset r-bar, so that training noise enters both and only the dataset differs. Gap: r-bar minus that value. Each cell: estimate, then m and h intervals. m: architecture bootstrap (architectures resampled with replacement; the same draw for all seeds and for the seed average), B = 2,000; h: hierarchical bootstrap (architectures and resampling units), B = 1,000; both 95% percentile intervals. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice.

Quantity	P9 (n 9)	N7 (n 7)	C16 (n 16)
Wound pairs, seed 0	0.033 m [−0.219, 0.377] h [−0.246, 0.397]	0.083 m [−0.314, 0.602] h [−0.294, 0.642]	0.035 m [−0.170, 0.264] h [−0.171, 0.333]
Wound pairs, seed 1	0.094 m [−0.137, 0.398] h [−0.219, 0.578]	0.482 m [0.000, 0.943] h [−0.098, 0.933]	0.247 m [−0.041, 0.539] h [−0.070, 0.512]
Wound pairs, seed 2	0.178 m [−0.292, 0.713] h [−0.287, 0.690]	0.298 m [−0.077, 0.712] h [−0.142, 0.804]	0.234 m [−0.087, 0.567] h [−0.113, 0.595]
Wound pairs, mean of single seeds	0.102 m [−0.178, 0.439] h [−0.195, 0.461]	0.288 m [0.010, 0.617] h [−0.028, 0.580]	0.172 m [−0.049, 0.397] h [−0.051, 0.397]
Wound pairs, seed average	0.122 m [−0.278, 0.623] h [−0.264, 0.688]	0.399 m [−0.111, 0.885] h [−0.250, 0.849]	0.170 m [−0.144, 0.497] h [−0.154, 0.516]
Seed average minus mean of single seeds	0.020 m [−0.212, 0.253] h [−0.175, 0.329]	0.111 m [−0.255, 0.421] h [−0.427, 0.441]	−0.002 m [−0.145, 0.172] h [−0.188, 0.185]
Same wound dataset, different seed (r-bar)	0.554 m [0.154, 0.802] h [0.199, 0.766]	0.268 m [−0.095, 0.590] h [−0.134, 0.617]	0.440 m [0.174, 0.643] h [0.186, 0.646]
Different wound dataset, different seed	0.059 m [−0.206, 0.413] h [−0.199, 0.454]	0.189 m [−0.159, 0.560] h [−0.170, 0.518]	0.090 m [−0.122, 0.325] h [−0.125, 0.334]
Gap: same dataset minus different dataset (both different seed)	0.495 m [0.091, 0.849] h [0.077, 0.812]	0.078 m [−0.130, 0.299] h [−0.161, 0.486]	0.350 m [0.110, 0.573] h [0.113, 0.612]
Colonoscopy pairs, mean of single seeds	0.481 m [0.223, 0.652] h [0.125, 0.622]	0.050 m [−0.167, 0.224] h [−0.141, 0.303]	0.389 m [0.229, 0.521] h [0.154, 0.481]
Colonoscopy pairs, seed average	0.300 m [−0.049, 0.695] h [−0.120, 0.716]	0.054 m [−0.200, 0.486] h [−0.208, 0.513]	0.346 m [0.060, 0.590] h [0.035, 0.566]
Modality difference, mean of single seeds	0.088 m [−0.019, 0.206] h [−0.054, 0.239]	0.026 m [−0.140, 0.248] h [−0.153, 0.223]	0.127 m [0.026, 0.231] h [−0.006, 0.225]

Quantity	P9 (n 9)	N7 (n 7)	C16 (n 16)
Modality difference, seed average	0.007 m [−0.103, 0.151] h [−0.158, 0.248]	0.088 m [−0.155, 0.406] h [−0.219, 0.317]	0.102 m [−0.009, 0.221] h [−0.070, 0.253]
Control, mean of single seeds	0.739 m [0.187, 0.983] h [0.084, 0.963]	0.821 m [0.372, 1.000] h [0.333, 1.000]	0.807 m [0.549, 0.925] h [0.471, 0.931]
Control, seed average	0.800 m [0.165, 1.000] h [−0.128, 1.000]	1.000 m [1.000, 1.000] h [0.412, 1.000]	0.879 m [0.584, 0.985] h [0.433, 0.985]

Table S4e. Single seeds, seed averages and the dataset gap (NSD). Cross-dataset agreement with single seeds and with seed averages. Seed s: every architecture represented by its seed-s model. Mean of single seeds: mean over seeds 0, 1 and 2. Seed average: the seed-averaged columns of the same architectures. Different wound dataset, different seed: seed s on one wound dataset against seed t on another (s different from t; 6 ordered seed pairs times 6 wound pairs), the counterpart of the same-dataset r-bar, so that training noise enters both and only the dataset differs. Gap: r-bar minus that value. Each cell: estimate, then m and h intervals. m: architecture bootstrap (architectures resampled with replacement; the same draw for all seeds and for the seed average), B = 2,000; h: hierarchical bootstrap (architectures and resampling units), B = 1,000; both 95% percentile intervals. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice.

Quantity	P9 (n 9)	N7 (n 7)	C16 (n 16)
Wound pairs, seed 0	0.414 m [−0.056, 0.778] h [−0.108, 0.764]	0.387 m [−0.123, 0.804] h [−0.157, 0.778]	0.387 m [0.059, 0.636] h [0.042, 0.621]
Wound pairs, seed 1	0.767 m [0.264, 0.974] h [0.107, 0.948]	0.298 m [−0.050, 0.673] h [−0.059, 0.784]	0.514 m [0.209, 0.776] h [0.132, 0.742]
Wound pairs, seed 2	0.572 m [−0.009, 0.915] h [−0.099, 0.890]	0.268 m [−0.069, 0.699] h [−0.193, 0.793]	0.426 m [0.104, 0.695] h [0.026, 0.684]
Wound pairs, mean of single seeds	0.584 m [0.186, 0.822] h [0.054, 0.771]	0.317 m [0.072, 0.588] h [−0.022, 0.580]	0.442 m [0.211, 0.617] h [0.170, 0.613]
Wound pairs, seed average	0.586 m [−0.074, 0.907] h [0.000, 0.921]	0.500 m [−0.098, 0.882] h [−0.157, 0.862]	0.554 m [0.190, 0.791] h [0.152, 0.770]
Seed average minus mean of single seeds	0.002 m [−0.316, 0.255] h [−0.165, 0.313]	0.183 m [−0.248, 0.587] h [−0.300, 0.538]	0.112 m [−0.099, 0.293] h [−0.084, 0.272]
Same wound dataset, different seed (r-bar)	0.667 m [0.333, 0.859] h [0.267, 0.865]	0.193 m [−0.186, 0.552] h [−0.167, 0.585]	0.458 m [0.178, 0.682] h [0.180, 0.678]
Different wound dataset, different seed	0.476 m [−0.005, 0.770] h [−0.035, 0.741]	0.184 m [−0.041, 0.488] h [−0.095, 0.490]	0.317 m [0.066, 0.542] h [0.046, 0.527]
Gap: same dataset minus different dataset (both different seed)	0.191 m [0.007, 0.475] h [0.002, 0.553]	0.010 m [−0.229, 0.230] h [−0.241, 0.376]	0.140 m [−0.013, 0.303] h [−0.005, 0.356]
Colonoscopy pairs, mean of single seeds	0.519 m [0.292, 0.670] h [0.240, 0.683]	0.027 m [−0.112, 0.208] h [−0.145, 0.214]	0.483 m [0.300, 0.598] h [0.254, 0.575]
Colonoscopy pairs, seed average	0.602 m [0.172, 0.875] h [0.119, 0.875]	−0.193 m [−0.241, −0.076] h [−0.238, 0.107]	0.551 m [0.198, 0.752] h [0.150, 0.741]
Modality difference, mean of single seeds	0.162 m [0.015, 0.306] h [0.013, 0.282]	0.068 m [−0.082, 0.233] h [−0.115, 0.222]	0.257 m [0.138, 0.356] h [0.114, 0.359]

Quantity	P9 (n 9)	N7 (n 7)	C16 (n 16)
Modality difference, seed average	0.101 m [−0.073, 0.313] h [−0.090, 0.345]	0.033 m [−0.138, 0.230] h [−0.196, 0.306]	0.332 m [0.157, 0.484] h [0.101, 0.511]
Control, mean of single seeds	0.772 m [0.304, 0.937] h [0.093, 0.948]	0.833 m [0.522, 1.000] h [0.333, 1.000]	0.819 m [0.550, 0.925] h [0.399, 0.919]
Control, seed average	0.783 m [0.166, 1.000] h [−0.011, 1.000]	0.679 m [−0.185, 1.000] h [−0.176, 1.000]	0.806 m [0.348, 0.970] h [0.281, 0.976]

Table S4f. Single-seed method sets. Point estimates. Single-seed sets P10_sk, N8_sk, M_sk and M19_sk as defined in the Notation (M_s0 is the set M). nnU-Net has one training and appears in every single-seed set that includes it. Set sizes differ because of the diverged trainings. Seed-averaged sets (M11s, M19, N8s) for comparison. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB). Control: AZH with its training slice. Agreement is the Spearman correlation of the per-method mean scores of two image sets over the methods of a method set.

Metric	Set	n	Wound pairs	Colonoscopy pairs	Modality difference (55)	Control
Dice	M_s0 (= M)	11	0.105	0.722	0.465	0.809
Dice	M_s1	11	0.291	0.174	0.024	0.673
Dice	M_s2	10	0.232	0.579	0.180	0.939
Dice	M11s	11	0.274	0.465	0.329	0.882
Dice	M19_s0	18	0.078	0.550	0.360	0.866
Dice	M19_s1	19	0.268	0.141	0.061	0.796
Dice	M19_s2	17	0.256	0.481	0.109	0.826
Dice	M19	19	0.206	0.403	0.225	0.916
Dice	N8_s0	7	0.083	−0.096	0.067	0.929
Dice	N8_s1	8	0.425	−0.067	0.049	0.905
Dice	N8_s2	7	0.298	0.329	−0.069	0.607
Dice	N8s	8	0.258	0.040	0.055	1.000
NSD	M_s0 (= M)	11	0.308	0.719	0.444	0.845
NSD	M_s1	11	0.621	0.403	0.138	0.782
NSD	M_s2	10	0.469	0.613	0.172	0.842
NSD	M11s	11	0.561	0.643	0.295	0.855
NSD	M19_s0	18	0.334	0.728	0.528	0.874
NSD	M19_s1	19	0.452	0.262	0.124	0.844
NSD	M19_s2	17	0.379	0.553	0.181	0.809
NSD	M19	19	0.479	0.577	0.360	0.865
NSD	N8_s0	7	0.387	−0.071	0.212	0.964
NSD	N8_s1	8	0.313	0.012	0.041	0.929
NSD	N8_s2	7	0.268	0.211	−0.052	0.643
NSD	N8s	8	0.393	−0.100	0.004	0.786

Table S4g. Reliabilities for the noise correction (C16). Used to correct agreement for seed noise and image-sampling noise. Model: observed method mean = true value + seed error + image-sampling error. R_img: split-half reliability over 500 random splits of the resampling units (Spearman-Brown median), for the seed-averaged columns and as the mean over seeds of the single-seed values. R_seed: seed reliability, r-bar for one seed and its Spearman-Brown value for 3 seeds. Additive rule: R = R_img + R_seed − 1; product rule: R = R_img R_seed. n/a: undefined. The additive rule over-corrects if part of the image error is specific to a seed; both rules are written for Pearson correlations and are approximate for Spearman.

Metric	Dataset	R_img, seed average	R_img, single seeds	R_seed, 1 seed	R_seed, 3 seeds	R additive, seed average	R additive, single seed	R product, seed average
Dice	Medetec	0.800	0.827	0.333	0.600	0.400	0.161	0.480
Dice	CO2Wounds-V2	0.858	0.854	0.264	0.518	0.376	0.117	0.444
Dice	FUSeg	0.821	0.738	0.589	0.811	0.633	0.327	0.667
Dice	AZH	0.932	0.889	0.575	0.802	0.735	0.464	0.748
Dice	Kvasir	0.939	0.941	0.534	0.775	0.714	0.476	0.728
Dice	Kvasir-sessile	0.915	0.854	0.610	0.824	0.740	0.464	0.755
Dice	CVC-ClinicDB	0.826	0.856	0.289	0.550	0.375	0.145	0.454
Dice	CVC-ColonDB	0.667	0.719	0.034	0.096	−0.237	−0.247	0.064
Dice	ETIS-Larib	0.819	0.727	0.640	0.842	0.662	0.367	0.690
Dice	PH2	0.912	0.930	0.538	0.778	0.690	0.468	0.709
Dice	TN3K	0.865	0.748	0.309	0.573	0.438	0.057	0.496
Dice	Control	0.955	0.924	0.832	0.937	0.893	0.756	0.895
NSD	Medetec	0.900	0.925	0.277	0.535	0.435	0.203	0.482
NSD	CO2Wounds-V2	0.883	0.889	0.244	0.492	0.376	0.133	0.435
NSD	FUSeg	0.960	0.942	0.749	0.900	0.860	0.691	0.864
NSD	AZH	0.941	0.926	0.560	0.792	0.733	0.485	0.745
NSD	Kvasir	0.965	0.932	0.630	0.837	0.802	0.563	0.807
NSD	Kvasir-sessile	0.947	0.908	0.712	0.881	0.828	0.620	0.835
NSD	CVC-ClinicDB	0.846	0.831	0.081	0.210	0.055	−0.088	0.178
NSD	CVC-ColonDB	0.809	0.674	0.114	0.278	0.087	−0.213	0.225
NSD	ETIS-Larib	0.844	0.843	0.375	0.642	0.486	0.218	0.542
NSD	PH2	0.842	0.769	0.409	0.675	0.516	0.178	0.568
NSD	TN3K	0.600	n/a	0.141	0.330	−0.070	n/a	0.198
NSD	Control	0.932	0.907	0.792	0.920	0.852	0.700	0.858

Table S4h. Wound-pair agreement corrected for seed and image noise (C16). Observed: agreement of the seed averages and the mean over single seeds. Corrected: r / sqrt(R_A R_B) with R for image noise only, seed noise only, both (additive rule), both (product rule), and the additive rule applied to single seeds. Values above 1 occur when the reliabilities are small; they show that the correction is fragile, not that agreement exceeds 1.

Metric	Pair	Observed, seed average	Observed, single seeds	Image noise only	Seed noise only	Both, additive	Both, product	Additive, from single seeds
Dice	Medetec with CO2Wounds-V2	0.676	0.522	0.817	1.213	1.745	1.465	3.801
Dice	Medetec with FUSeg	−0.206	0.000	−0.254	−0.295	−0.409	−0.364	0.000
Dice	Medetec with AZH	0.162	0.202	0.187	0.233	0.299	0.270	0.740
Dice	CO2Wounds-V2 with FUSeg	−0.006	−0.037	−0.007	−0.009	−0.012	−0.011	−0.190
Dice	CO2Wounds-V2 with AZH	0.224	0.287	0.250	0.347	0.425	0.388	1.231
Dice	FUSeg with AZH	0.171	0.059	0.195	0.211	0.250	0.242	0.151
NSD	Medetec with CO2Wounds-V2	0.800	0.728	0.897	1.559	1.979	1.748	4.428
NSD	Medetec with FUSeg	0.297	0.108	0.320	0.428	0.486	0.461	0.288
NSD	Medetec with AZH	0.471	0.473	0.512	0.723	0.833	0.785	1.506
NSD	CO2Wounds-V2 with FUSeg	0.629	0.227	0.683	0.946	1.108	1.027	0.749
NSD	CO2Wounds-V2 with AZH	0.594	0.640	0.652	0.951	1.132	1.044	2.515
NSD	FUSeg with AZH	0.535	0.477	0.563	0.634	0.674	0.667	0.824

Table S4i. Corrected agreement by pair category. Corrected cells: raw mean / mean after capping each pair at −1 and 1; in brackets the number of pairs where the correction is defined (R above 0 for both datasets) of all pairs, and the number of pairs with absolute corrected value above 1. Columns as in Table S4h.

Metric	Set	Category	Observed, seed average	Observed, single seeds	Image noise only	Seed noise only	Both, additive	Both, product	Additive, from single seeds
Dice	P9	wound pairs (6)	0.122	0.102	0.136 / 0.136 (6 of 6; 0)	0.176 / 0.176 (6 of 6; 0)	0.208 / 0.192 (6 of 6; 1)	0.196 / 0.189 (6 of 6; 1)	0.313 / 0.214 (6 of 6; 1)
Dice	P9	colonoscopy pairs (10)	0.300	0.481	0.381 / 0.359 (10 of 10; 2)	0.337 / 0.337 (6 of 10; 0)	0.604 / 0.590 (3 of 10; 1)	0.388 / 0.388 (6 of 10; 0)	2.396 / 1.000 (3 of 10; 3)
Dice	P9	control AZH with control	0.800	0.739	0.866 / 0.866 (1 of 1; 0)	0.892 / 0.892 (1 of 1; 0)	0.977 / 0.977 (1 of 1; 0)	0.966 / 0.966 (1 of 1; 0)	1.206 / 1.000 (1 of 1; 1)
Dice	N7	wound pairs (6)	0.399	0.288	0.566 / 0.566 (6 of 6; 0)	0.842 / 0.803 (3 of 6; 1)	0.823 / 0.823 (1 of 6; 0)	1.163 / 0.929 (3 of 6; 2)	0.735 / 0.735 (1 of 6; 0)
Dice	N7	colonoscopy pairs (10)	0.054	0.050	0.076 / 0.045 (10 of 10; 3)	−0.136 / −0.148 (6 of 10; 3)	−0.207 / −0.226 (6 of 10; 4)	−0.161 / −0.209 (6 of 10; 4)	0.229 / −0.333 (6 of 10; 6)
Dice	N7	control AZH with control	1.000	0.821	1.038 / 1.000 (1 of 1; 1)	1.180 / 1.000 (1 of 1; 1)	1.234 / 1.000 (1 of 1; 1)	1.225 / 1.000 (1 of 1; 1)	1.393 / 1.000 (1 of 1; 1)
Dice	C16	wound pairs (6)	0.170	0.172	0.198 / 0.198 (6 of 6; 0)	0.283 / 0.248 (6 of 6; 1)	0.383 / 0.259 (6 of 6; 1)	0.332 / 0.254 (6 of 6; 1)	0.956 / 0.450 (6 of 6; 2)
Dice	C16	colonoscopy pairs (10)	0.346	0.389	0.418 / 0.418 (10 of 10; 0)	0.796 / 0.614 (10 of 10; 3)	0.602 / 0.549 (6 of 10; 2)	1.003 / 0.660 (10 of 10; 4)	1.316 / 0.872 (6 of 10; 4)
Dice	C16	control AZH with control	0.879	0.807	0.932 / 0.932 (1 of 1; 0)	1.014 / 1.000 (1 of 1; 1)	1.086 / 1.000 (1 of 1; 1)	1.075 / 1.000 (1 of 1; 1)	1.362 / 1.000 (1 of 1; 1)
NSD	P9	wound pairs (6)	0.586	0.584	0.629 / 0.629 (6 of 6; 0)	0.732 / 0.704 (6 of 6; 1)	0.812 / 0.756 (6 of 6; 1)	0.787 / 0.741 (6 of 6; 1)	1.334 / 0.892 (6 of 6; 3)
NSD	P9	colonoscopy pairs (10)	0.602	0.519	0.656 / 0.651 (10 of 10; 1)	1.417 / 0.946 (10 of 10; 8)	3.070 / 0.965 (10 of 10; 9)	1.557 / 0.960 (10 of 10; 8)	1.253 / 0.927 (3 of 10; 2)

Metric	Set	Category	Observed, seed average	Observed, single seeds	Image noise only	Seed noise only	Both, additive	Both, product	Additive, from single seeds
NSD	P9	control AZH with control	0.783	0.772	0.827 / 0.827 (1 of 1; 0)	0.822 / 0.822 (1 of 1; 0)	0.871 / 0.871 (1 of 1; 0)	0.869 / 0.869 (1 of 1; 0)	0.979 / 0.979 (1 of 1; 0)
NSD	N7	wound pairs (6)	0.500	0.317	0.562 / 0.562 (6 of 6; 0)	1.275 / 1.000 (3 of 6; 3)	2.120 / 1.000 (3 of 6; 3)	1.424 / 1.000 (3 of 6; 3)	0.697 / 0.697 (1 of 6; 0)
NSD	N7	colonoscopy pairs (10)	−0.193	0.027	−0.337 / −0.322 (6 of 10; 3)	−0.265 / −0.256 (3 of 10; 2)	−0.445 / −0.333 (3 of 10; 3)	−0.446 / −0.389 (6 of 10; 5)	−7.704 / −0.333 (3 of 10; 3)
NSD	N7	control AZH with control	0.679	0.833	0.781 / 0.781 (1 of 1; 0)	1.019 / 1.000 (1 of 1; 1)	1.270 / 1.000 (1 of 1; 1)	1.173 / 1.000 (1 of 1; 1)	2.609 / 1.000 (1 of 1; 1)
NSD	C16	wound pairs (6)	0.554	0.442	0.604 / 0.604 (6 of 6; 0)	0.873 / 0.780 (6 of 6; 1)	1.035 / 0.832 (6 of 6; 3)	0.955 / 0.819 (6 of 6; 3)	1.718 / 0.810 (6 of 6; 3)
NSD	C16	colonoscopy pairs (10)	0.551	0.483	0.625 / 0.625 (10 of 10; 0)	1.209 / 0.901 (10 of 10; 6)	2.612 / 0.921 (10 of 10; 8)	1.392 / 0.914 (10 of 10; 8)	1.298 / 1.000 (3 of 10; 3)
NSD	C16	control AZH with control	0.806	0.819	0.860 / 0.860 (1 of 1; 0)	0.944 / 0.944 (1 of 1; 0)	1.020 / 1.000 (1 of 1; 1)	1.008 / 1.000 (1 of 1; 1)	1.405 / 1.000 (1 of 1; 1)

Table S4j. Spread of modern model means against seed noise. Dice and NSD. M19 columns: mean, range and SD of the 19 method means (seed averages) and the median gap between neighbours after sorting. Between SD: SD of the 16 seed-averaged C16 means. Seed SD: SD over the 3 seeds of one architecture's mean, pooled over C16 as the root mean variance, and its median; SE of a 3-seed average = pooled seed SD / sqrt(3). ICC: share of the between-architecture variance that is not seed noise, for one seed and for a 3-seed average (negative when the seed noise exceeds the observed spread). Image SE: median over M19 of the SD of a method mean over 1,000 resamples of the units; it is largely shared by all methods, so it moves rankings less than its size suggests.

Metric	Dataset	M19 mean	M19 range	M19 SD	Median neighbour gap	Between SD (C16)	Seed SD, pooled	Seed SD, median	SE of 3-seed average	Between / seed SD	ICC, one seed	ICC, 3-seed average	Image SE, median
Dice	Medetec	0.709	0.203	0.045	0.005	0.043	0.033	0.025	0.019	1.29	0.572	0.801	0.037
Dice	CO2Wounds-V2	0.438	0.069	0.022	0.004	0.020	0.024	0.021	0.014	0.84	0.268	0.524	0.028
Dice	FUSeg	0.798	0.090	0.019	0.001	0.019	0.010	0.006	0.006	1.93	0.773	0.911	0.022
Dice	AZH	0.582	0.205	0.047	0.006	0.049	0.031	0.018	0.018	1.61	0.692	0.871	0.037
Dice	Kvasir	0.063	0.093	0.027	0.003	0.029	0.027	0.021	0.016	1.07	0.448	0.709	0.012
Dice	Kvasir-sessile	0.022	0.044	0.012	0.001	0.013	0.011	0.007	0.006	1.18	0.515	0.761	0.007
Dice	CVC-ClinicDB	0.068	0.043	0.012	0.002	0.012	0.016	0.011	0.009	0.80	0.237	0.483	0.014
Dice	CVC-ColonDB	0.007	0.013	0.004	0.000	0.004	0.008	0.006	0.004	0.49	−0.107	−0.408	0.002
Dice	ETIS-Larib	0.023	0.016	0.005	0.001	0.005	0.004	0.003	0.002	1.44	0.634	0.839	0.009
Dice	PH2	0.048	0.434	0.096	0.002	0.015	0.016	0.014	0.009	0.94	0.358	0.626	0.007
Dice	TN3K	0.015	0.140	0.031	0.001	0.007	0.010	0.004	0.006	0.67	0.105	0.261	0.002
Dice	Control	0.480	0.337	0.087	0.010	0.094	0.040	0.026	0.023	2.34	0.837	0.939	0.043
NSD	Medetec	0.403	0.209	0.049	0.006	0.046	0.050	0.039	0.029	0.92	0.340	0.607	0.041
NSD	CO2Wounds-V2	0.252	0.068	0.018	0.002	0.019	0.024	0.015	0.014	0.79	0.221	0.460	0.021
NSD	FUSeg	0.578	0.276	0.069	0.004	0.066	0.030	0.020	0.017	2.17	0.815	0.929	0.024
NSD	AZH	0.491	0.241	0.058	0.009	0.057	0.038	0.023	0.022	1.51	0.661	0.854	0.035
NSD	Kvasir	0.032	0.055	0.015	0.002	0.016	0.012	0.008	0.007	1.30	0.575	0.802	0.006
NSD	Kvasir-sessile	0.014	0.033	0.009	0.001	0.010	0.007	0.004	0.004	1.33	0.591	0.812	0.004
NSD	CVC-ClinicDB	0.037	0.025	0.007	0.001	0.008	0.011	0.009	0.007	0.68	0.112	0.275	0.007
NSD	CVC-ColonDB	0.007	0.014	0.004	0.000	0.004	0.006	0.004	0.003	0.70	0.132	0.314	0.002
NSD	ETIS-Larib	0.019	0.025	0.007	0.001	0.006	0.006	0.004	0.004	1.01	0.408	0.674	0.005
NSD	PH2	0.011	0.099	0.022	0.000	0.003	0.004	0.003	0.002	0.93	0.344	0.611	0.002
NSD	TN3K	0.007	0.072	0.016	0.000	0.002	0.005	0.002	0.003	0.52	−0.063	−0.215	0.002
NSD	Control	0.409	0.311	0.086	0.010	0.088	0.038	0.025	0.022	2.34	0.837	0.939	0.042

Table S5a. Benchmark validity by method set (mean over wound references). Generator B at texture amplitude 40 and 11. Benchmark agreement (bench) with a wound reference R: Spearman agreement of mean Dice on generator B with R, averaged over the 3 synthetic seeds (2024, 7, 101); each synthetic set has 45 images. Other: mean agreement of the other 3 wound datasets with R. Diff = bench minus other: positive means the benchmark ranks the methods more like R than another wound dataset does. Mean: average over the 4 wound references (other then equals the mean of the 6 wound pairs). Dice only. Method sets as defined in the Notation. Each cell: estimate [interval]. Intervals: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000, 95% percentile.

Set	n	Bench, amplitude 40	Diff, amplitude 40	Bench, amplitude 11	Diff, amplitude 11	Other (wound with wound)
U	18	0.526 [0.217, 0.740]	0.175 [−0.071, 0.323]	0.450 [0.122, 0.673]	0.099 [−0.124, 0.258]	0.351 [0.122, 0.594]
F	14	0.229 [−0.038, 0.450]	0.065 [−0.419, 0.414]	0.202 [−0.106, 0.437]	0.038 [−0.444, 0.405]	0.164 [−0.152, 0.531]
K	14	0.112 [−0.350, 0.536]	−0.250 [−0.649, 0.082]	0.142 [−0.310, 0.559]	−0.221 [−0.594, 0.089]	0.363 [0.048, 0.636]
M	11	0.336 [−0.068, 0.593]	0.232 [−0.186, 0.424]	0.355 [−0.058, 0.618]	0.251 [−0.160, 0.446]	0.105 [−0.121, 0.377]
M11s	11	0.408 [0.014, 0.717]	0.134 [−0.186, 0.364]	0.472 [0.082, 0.773]	0.198 [−0.026, 0.377]	0.274 [−0.111, 0.650]
M19	19	0.205 [−0.167, 0.485]	−0.001 [−0.406, 0.299]	0.287 [−0.100, 0.563]	0.081 [−0.329, 0.345]	0.206 [−0.070, 0.491]
P10s	10	0.302 [−0.052, 0.608]	0.100 [−0.247, 0.371]	0.381 [0.014, 0.744]	0.179 [−0.104, 0.390]	0.202 [−0.212, 0.632]
N8s	8	−0.127 [−0.611, 0.555]	−0.385 [−1.173, 0.307]	0.024 [−0.547, 0.632]	−0.234 [−1.123, 0.433]	0.258 [−0.149, 0.722]
U+F	32	0.346 [0.071, 0.573]	−0.178 [−0.424, 0.040]	0.137 [−0.168, 0.402]	−0.387 [−0.668, −0.114]	0.523 [0.294, 0.694]
U+K	32	0.319 [0.063, 0.540]	0.049 [−0.112, 0.174]	0.267 [0.017, 0.481]	−0.003 [−0.149, 0.142]	0.270 [0.105, 0.432]
U+M	29	0.720 [0.525, 0.843]	−0.077 [−0.258, 0.035]	0.424 [0.090, 0.684]	−0.372 [−0.699, −0.120]	0.796 [0.742, 0.858]
U+M11s	29	0.687 [0.447, 0.842]	−0.118 [−0.343, 0.031]	0.388 [0.024, 0.658]	−0.418 [−0.766, −0.152]	0.806 [0.748, 0.868]
U+M19	37	0.646 [0.438, 0.804]	−0.172 [−0.378, −0.015]	0.318 [−0.012, 0.596]	−0.500 [−0.819, −0.231]	0.818 [0.773, 0.866]
ALL57	57	0.595 [0.466, 0.699]	−0.091 [−0.199, −0.004]	0.431 [0.255, 0.576]	−0.255 [−0.420, −0.117]	0.686 [0.594, 0.764]
ALL	65	0.637 [0.512, 0.736]	−0.133 [−0.248, −0.041]	0.443 [0.275, 0.596]	−0.327 [−0.489, −0.175]	0.770 [0.711, 0.825]

Table S5b. Benchmark validity per wound reference. Generator B at texture amplitude 40 and 11, per wound reference R. Benchmark agreement (bench) with a wound reference R: Spearman agreement of mean Dice on generator B with R, averaged over the 3 synthetic seeds (2024, 7, 101); each synthetic set has 45 images. Other: mean agreement of the other 3 wound datasets with R. Diff = bench minus other: positive means the benchmark ranks the methods more like R than another wound dataset does. Mean: average over the 4 wound references (other then equals the mean of the 6 wound pairs). Dice only. Method sets as defined in the Notation. Intervals: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000, 95% percentile. The four references share data, so their values are not independent.

Set	Reference	Bench, amplitude 40	Diff, amplitude 40	Bench, amplitude 11	Diff, amplitude 11	Other
U	Medetec	0.591 [0.057, 0.924]	0.017 [−0.421, 0.323]	0.485 [0.014, 0.823]	−0.089 [−0.455, 0.225]	0.574 [0.393, 0.736]
U	CO2Wounds-V2	0.805 [0.557, 0.908]	0.530 [0.134, 0.926]	0.480 [0.119, 0.730]	0.205 [−0.213, 0.613]	0.275 [−0.180, 0.632]
U	FUSeg	0.390 [−0.139, 0.805]	0.037 [−0.403, 0.412]	0.357 [−0.174, 0.761]	0.004 [−0.402, 0.377]	0.353 [0.000, 0.616]
U	AZH	0.318 [−0.229, 0.730]	0.117 [−0.238, 0.389]	0.477 [0.033, 0.802]	0.277 [−0.119, 0.701]	0.201 [−0.214, 0.577]
F	Medetec	0.046 [−0.485, 0.575]	−0.189 [−0.690, 0.370]	−0.010 [−0.546, 0.547]	−0.245 [−0.761, 0.340]	0.235 [−0.174, 0.627]
F	CO2Wounds-V2	0.468 [−0.021, 0.818]	0.297 [−0.399, 0.944]	0.508 [−0.014, 0.818]	0.337 [−0.373, 0.950]	0.171 [−0.245, 0.576]
F	FUSeg	0.097 [−0.639, 0.753]	0.132 [−0.537, 0.750]	0.073 [−0.627, 0.730]	0.107 [−0.541, 0.723]	−0.034 [−0.499, 0.451]
F	AZH	0.304 [−0.234, 0.707]	0.019 [−0.601, 0.509]	0.238 [−0.314, 0.699]	−0.047 [−0.646, 0.471]	0.285 [0.002, 0.583]
K	Medetec	0.165 [−0.433, 0.645]	−0.308 [−0.776, 0.137]	0.152 [−0.430, 0.631]	−0.321 [−0.779, 0.148]	0.473 [0.157, 0.699]
K	CO2Wounds-V2	0.064 [−0.641, 0.660]	−0.456 [−1.037, 0.120]	0.119 [−0.543, 0.667]	−0.400 [−0.915, 0.092]	0.519 [0.211, 0.741]
K	FUSeg	−0.338 [−0.835, 0.272]	−0.734 [−1.141, −0.282]	−0.311 [−0.797, 0.297]	−0.708 [−1.083, −0.287]	0.396 [0.003, 0.688]
K	AZH	0.559 [−0.028, 0.882]	0.497 [−0.371, 1.212]	0.607 [0.041, 0.936]	0.545 [−0.301, 1.275]	0.062 [−0.487, 0.528]
M	Medetec	0.297 [−0.450, 0.846]	0.100 [−0.562, 0.570]	0.182 [−0.586, 0.864]	−0.015 [−0.602, 0.462]	0.197 [−0.276, 0.578]
M	CO2Wounds-V2	0.445 [−0.197, 0.759]	0.364 [−0.178, 0.798]	0.624 [0.020, 0.875]	0.542 [0.097, 0.956]	0.082 [−0.337, 0.467]
M	FUSeg	0.279 [−0.404, 0.775]	0.245 [−0.319, 0.689]	0.273 [−0.545, 0.753]	0.239 [−0.392, 0.664]	0.033 [−0.442, 0.412]
M	AZH	0.324 [−0.443, 0.843]	0.218 [−0.586, 0.754]	0.342 [−0.409, 0.871]	0.236 [−0.523, 0.766]	0.106 [−0.286, 0.429]
M11s	Medetec	0.170 [−0.587, 0.738]	−0.209 [−0.717, 0.237]	0.273 [−0.456, 0.833]	−0.106 [−0.565, 0.307]	0.379 [−0.091, 0.733]
M11s	CO2Wounds-V2	0.664 [0.118, 0.925]	0.348 [−0.118, 0.750]	0.755 [0.317, 0.971]	0.439 [0.071, 0.777]	0.315 [−0.132, 0.692]
M11s	FUSeg	0.191 [−0.474, 0.769]	0.115 [−0.416, 0.664]	0.255 [−0.441, 0.766]	0.179 [−0.310, 0.639]	0.076 [−0.526, 0.590]
M11s	AZH	0.609 [−0.059, 0.962]	0.282 [−0.332, 0.811]	0.606 [−0.067, 0.984]	0.279 [−0.298, 0.813]	0.327 [−0.044, 0.678]
M19	Medetec	0.073 [−0.405, 0.499]	−0.222 [−0.608, 0.173]	0.168 [−0.352, 0.576]	−0.126 [−0.547, 0.246]	0.294 [−0.024, 0.567]
M19	CO2Wounds-V2	0.472 [−0.007, 0.838]	0.181 [−0.325, 0.616]	0.556 [0.089, 0.866]	0.264 [−0.236, 0.674]	0.291 [0.003, 0.565]
M19	FUSeg	0.046 [−0.402, 0.466]	0.058 [−0.469, 0.559]	0.070 [−0.407, 0.500]	0.082 [−0.448, 0.546]	−0.012 [−0.419, 0.406]
M19	AZH	0.231 [−0.301, 0.645]	−0.019 [−0.504, 0.403]	0.354 [−0.162, 0.748]	0.104 [−0.377, 0.526]	0.250 [−0.035, 0.510]
P10s	Medetec	−0.038 [−0.732, 0.604]	−0.335 [−0.788, 0.130]	0.091 [−0.636, 0.755]	−0.206 [−0.663, 0.235]	0.297 [−0.228, 0.727]
P10s	CO2Wounds-V2	0.552 [−0.113, 0.872]	0.327 [−0.268, 0.811]	0.673 [0.147, 0.950]	0.448 [−0.031, 0.844]	0.224 [−0.220, 0.654]
P10s	FUSeg	0.216 [−0.480, 0.819]	0.141 [−0.426, 0.807]	0.285 [−0.454, 0.819]	0.210 [−0.306, 0.771]	0.075 [−0.577, 0.620]
P10s	AZH	0.479 [−0.269, 0.938]	0.267 [−0.501, 0.963]	0.475 [−0.285, 0.966]	0.263 [−0.475, 0.950]	0.212 [−0.169, 0.662]
N8s	Medetec	−0.254 [−0.951, 0.615]	−0.500 [−1.282, 0.413]	−0.190 [−0.924, 0.637]	−0.437 [−1.258, 0.414]	0.246 [−0.291, 0.744]
N8s	CO2Wounds-V2	0.063 [−0.612, 0.895]	−0.294 [−1.290, 0.650]	0.190 [−0.583, 0.975]	−0.167 [−1.167, 0.720]	0.357 [−0.224, 0.794]
N8s	FUSeg	−0.103 [−0.823, 0.630]	−0.063 [−1.324, 1.114]	−0.032 [−0.990, 0.631]	0.008 [−1.267, 1.000]	−0.040 [−0.713, 0.778]
N8s	AZH	−0.214 [−0.882, 0.646]	−0.683 [−1.392, 0.160]	0.127 [−0.744, 0.924]	−0.341 [−1.297, 0.530]	0.468 [−0.004, 0.795]
U+F	Medetec	0.343 [−0.040, 0.656]	−0.159 [−0.555, 0.242]	0.209 [−0.193, 0.544]	−0.294 [−0.743, 0.186]	0.503 [0.234, 0.708]
U+F	CO2Wounds-V2	0.625 [0.386, 0.780]	0.131 [−0.183, 0.418]	0.311 [−0.009, 0.576]	−0.183 [−0.533, 0.147]	0.494 [0.225, 0.708]
U+F	FUSeg	0.206 [−0.146, 0.515]	−0.339 [−0.686, −0.033]	−0.036 [−0.419, 0.318]	−0.581 [−0.945, −0.219]	0.545 [0.312, 0.719]

Set	Reference	Bench, amplitude 40	Diff, amplitude 40	Bench, amplitude 11	Diff, amplitude 11	Other
U+F	AZH	0.208 [−0.204, 0.551]	−0.343 [−0.650, −0.080]	0.062 [−0.301, 0.376]	−0.489 [−0.793, −0.211]	0.551 [0.324, 0.732]
U+K	Medetec	0.444 [0.061, 0.724]	−0.052 [−0.350, 0.220]	0.382 [−0.011, 0.664]	−0.114 [−0.437, 0.168]	0.496 [0.356, 0.606]
U+K	CO2Wounds-V2	0.164 [−0.234, 0.492]	−0.125 [−0.458, 0.185]	−0.011 [−0.390, 0.339]	−0.300 [−0.579, −0.006]	0.289 [−0.003, 0.524]
U+K	FUSeg	−0.026 [−0.434, 0.341]	−0.362 [−0.658, −0.089]	−0.056 [−0.411, 0.300]	−0.393 [−0.659, −0.104]	0.337 [0.137, 0.489]
U+K	AZH	0.695 [0.488, 0.849]	0.735 [0.459, 1.018]	0.753 [0.554, 0.886]	0.794 [0.525, 1.062]	−0.041 [−0.320, 0.230]
U+M	Medetec	0.751 [0.569, 0.890]	−0.103 [−0.265, 0.022]	0.413 [0.031, 0.741]	−0.442 [−0.803, −0.128]	0.855 [0.806, 0.899]
U+M	CO2Wounds-V2	0.812 [0.630, 0.906]	0.035 [−0.166, 0.182]	0.473 [0.116, 0.707]	−0.305 [−0.661, −0.069]	0.777 [0.673, 0.871]
U+M	FUSeg	0.657 [0.372, 0.859]	−0.136 [−0.394, 0.047]	0.354 [−0.066, 0.711]	−0.440 [−0.833, −0.096]	0.793 [0.706, 0.866]
U+M	AZH	0.658 [0.399, 0.832]	−0.103 [−0.306, 0.034]	0.457 [0.136, 0.687]	−0.303 [−0.622, −0.085]	0.761 [0.656, 0.850]
U+M11s	Medetec	0.710 [0.486, 0.867]	−0.155 [−0.364, −0.006]	0.378 [−0.020, 0.705]	−0.487 [−0.865, −0.172]	0.865 [0.814, 0.908]
U+M11s	CO2Wounds-V2	0.801 [0.579, 0.918]	0.011 [−0.231, 0.184]	0.435 [0.063, 0.688]	−0.355 [−0.716, −0.097]	0.790 [0.683, 0.884]
U+M11s	FUSeg	0.601 [0.278, 0.848]	−0.194 [−0.497, 0.035]	0.303 [−0.134, 0.677]	−0.492 [−0.913, −0.132]	0.795 [0.707, 0.872]
U+M11s	AZH	0.638 [0.348, 0.839]	−0.135 [−0.388, 0.034]	0.435 [0.103, 0.677]	−0.338 [−0.670, −0.107]	0.773 [0.667, 0.865]
U+M19	Medetec	0.642 [0.418, 0.809]	−0.214 [−0.419, −0.054]	0.303 [−0.062, 0.615]	−0.553 [−0.912, −0.251]	0.856 [0.809, 0.899]
U+M19	CO2Wounds-V2	0.755 [0.557, 0.887]	−0.066 [−0.274, 0.080]	0.390 [0.051, 0.659]	−0.431 [−0.761, −0.172]	0.821 [0.757, 0.880]
U+M19	FUSeg	0.579 [0.302, 0.791]	−0.210 [−0.482, −0.007]	0.233 [−0.131, 0.571]	−0.556 [−0.903, −0.230]	0.789 [0.723, 0.857]
U+M19	AZH	0.609 [0.360, 0.794]	−0.198 [−0.418, −0.023]	0.347 [0.018, 0.600]	−0.460 [−0.769, −0.222]	0.807 [0.745, 0.864]
ALL57	Medetec	0.618 [0.457, 0.755]	−0.101 [−0.259, 0.050]	0.470 [0.249, 0.644]	−0.249 [−0.482, −0.055]	0.719 [0.610, 0.806]
ALL57	CO2Wounds-V2	0.577 [0.415, 0.703]	−0.072 [−0.224, 0.070]	0.384 [0.165, 0.568]	−0.265 [−0.476, −0.078]	0.649 [0.522, 0.760]
ALL57	FUSeg	0.493 [0.303, 0.649]	−0.233 [−0.397, −0.084]	0.296 [0.049, 0.503]	−0.430 [−0.652, −0.226]	0.727 [0.632, 0.802]
ALL57	AZH	0.692 [0.549, 0.805]	0.042 [−0.097, 0.158]	0.574 [0.413, 0.704]	−0.076 [−0.240, 0.065]	0.650 [0.561, 0.732]
ALL	Medetec	0.651 [0.504, 0.764]	−0.144 [−0.282, −0.022]	0.472 [0.287, 0.635]	−0.323 [−0.510, −0.144]	0.795 [0.728, 0.854]
ALL	CO2Wounds-V2	0.639 [0.505, 0.747]	−0.108 [−0.246, 0.010]	0.421 [0.218, 0.593]	−0.327 [−0.520, −0.149]	0.748 [0.662, 0.826]
ALL	FUSeg	0.552 [0.379, 0.699]	−0.241 [−0.404, −0.103]	0.325 [0.109, 0.533]	−0.468 [−0.671, −0.263]	0.793 [0.734, 0.845]
ALL	AZH	0.706 [0.569, 0.809]	−0.040 [−0.172, 0.066]	0.555 [0.392, 0.695]	−0.191 [−0.352, −0.049]	0.746 [0.688, 0.805]

Table S5c. Benchmark validity with hierarchical intervals (K and modern sets). Same quantities as Tables S5a and S5b with hierarchical and data-only intervals. Hierarchical bootstrap: methods within the group plus, independently per dataset, resampling units of the clinical sets and images of the synthetic sets, B = 2,000; data-only bootstrap: units and images only, methods fixed, B = 2,000 (context, not the primary interval). Only K and the modern sets have per-image synthetic scores; U and F synthetic means exist per method only, so sets containing them get the method bootstrap only. Benchmark agreement (bench) with a wound reference R: Spearman agreement of mean Dice on generator B with R, averaged over the 3 synthetic seeds (2024, 7, 101); each synthetic set has 45 images. Other: mean agreement of the other 3 wound datasets with R. Diff = bench minus other: positive means the benchmark ranks the methods more like R than another wound dataset does. Mean: average over the 4 wound references (other then equals the mean of the 6 wound pairs). Dice only. Method sets as defined in the Notation.

Set	Reference	Bench 40 (h)	Diff 40 (h)	Diff 40 (data-only)	Bench 11 (h)	Diff 11 (h)	Other (h)
K	Medetec	0.165 [−0.481, 0.741]	−0.308 [−0.782, 0.348]	−0.308 [−0.463, 0.054]	0.152 [−0.378, 0.760]	−0.321 [−0.696, 0.369]	0.473 [0.021, 0.720]
K	CO2Wounds-V2	0.064 [−0.628, 0.679]	−0.456 [−0.954, 0.174]	−0.456 [−0.542, −0.226]	0.119 [−0.492, 0.717]	−0.400 [−0.822, 0.202]	0.519 [0.126, 0.736]
K	FUSeg	−0.338 [−0.814, 0.453]	−0.734 [−1.028, −0.082]	−0.734 [−0.743, −0.396]	−0.311 [−0.777, 0.475]	−0.708 [−0.961, −0.071]	0.396 [−0.136, 0.703]
K	AZH	0.559 [−0.032, 0.899]	0.497 [−0.246, 1.188]	0.497 [0.300, 0.740]	0.607 [−0.004, 0.906]	0.545 [−0.202, 1.165]	0.062 [−0.482, 0.549]
K	mean of 4	0.112 [−0.314, 0.584]	−0.250 [−0.574, 0.161]	−0.250 [−0.293, −0.034]	0.142 [−0.251, 0.607]	−0.221 [−0.495, 0.178]	0.363 [−0.027, 0.652]
M	Medetec	0.297 [−0.507, 0.821]	0.100 [−0.552, 0.598]	0.100 [−0.206, 0.282]	0.182 [−0.571, 0.853]	−0.015 [−0.558, 0.545]	0.197 [−0.284, 0.579]
M	CO2Wounds-V2	0.445 [−0.395, 0.782]	0.364 [−0.366, 0.792]	0.364 [−0.018, 0.503]	0.624 [−0.152, 0.872]	0.542 [−0.115, 0.927]	0.082 [−0.387, 0.492]
M	FUSeg	0.279 [−0.502, 0.778]	0.245 [−0.476, 0.781]	0.245 [−0.064, 0.488]	0.273 [−0.527, 0.774]	0.239 [−0.466, 0.732]	0.033 [−0.409, 0.492]
M	AZH	0.324 [−0.486, 0.816]	0.218 [−0.503, 0.819]	0.218 [−0.027, 0.449]	0.342 [−0.522, 0.837]	0.236 [−0.544, 0.858]	0.106 [−0.372, 0.476]
M	mean of 4	0.336 [−0.104, 0.575]	0.232 [−0.224, 0.412]	0.232 [0.047, 0.301]	0.355 [−0.076, 0.609]	0.251 [−0.194, 0.423]	0.105 [−0.192, 0.431]
M11s	Medetec	0.170 [−0.608, 0.754]	−0.209 [−0.723, 0.341]	−0.209 [−0.361, 0.018]	0.273 [−0.535, 0.835]	−0.106 [−0.636, 0.386]	0.379 [−0.119, 0.738]
M11s	CO2Wounds-V2	0.664 [−0.152, 0.945]	0.348 [−0.269, 0.836]	0.348 [−0.003, 0.591]	0.755 [−0.062, 0.960]	0.439 [−0.208, 0.847]	0.315 [−0.232, 0.689]
M11s	FUSeg	0.191 [−0.601, 0.770]	0.115 [−0.514, 0.651]	0.115 [−0.194, 0.264]	0.255 [−0.587, 0.806]	0.179 [−0.450, 0.630]	0.076 [−0.502, 0.643]
M11s	AZH	0.609 [−0.156, 0.931]	0.282 [−0.452, 0.881]	0.282 [0.039, 0.461]	0.606 [−0.145, 0.962]	0.279 [−0.416, 0.917]	0.327 [−0.211, 0.678]
M11s	mean of 4	0.408 [−0.051, 0.699]	0.134 [−0.271, 0.376]	0.134 [−0.039, 0.231]	0.472 [−0.010, 0.738]	0.198 [−0.206, 0.394]	0.274 [−0.154, 0.643]
M19	Medetec	0.073 [−0.390, 0.533]	−0.222 [−0.623, 0.266]	−0.222 [−0.364, 0.019]	0.168 [−0.361, 0.605]	−0.126 [−0.553, 0.325]	0.294 [−0.073, 0.563]
M19	CO2Wounds-V2	0.472 [−0.102, 0.786]	0.181 [−0.371, 0.637]	0.181 [−0.033, 0.411]	0.556 [−0.001, 0.833]	0.264 [−0.267, 0.695]	0.291 [−0.090, 0.529]
M19	FUSeg	0.046 [−0.441, 0.514]	0.058 [−0.472, 0.551]	0.058 [−0.164, 0.246]	0.070 [−0.475, 0.532]	0.082 [−0.442, 0.503]	−0.012 [−0.408, 0.435]
M19	AZH	0.231 [−0.352, 0.630]	−0.019 [−0.533, 0.446]	−0.019 [−0.205, 0.146]	0.354 [−0.264, 0.751]	0.104 [−0.439, 0.577]	0.250 [−0.131, 0.531]
M19	mean of 4	0.205 [−0.127, 0.478]	−0.001 [−0.361, 0.270]	−0.001 [−0.124, 0.117]	0.287 [−0.065, 0.547]	0.081 [−0.292, 0.328]	0.206 [−0.107, 0.483]
P10s	Medetec	−0.038 [−0.683, 0.673]	−0.335 [−0.921, 0.249]	−0.335 [−0.509, −0.093]	0.091 [−0.650, 0.808]	−0.206 [−0.823, 0.298]	0.297 [−0.168, 0.757]
P10s	CO2Wounds-V2	0.552 [−0.304, 0.923]	0.327 [−0.433, 0.889]	0.327 [−0.040, 0.598]	0.673 [−0.183, 0.925]	0.448 [−0.327, 0.889]	0.224 [−0.303, 0.665]
P10s	FUSeg	0.216 [−0.559, 0.839]	0.141 [−0.560, 0.806]	0.141 [−0.166, 0.283]	0.285 [−0.530, 0.858]	0.210 [−0.459, 0.788]	0.075 [−0.500, 0.681]
P10s	AZH	0.479 [−0.379, 0.910]	0.267 [−0.684, 0.939]	0.267 [−0.012, 0.396]	0.475 [−0.370, 0.951]	0.263 [−0.615, 0.931]	0.212 [−0.287, 0.705]
P10s	mean of 4	0.302 [−0.146, 0.653]	0.100 [−0.395, 0.367]	0.100 [−0.102, 0.199]	0.381 [−0.116, 0.748]	0.179 [−0.314, 0.393]	0.202 [−0.194, 0.665]
N8s	Medetec	−0.254 [−0.924, 0.654]	−0.500 [−1.243, 0.599]	−0.500 [−0.770, 0.024]	−0.190 [−0.924, 0.692]	−0.437 [−1.220, 0.517]	0.246 [−0.435, 0.692]
N8s	CO2Wounds-V2	0.063 [−0.772, 0.863]	−0.294 [−1.274, 0.867]	−0.294 [−0.675, 0.461]	0.190 [−0.744, 0.924]	−0.167 [−1.256, 0.840]	0.357 [−0.366, 0.744]
N8s	FUSeg	−0.103 [−0.811, 0.713]	−0.063 [−1.155, 1.210]	−0.063 [−0.341, 0.460]	−0.032 [−0.842, 0.684]	0.008 [−1.163, 1.023]	−0.040 [−0.730, 0.643]
N8s	AZH	−0.214 [−0.885, 0.714]	−0.683 [−1.308, 0.444]	−0.683 [−0.825, −0.127]	0.127 [−0.722, 0.906]	−0.341 [−1.218, 0.709]	0.468 [−0.205, 0.744]
N8s	mean of 4	−0.127 [−0.593, 0.500]	−0.385 [−1.042, 0.380]	−0.385 [−0.550, 0.018]	0.024 [−0.541, 0.593]	−0.234 [−1.029, 0.419]	0.258 [−0.222, 0.658]

Table S5d. Benchmark validity in single-seed method sets (amplitude 40). Single-seed sets P10_sk, N8_sk and M19_sk as defined in the Notation; M is the original architectures at seed 0 plus nnU-Net. Seed-averaged sets for comparison. Bench, other, diff: means over the 4 wound references, estimate [interval]; the last 4 columns give bench per reference. Benchmark agreement (bench) with a wound reference R: Spearman agreement of mean Dice on generator B with R, averaged over the 3 synthetic seeds (2024, 7, 101); each synthetic set has 45 images. Other: mean agreement of the other 3 wound datasets with R. Diff = bench minus other: positive means the benchmark ranks the methods more like R than another wound dataset does. Mean: average over the 4 wound references (other then equals the mean of the 6 wound pairs). Dice only. Intervals: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000, 95% percentile.

Set	n	Bench	Other	Diff	Bench, Medetec	Bench, CO2Wounds-V2	Bench, FUSeg	Bench, AZH
P10s	10	0.302 [−0.052, 0.608]	0.202 [−0.212, 0.632]	0.100 [−0.247, 0.371]	−0.038	0.552	0.216	0.479
P10_s0	10	0.233 [−0.178, 0.459]	0.055 [−0.176, 0.344]	0.179 [−0.311, 0.409]	0.160	0.309	0.366	0.099
P10_s1	10	0.398 [0.030, 0.692]	0.285 [−0.056, 0.628]	0.113 [−0.156, 0.354]	0.277	0.560	0.176	0.580
P10_s2	9	0.292 [−0.135, 0.689]	0.178 [−0.293, 0.696]	0.114 [−0.241, 0.424]	0.039	0.622	0.044	0.461
N8s	8	−0.127 [−0.611, 0.555]	0.258 [−0.149, 0.722]	−0.385 [−1.173, 0.307]	−0.254	0.063	−0.103	−0.214
N8_s0	7	0.190 [−0.193, 0.667]	0.083 [−0.314, 0.588]	0.107 [−0.490, 0.403]	−0.333	0.786	0.143	0.167
N8_s1	8	−0.149 [−0.712, 0.655]	0.425 [0.026, 0.861]	−0.573 [−1.462, 0.296]	−0.262	−0.159	0.016	−0.190
N8_s2	7	−0.036 [−0.676, 0.566]	0.298 [−0.062, 0.698]	−0.333 [−1.209, 0.333]	−0.119	0.238	−0.095	−0.167
M19	19	0.205 [−0.167, 0.485]	0.206 [−0.070, 0.491]	−0.001 [−0.406, 0.299]	0.073	0.472	0.046	0.231
M19_s0	18	0.251 [0.000, 0.436]	0.078 [−0.117, 0.282]	0.173 [−0.077, 0.361]	0.117	0.489	0.183	0.214
M19_s1	19	0.196 [−0.200, 0.501]	0.268 [0.011, 0.520]	−0.071 [−0.563, 0.287]	0.150	0.281	0.068	0.285
M19_s2	17	0.228 [−0.161, 0.540]	0.256 [−0.059, 0.571]	−0.028 [−0.446, 0.274]	0.141	0.537	0.011	0.223
M	11	0.336 [−0.068, 0.593]	0.105 [−0.121, 0.377]	0.232 [−0.186, 0.424]	0.297	0.445	0.279	0.324
M11s	11	0.408 [0.014, 0.717]	0.274 [−0.111, 0.650]	0.134 [−0.186, 0.364]	0.170	0.664	0.191	0.609

Table S5e. Texture effect (amplitude 40 minus amplitude 11). Texture effect: benchmark agreement at texture amplitude 40 minus that at amplitude 11 (each averaged over the 3 synthetic seeds), per wound reference and as the mean over the 4 references, estimate [interval]. Intervals: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000, 95% percentile. Last column: the mean with the hierarchical interval where per-image synthetic scores exist (n/a otherwise). Hierarchical bootstrap: methods within the group plus, independently per dataset, resampling units of the clinical sets and images of the synthetic sets, B = 2,000; data-only bootstrap: units and images only, methods fixed, B = 2,000 (context, not the primary interval).

Set	Medetec	CO2Wounds-V2	FUSeg	AZH	Mean of 4	Mean of 4 (h)
U	0.107 [−0.216, 0.406]	0.325 [0.093, 0.600]	0.033 [−0.283, 0.355]	−0.160 [−0.587, 0.239]	0.076 [−0.109, 0.245]	n/a
F	0.056 [−0.047, 0.152]	−0.040 [−0.141, 0.064]	0.025 [−0.084, 0.136]	0.066 [−0.079, 0.322]	0.027 [−0.061, 0.116]	n/a
K	0.013 [−0.252, 0.266]	−0.056 [−0.346, 0.153]	−0.026 [−0.248, 0.164]	−0.048 [−0.385, 0.213]	−0.029 [−0.197, 0.071]	−0.029 [−0.237, 0.101]
M	0.115 [−0.240, 0.483]	−0.179 [−0.493, 0.032]	0.006 [−0.385, 0.337]	−0.018 [−0.385, 0.338]	−0.019 [−0.144, 0.072]	−0.019 [−0.218, 0.127]
M11s	−0.103 [−0.297, 0.022]	−0.091 [−0.280, 0.025]	−0.064 [−0.313, 0.147]	0.003 [−0.146, 0.103]	−0.064 [−0.226, 0.012]	−0.064 [−0.212, 0.063]
M19	−0.096 [−0.283, 0.064]	−0.084 [−0.234, 0.030]	−0.025 [−0.185, 0.148]	−0.123 [−0.353, 0.063]	−0.082 [−0.186, 0.004]	−0.082 [−0.208, 0.033]
P10s	−0.129 [−0.346, 0.029]	−0.121 [−0.348, 0.029]	−0.069 [−0.335, 0.179]	0.004 [−0.189, 0.146]	−0.079 [−0.247, 0.017]	−0.079 [−0.280, 0.077]
N8s	−0.063 [−0.487, 0.301]	−0.127 [−0.650, 0.324]	−0.071 [−0.498, 0.380]	−0.341 [−0.869, 0.000]	−0.151 [−0.405, 0.000]	−0.151 [−0.427, 0.097]
U+F	0.135 [−0.046, 0.308]	0.314 [0.131, 0.516]	0.242 [0.094, 0.417]	0.146 [−0.065, 0.368]	0.209 [0.084, 0.360]	n/a
U+K	0.062 [−0.098, 0.210]	0.175 [0.031, 0.314]	0.031 [−0.139, 0.196]	−0.059 [−0.206, 0.086]	0.052 [−0.042, 0.141]	n/a
U+M	0.338 [0.103, 0.617]	0.340 [0.117, 0.629]	0.304 [0.058, 0.605]	0.201 [−0.001, 0.441]	0.295 [0.096, 0.554]	n/a
U+M11s	0.332 [0.051, 0.655]	0.365 [0.097, 0.689]	0.299 [−0.010, 0.645]	0.203 [−0.018, 0.456]	0.300 [0.051, 0.587]	n/a
U+M19	0.339 [0.095, 0.612]	0.366 [0.124, 0.630]	0.347 [0.069, 0.637]	0.263 [0.077, 0.496]	0.328 [0.095, 0.587]	n/a
ALL57	0.148 [0.045, 0.269]	0.193 [0.071, 0.332]	0.197 [0.078, 0.339]	0.118 [0.028, 0.228]	0.164 [0.068, 0.279]	n/a
ALL	0.179 [0.064, 0.304]	0.218 [0.084, 0.365]	0.227 [0.086, 0.373]	0.151 [0.053, 0.260]	0.194 [0.079, 0.318]	n/a

Table S5f. Generator families. Agreement of each generator family with the clinical wound data: Spearman agreement of mean Dice on the synthetic set with each of the 4 wound datasets, averaged over the 4 datasets and over the 3 synthetic seeds. nrA: generator A, uniform drape; nrB0: generator B, texture ablated; nrC0: generator C, no noise; nrC34: generator C, noise 3.4. Last column: nrB0 minus nrC34, estimate [interval]. Intervals: method bootstrap (methods resampled with replacement within each group, pooled sets keep their composition), B = 2,000, 95% percentile.

Set	Generator A (nrA)	Generator B, texture ablated (nrB0)	Generator C, no noise (nrC0)	Generator C, noise 3.4 (nrC34)	nrB0 minus nrC34
U	0.309	0.402	0.304	0.287	0.115 [−0.138, 0.357]
F	−0.006	0.174	0.286	0.303	−0.129 [−0.458, 0.140]
K	0.357	0.247	0.386	0.387	−0.141 [−0.412, 0.057]
M	0.284	0.306	0.303	0.321	−0.015 [−0.234, 0.182]
M11s	0.254	0.445	0.545	0.541	−0.095 [−0.393, 0.158]
M19	0.176	0.267	0.393	0.447	−0.180 [−0.524, 0.056]
P10s	0.114	0.348	0.485	0.480	−0.131 [−0.455, 0.172]
N8s	0.117	0.020	0.202	0.323	−0.304 [−0.923, 0.119]
U+F	−0.224	0.132	0.076	0.067	0.065 [−0.122, 0.247]
U+K	0.269	0.273	0.275	0.273	−0.001 [−0.162, 0.172]
U+M	−0.599	0.321	0.159	0.134	0.187 [−0.066, 0.440]
U+M11s	−0.600	0.270	0.121	0.105	0.164 [−0.096, 0.409]
U+M19	−0.663	0.203	0.096	0.083	0.121 [−0.140, 0.391]
ALL57	−0.521	0.387	0.220	0.224	0.163 [0.014, 0.312]
ALL	−0.633	0.383	0.193	0.196	0.187 [0.020, 0.353]

Table S5g. Leave one method out of M19: benchmark validity (amplitude 40). Bench, wound pairs (other) and diff as means over the 4 wound references, after dropping one method from M19 (18 remaining). Point estimates. Method keys as in Table S3b.

Dropped method	Bench	Wound pairs (6)	Diff
none (all 19)	**0.205**	**0.206**	**−0.001**
unet_r18	0.192	0.202	−0.009
unetpp_r18	0.230	0.239	−0.008
deeplabv3p_r18	0.199	0.246	−0.047
fpn_r18	0.210	0.216	−0.006
pspnet_r18	0.118	0.104	0.014
linknet_r18	0.205	0.197	0.007
manet_r18	0.214	0.169	0.045
segformer_mitb0	0.204	0.222	−0.018
unet_mbv2	0.223	0.258	−0.035
unet_effb0	0.194	0.221	−0.028
unet_r50	0.213	0.228	−0.015
deeplabv3p_r50	0.208	0.194	0.014
upernet_convnext_t	0.234	0.214	0.020
unet_swin_t	0.223	0.226	−0.004
segformer_mitb2	0.147	0.174	−0.028

Dropped method	Bench	Wound pairs (6)	Diff
unet_effb3	0.316	0.154	0.161
fpn_regnety016	0.222	0.210	0.012
manet_r50	0.214	0.256	−0.042
nnunet_2d	0.125	0.168	−0.044

Table S6a. Robustness to post-processing (modern sets). Scores from the rescoring of the trained models (each model's probability maps scored again under every variant): M19 as in the main analysis; K11 as defined in the Notation (K without its 3 pixel classifiers, which were not rescored). Post-processing P0: Gaussian smoothing of the probability map, threshold, small-component removal (main analysis); P1: threshold, closing and opening with a 5 px ellipse, small-component removal; P2: threshold and small-component removal only. nnU-Net values in the rescoring come from the second-session re-prediction (Table S9b). Intervals: method bootstrap, B = 2,000, 95% percentile, the same draws for every condition, so differences between conditions get paired intervals. Last column: change in wound-pair agreement from P0 with its paired interval. NSD at 2 px. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Colonoscopy pairs: mean over the 10 pairs of the 5 colonoscopy datasets. Modality difference (55 pairs): mean agreement of the 16 same-modality pairs minus the mean of the 39 across-modality pairs of the 11 clinical references; 52-pair version: the same without the 3 same-source pairs (FUSeg with AZH; Kvasir with Kvasir-sessile; CVC-ClinicDB with CVC-ColonDB).

Set	Metric	Variant	Wound pairs	Colonoscopy pairs	Modality difference (55)	Wound pairs minus P0
M19	Dice	P0	0.206 [−0.070, 0.491]	0.403 [0.154, 0.630]	0.225 [0.048, 0.457]	
M19	Dice	P1	0.256 [−0.021, 0.522]	0.414 [0.156, 0.633]	0.229 [0.062, 0.433]	0.051 [−0.072, 0.165]
M19	Dice	P2	0.255 [−0.018, 0.517]	0.425 [0.167, 0.639]	0.243 [0.069, 0.457]	0.049 [−0.066, 0.165]
M19	NSD	P0	0.479 [0.169, 0.725]	0.577 [0.302, 0.752]	0.360 [0.179, 0.522]	
M19	NSD	P1	0.526 [0.245, 0.740]	0.601 [0.333, 0.764]	0.369 [0.192, 0.543]	0.048 [−0.132, 0.225]
M19	NSD	P2	0.522 [0.237, 0.739]	0.618 [0.347, 0.782]	0.378 [0.195, 0.551]	0.044 [−0.150, 0.231]
K11	Dice	P0	0.258 [−0.053, 0.511]	0.713 [0.348, 0.877]	0.459 [0.134, 0.725]	
K11	Dice	P1	0.270 [−0.028, 0.507]	0.738 [0.422, 0.891]	0.486 [0.166, 0.750]	0.012 [−0.020, 0.045]
K11	Dice	P2	0.267 [−0.036, 0.507]	0.738 [0.422, 0.891]	0.482 [0.159, 0.750]	0.009 [−0.023, 0.040]
K11	NSD	P0	0.182 [−0.174, 0.592]	0.863 [0.644, 0.938]	0.605 [0.308, 0.837]	
K11	NSD	P1	0.114 [−0.199, 0.511]	0.830 [0.550, 0.938]	0.595 [0.274, 0.869]	−0.068 [−0.206, 0.012]
K11	NSD	P2	0.174 [−0.192, 0.592]	0.838 [0.556, 0.944]	0.610 [0.250, 0.919]	−0.008 [−0.159, 0.174]

Table S6b. Robustness to the NSD tolerance (modern sets, P0). NSD at a boundary tolerance of 1, 2 (main analysis), 3 and 5 px and at 1% of the image diagonal; Dice for reference. Last column: change in wound-pair agreement from NSD at 2 px, paired interval. Scores from the rescoring of the trained models (each model's probability maps scored again under every variant): M19 as in the main analysis; K11 as defined in the Notation (K without its 3 pixel classifiers, which were not rescored). Intervals: method bootstrap, B = 2,000, 95% percentile.

Set	Metric	Wound pairs	Colonoscopy pairs	Modality difference (55)	Wound pairs minus NSD 2 px
M19	Dice	0.206 [−0.070, 0.491]	0.403 [0.154, 0.630]	0.225 [0.048, 0.457]	−0.273 [−0.520, 0.003]
M19	NSD 1 px	0.504 [0.201, 0.750]	0.561 [0.278, 0.745]	0.309 [0.149, 0.455]	0.026 [−0.080, 0.156]
M19	NSD 2 px	0.479 [0.169, 0.725]	0.577 [0.302, 0.752]	0.360 [0.179, 0.522]	
M19	NSD 3 px	0.392 [0.080, 0.662]	0.587 [0.328, 0.767]	0.383 [0.190, 0.562]	−0.087 [−0.265, 0.077]
M19	NSD 5 px	0.309 [0.001, 0.585]	0.626 [0.379, 0.783]	0.453 [0.252, 0.637]	−0.170 [−0.418, 0.095]
M19	NSD 1% of diagonal	0.434 [0.097, 0.709]	0.595 [0.339, 0.769]	0.414 [0.214, 0.599]	−0.045 [−0.268, 0.197]
K11	Dice	0.258 [−0.053, 0.511]	0.713 [0.348, 0.877]	0.459 [0.134, 0.725]	0.076 [−0.326, 0.401]
K11	NSD 1 px	0.162 [−0.198, 0.588]	0.862 [0.642, 0.946]	0.609 [0.328, 0.852]	−0.020 [−0.175, 0.191]
K11	NSD 2 px	0.182 [−0.174, 0.592]	0.863 [0.644, 0.938]	0.605 [0.308, 0.837]	
K11	NSD 3 px	0.214 [−0.125, 0.592]	0.835 [0.566, 0.928]	0.566 [0.265, 0.790]	0.032 [−0.130, 0.198]
K11	NSD 5 px	0.230 [−0.119, 0.595]	0.830 [0.537, 0.924]	0.593 [0.242, 0.863]	0.048 [−0.109, 0.185]
K11	NSD 1% of diagonal	0.232 [−0.101, 0.595]	0.821 [0.527, 0.922]	0.567 [0.241, 0.819]	0.050 [−0.114, 0.215]

Table S6c. Threshold protocol: default and oracle (Dice, P0). Default: each model at its own threshold tuned on held-out FUSeg training images (main analysis). Oracle: for each individual model and dataset, the grid threshold with the best mean Dice on that dataset (chosen on the scored images, an optimistic upper bound); seed averages average their members' oracle scores. Grid: 0.05 to 0.90 in steps of 0.05. Last column: change in wound-pair agreement from the default, paired interval. Scores from the rescoring of the trained models (each model's probability maps scored again under every variant): M19 as in the main analysis; K11 as defined in the Notation (K without its 3 pixel classifiers, which were not rescored). Intervals: method bootstrap, B = 2,000, 95% percentile.

Set	Threshold	Wound pairs	Colonoscopy pairs	Modality difference (55)	Wound pairs minus default
M19	default (own FUSeg-tuned threshold)	0.206 [−0.070, 0.491]	0.403 [0.154, 0.630]	0.225 [0.048, 0.457]	
M19	oracle per dataset	0.115 [−0.136, 0.391]	0.542 [0.273, 0.730]	0.191 [0.054, 0.338]	−0.090 [−0.280, 0.091]
K11	default (own FUSeg-tuned threshold)	0.258 [−0.053, 0.511]	0.713 [0.348, 0.877]	0.459 [0.134, 0.725]	
K11	oracle per dataset	0.114 [−0.167, 0.408]	0.626 [0.243, 0.831]	0.347 [0.020, 0.658]	−0.144 [−0.416, 0.163]

Table S6d. One common threshold for all models (Dice, P0). Every model thresholded at the same grid value. Mean wound Dice: M19 group mean Dice averaged over the 4 wound datasets. Calibration rho: mean over the 4 wound datasets of the Spearman correlation between the M19 method means and each method's default (FUSeg-tuned) threshold; a high value means that the ranking follows how confident each model's probabilities are. Minus default: change in M19 wound-pair agreement from the default protocol, paired interval (4 decimals where a bound rounds to zero at 3). Intervals: method bootstrap, B = 2,000, 95% percentile.

Threshold	M19 wound pairs	Minus default	M19 modality difference	Mean wound Dice	Calibration rho	K11 wound pairs	K11 modality difference
default	0.206 [−0.070, 0.491]		0.225 [0.048, 0.457]	0.632	0.272	0.258 [−0.053, 0.511]	0.459 [0.134, 0.725]
0.05	0.361 [0.040, 0.636]	0.155 [−0.178, 0.516]	0.363 [0.078, 0.681]	0.647	0.043	0.080 [−0.092, 0.248]	0.415 [0.130, 0.648]
0.10	0.204 [−0.078, 0.496]	−0.001 [−0.260, 0.278]	0.215 [0.088, 0.350]	0.655	0.305	0.082 [−0.132, 0.357]	0.400 [0.113, 0.595]
0.15	0.213 [−0.060, 0.512]	0.007 [−0.209, 0.227]	0.198 [0.078, 0.320]	0.653	0.364	0.050 [−0.192, 0.321]	0.374 [0.054, 0.581]
0.20	0.251 [−0.047, 0.543]	0.045 [−0.148, 0.227]	0.207 [0.086, 0.346]	0.650	0.418	0.094 [−0.180, 0.377]	0.362 [0.036, 0.596]
0.25	0.273 [−0.017, 0.565]	0.067 [−0.117, 0.230]	0.207 [0.080, 0.366]	0.645	0.438	0.105 [−0.166, 0.372]	0.325 [0.044, 0.552]
0.30	0.282 [−0.003, 0.564]	0.076 [−0.098, 0.240]	0.225 [0.076, 0.411]	0.639	0.458	0.171 [−0.160, 0.452]	0.361 [0.081, 0.593]
0.35	0.268 [−0.013, 0.541]	0.062 [−0.112, 0.222]	0.185 [0.050, 0.340]	0.633	0.478	0.164 [−0.137, 0.468]	0.376 [0.091, 0.640]
0.40	0.312 [0.035, 0.575]	0.106 [−0.065, 0.264]	0.193 [0.051, 0.365]	0.625	0.498	0.152 [−0.140, 0.442]	0.357 [0.080, 0.620]
0.45	0.369 [0.099, 0.607]	0.1635 [0.0003, 0.3203]	0.206 [0.053, 0.385]	0.617	0.515	0.129 [−0.145, 0.404]	0.319 [0.046, 0.591]
0.50	0.404 [0.139, 0.627]	0.198 [0.033, 0.347]	0.218 [0.062, 0.401]	0.608	0.545	0.141 [−0.133, 0.421]	0.320 [0.016, 0.634]
0.55	0.439 [0.156, 0.667]	0.233 [0.079, 0.393]	0.195 [0.043, 0.374]	0.599	0.563	0.117 [−0.156, 0.389]	0.322 [0.019, 0.639]
0.60	0.463 [0.184, 0.687]	0.257 [0.087, 0.408]	0.203 [0.035, 0.400]	0.588	0.558	0.118 [−0.174, 0.406]	0.316 [0.021, 0.616]
0.65	0.495 [0.204, 0.707]	0.289 [0.113, 0.442]	0.200 [0.034, 0.405]	0.576	0.588	0.162 [−0.152, 0.490]	0.301 [0.007, 0.620]
0.70	0.521 [0.228, 0.730]	0.315 [0.116, 0.518]	0.202 [0.039, 0.408]	0.564	0.599	0.174 [−0.174, 0.517]	0.305 [0.010, 0.628]
0.75	0.565 [0.266, 0.761]	0.360 [0.141, 0.580]	0.201 [0.035, 0.399]	0.549	0.627	0.142 [−0.199, 0.485]	0.288 [0.012, 0.611]
0.80	0.569 [0.279, 0.762]	0.363 [0.149, 0.594]	0.193 [0.033, 0.381]	0.531	0.613	0.117 [−0.183, 0.433]	0.269 [0.010, 0.581]
0.85	0.614 [0.328, 0.805]	0.408 [0.188, 0.638]	0.197 [0.039, 0.383]	0.510	0.616	0.153 [−0.185, 0.476]	0.286 [0.022, 0.573]
0.90	0.625 [0.333, 0.815]	0.419 [0.170, 0.654]	0.196 [0.047, 0.374]	0.483	0.579	0.068 [−0.210, 0.380]	0.245 [0.024, 0.479]

Table S6e. Oracle thresholds per dataset (M19). Group mean Dice at the default and at the oracle threshold; share of the individual M19 models whose oracle threshold differs from their default; share at the lowest grid value (0.05, where the oracle is limited by the grid); median oracle threshold; Spearman correlation of the M19 method means at default and oracle thresholds.

Dataset	Mean Dice, default	Mean Dice, oracle	Share with other threshold	Share at grid minimum	Median oracle threshold	Rank correlation default with oracle
Medetec	0.709	0.773	100.0%	92.3%	0.05	0.860
CO2Wounds-V2	0.438	0.504	100.0%	96.2%	0.05	0.777
FUSeg	0.798	0.802	78.8%	0.0%	0.40	0.940
AZH	0.582	0.611	98.1%	40.4%	0.10	0.893
Kvasir	0.063	0.096	100.0%	98.1%	0.05	0.926
Kvasir-sessile	0.022	0.040	100.0%	98.1%	0.05	0.975
CVC-ClinicDB	0.068	0.094	100.0%	100.0%	0.05	0.798
CVC-ColonDB	0.007	0.017	100.0%	100.0%	0.05	0.830
ETIS-Larib	0.023	0.030	100.0%	100.0%	0.05	0.840
PH2	0.048	0.075	100.0%	100.0%	0.05	0.928
TN3K	0.015	0.027	100.0%	96.2%	0.05	0.963
Control	0.480	0.513	96.2%	53.8%	0.05	0.940

Table S6f. Resolution: FUSeg at native size and at 256 px (P0). FUSeg at 256 px: the same FUSeg validation images and masks resampled so that the long side is 256 px (image by area averaging, mask by nearest neighbour), with the FUSeg exclusions. Shift: 256 px minus native, mean over the methods of the set [smallest, largest]. Rank correlation: Spearman correlation of the method means at the two resolutions. Wound pairs computed with FUSeg at native size or at 256 px, and the change with its paired interval. Intervals: method bootstrap, B = 2,000, 95% percentile.

Set	Metric	Mean, native	Mean, 256 px	Shift [min, max]	Rank correlation	Wound pairs, native	Wound pairs, 256 px	Change
M19	Dice	0.798	0.781	−0.017 [−0.077, −0.005]	0.918	0.206 [−0.070, 0.491]	0.165 [−0.093, 0.438]	−0.040 [−0.187, 0.067]
M19	NSD	0.578	0.745	+0.166 [−0.014, +0.212]	0.682	0.479 [0.169, 0.725]	0.485 [0.184, 0.723]	0.007 [−0.100, 0.119]
K11	Dice	0.080	0.080	0.000 [−0.001, +0.002]	0.945	0.258 [−0.053, 0.511]	0.192 [−0.104, 0.482]	−0.065 [−0.242, 0.000]
K11	NSD	0.019	0.035	+0.016 [+0.008, +0.022]	0.909	0.182 [−0.174, 0.592]	0.177 [−0.167, 0.573]	−0.005 [−0.142, 0.097]

Table S6g. Composition effect in pooled sets under every condition (wound domain). Pooled sets U+M19 (37), U+F (32), U+K11 (29): U+M19 and U+F as in the main analysis; U+K11 is U pooled with K11 (Notation), so it differs from U+K. Scores from the rescoring of every method (untrained and from-scratch methods rescored on a local computer, the other trained models in the second GPU session, nnU-Net re-predicted). Conditions: post-processing P0 (main), P1, P2 for Dice and NSD at 2 px; NSD at 1, 3, 5 px and at 1% of the image diagonal (P0); FUSeg replaced by FUSeg at 256 px (P0). For untrained methods P1 equals P0, because their main pipeline already applies closing and opening. Stratified agreement: methods ranked within their own group, ranks turned into within-group percentiles, Pearson correlation of the pooled percentiles. Size-weighted within: the size-weighted mean of the within-group agreements. Intervals: method bootstrap (methods resampled within each group), B = 2,000, 95% percentile, the same draws for every condition (paired changes). Gap: pooled wound-pair agreement minus the higher of the two within-group values (paired interval). Criteria: C1, gap interval above 0; C2, interval of the pooled wound with non-wound agreement below 0; C3, stratified agreement within 0.05 of the size-weighted within-group value. Holds: all three. Wound pairs: mean over the 6 pairs of the 4 wound datasets. Wound with non-wound: mean over the 28 pairs of a wound dataset with a non-wound dataset.

Set	Condition	Pooled wound pairs	Within U	Within other group	Gap	Pooled wound with non-wound	Stratified wound pairs	Size-weighted within	C1 / C2 / C3	Holds
U+M19	Dice, P0 (main)	0.818 [0.773, 0.866]	0.351	M19 0.206	0.468 [0.258, 0.621]	−0.561 [−0.664, −0.446]	0.276 [0.087, 0.459]	0.276	yes / yes / yes	yes
U+M19	Dice, P1	0.825 [0.780, 0.871]	0.351	M19 0.256	0.474 [0.255, 0.612]	−0.559 [−0.662, −0.446]	0.302 [0.116, 0.482]	0.302	yes / yes / yes	yes
U+M19	Dice, P2	0.829 [0.783, 0.875]	0.384	M19 0.255	0.444 [0.232, 0.598]	−0.580 [−0.683, −0.456]	0.318 [0.127, 0.498]	0.318	yes / yes / yes	yes
U+M19	NSD 2 px, P0 (main)	0.885 [0.834, 0.928]	0.622	M19 0.479	0.263 [0.070, 0.434]	−0.436 [−0.576, −0.312]	0.548 [0.338, 0.720]	0.548	yes / yes / yes	yes
U+M19	NSD 2 px, P1	0.888 [0.841, 0.927]	0.622	M19 0.526	0.267 [0.074, 0.418]	−0.449 [−0.579, −0.331]	0.573 [0.379, 0.725]	0.573	yes / yes / yes	yes
U+M19	NSD 2 px, P2	0.894 [0.847, 0.932]	0.640	M19 0.522	0.254 [0.077, 0.409]	−0.430 [−0.566, −0.309]	0.579 [0.388, 0.729]	0.579	yes / yes / yes	yes
U+M19	NSD 1 px	0.879 [0.820, 0.927]	0.606	M19 0.504	0.272 [0.066, 0.423]	−0.435 [−0.573, −0.312]	0.554 [0.341, 0.723]	0.554	yes / yes / yes	yes
U+M19	NSD 3 px	0.876 [0.825, 0.921]	0.632	M19 0.392	0.244 [0.046, 0.443]	−0.443 [−0.580, −0.321]	0.509 [0.302, 0.687]	0.509	yes / yes / yes	yes
U+M19	NSD 5 px	0.861 [0.806, 0.908]	0.628	M19 0.309	0.232 [0.044, 0.449]	−0.452 [−0.584, −0.327]	0.464 [0.256, 0.640]	0.464	yes / yes / yes	yes
U+M19	NSD 1% of diagonal	0.879 [0.825, 0.927]	0.615	M19 0.434	0.264 [0.062, 0.450]	−0.446 [−0.585, −0.326]	0.522 [0.302, 0.707]	0.522	yes / yes / yes	yes
U+M19	Dice, FUSeg at 256 px	0.819 [0.775, 0.868]	0.404	M19 0.165	0.415 [0.207, 0.600]	−0.557 [−0.661, −0.442]	0.281 [0.096, 0.469]	0.281	yes / yes / yes	yes
U+M19	NSD 2 px, FUSeg at 256 px	0.886 [0.834, 0.927]	0.626	M19 0.485	0.260 [0.073, 0.431]	−0.436 [−0.575, −0.311]	0.554 [0.337, 0.717]	0.554	yes / yes / yes	yes
U+F	Dice, P0 (main)	0.523 [0.294, 0.694]	0.351	F 0.164	0.173 [−0.057, 0.353]	−0.265 [−0.441, −0.093]	0.269 [0.074, 0.463]	0.269	no / yes / yes	no
U+F	Dice, P1	0.516 [0.286, 0.693]	0.351	F 0.183	0.166 [−0.049, 0.345]	−0.229 [−0.414, −0.055]	0.277 [0.090, 0.463]	0.277	no / yes / yes	no
U+F	Dice, P2	0.538 [0.308, 0.710]	0.384	F 0.190	0.154 [−0.056, 0.343]	−0.253 [−0.436, −0.076]	0.300 [0.112, 0.489]	0.299	no / yes / yes	no
U+F	NSD 2 px, P0 (main)	0.555 [0.340, 0.729]	0.622	F 0.314	−0.067 [−0.276, 0.140]	−0.105 [−0.273, 0.055]	0.487 [0.270, 0.666]	0.487	no / no / yes	no
U+F	NSD 2 px, P1	0.501 [0.277, 0.671]	0.622	F 0.146	−0.121 [−0.329, 0.113]	−0.083 [−0.241, 0.064]	0.414 [0.206, 0.603]	0.413	no / no / yes	no
U+F	NSD 2 px, P2	0.478 [0.268, 0.647]	0.640	F 0.168	−0.161 [−0.384, 0.098]	−0.052 [−0.204, 0.084]	0.434 [0.216, 0.620]	0.434	no / no / yes	no
U+F	NSD 1 px	0.524 [0.304, 0.701]	0.606	F 0.345	−0.083 [−0.304, 0.118]	−0.095 [−0.265, 0.056]	0.492 [0.271, 0.671]	0.492	no / no / yes	no
U+F	NSD 3 px	0.591 [0.372, 0.753]	0.632	F 0.267	−0.041 [−0.253, 0.179]	−0.112 [−0.278, 0.049]	0.472 [0.250, 0.647]	0.472	no / no / yes	no
U+F	NSD 5 px	0.621 [0.411, 0.775]	0.628	F 0.178	−0.007 [−0.213, 0.232]	−0.093 [−0.269, 0.076]	0.432 [0.218, 0.598]	0.431	no / no / yes	no
U+F	NSD 1% of diagonal	0.579 [0.363, 0.742]	0.615	F 0.205	−0.036 [−0.246, 0.192]	−0.115 [−0.281, 0.044]	0.436 [0.216, 0.612]	0.436	no / no / yes	no
U+F	Dice, FUSeg at 256 px	0.539 [0.313, 0.707]	0.404	F 0.160	0.135 [−0.088, 0.334]	−0.259 [−0.439, −0.084]	0.297 [0.096, 0.492]	0.297	no / yes / yes	no
U+F	NSD 2 px, FUSeg at 256 px	0.533 [0.320, 0.706]	0.626	F 0.176	−0.093 [−0.289, 0.142]	−0.115 [−0.276, 0.036]	0.429 [0.222, 0.610]	0.429	no / no / yes	no
U+K11	Dice, P0 (main)	0.200 [0.067, 0.360]	0.351	K11 0.258	−0.150 [−0.381, −0.021]	0.080 [−0.047, 0.195]	0.315 [0.116, 0.488]	0.315	no / no / yes	no
U+K11	Dice, P1	0.200 [0.066, 0.359]	0.351	K11 0.270	−0.151 [−0.376, −0.027]	0.080 [−0.046, 0.193]	0.320 [0.123, 0.488]	0.320	no / no / yes	no
U+K11	Dice, P2	0.220 [0.084, 0.374]	0.384	K11 0.267	−0.164 [−0.373, −0.036]	0.078 [−0.043, 0.190]	0.340 [0.141, 0.511]	0.340	no / no / yes	no
U+K11	NSD 2 px, P0 (main)	0.554 [0.324, 0.739]	0.622	K11 0.182	−0.068 [−0.199, 0.076]	0.118 [−0.082, 0.292]	0.455 [0.227, 0.660]	0.455	no / no / yes	no
U+K11	NSD 2 px, P1	0.553 [0.324, 0.739]	0.622	K11 0.114	−0.068 [−0.193, 0.075]	0.105 [−0.098, 0.285]	0.430 [0.208, 0.630]	0.429	no / no / yes	no
U+K11	NSD 2 px, P2	0.586 [0.374, 0.745]	0.640	K11 0.174	−0.054 [−0.193, 0.107]	0.060 [−0.143, 0.236]	0.464 [0.240, 0.650]	0.463	no / no / yes	no
U+K11	NSD 1 px	0.566 [0.344, 0.738]	0.606	K11 0.162	−0.040 [−0.197, 0.126]	0.097 [−0.112, 0.282]	0.438 [0.208, 0.645]	0.438	no / no / yes	no

Set	Condition	Pooled wound pairs	Within U	Within other group	Gap	Pooled wound with non-wound	Stratified wound pairs	Size-weighted within	C1 / C2 / C3	Holds
U+K11	NSD 3 px	0.548 [0.321, 0.733]	0.632	K11 0.214	−0.084 [−0.223, 0.059]	0.147 [−0.052, 0.315]	0.474 [0.250, 0.672]	0.473	no / no / yes	no
U+K11	NSD 5 px	0.549 [0.303, 0.735]	0.628	K11 0.230	−0.080 [−0.215, 0.050]	0.188 [−0.010, 0.349]	0.478 [0.242, 0.674]	0.477	no / no / yes	no
U+K11	NSD 1% of diagonal	0.544 [0.304, 0.733]	0.615	K11 0.232	−0.071 [−0.194, 0.064]	0.161 [−0.044, 0.329]	0.470 [0.242, 0.666]	0.469	no / no / yes	no
U+K11	Dice, FUSeg at 256 px	0.226 [0.080, 0.381]	0.404	K11 0.192	−0.177 [−0.391, −0.019]	0.090 [−0.034, 0.202]	0.324 [0.126, 0.506]	0.323	no / no / yes	no
U+K11	NSD 2 px, FUSeg at 256 px	0.561 [0.328, 0.747]	0.626	K11 0.177	−0.065 [−0.183, 0.078]	0.142 [−0.059, 0.313]	0.456 [0.231, 0.659]	0.456	no / no / yes	no

Table S6h. Composition criteria in the colonoscopy domain (mirror view). The criteria of Table S6g applied with the 10 colonoscopy pairs and the 30 pairs of a colonoscopy dataset with a non-colonoscopy dataset in place of the wound pairs; relevant mainly for U+K11, whose trained methods were trained on colonoscopy images. Pooled sets U+M19 (37), U+F (32), U+K11 (29): U+M19 and U+F as in the main analysis; U+K11 is U pooled with K11 (Notation), so it differs from U+K. Scores from the rescoring of every method (untrained and from-scratch methods rescored on a local computer, the other trained models in the second GPU session, nnU-Net re-predicted). Intervals as in Table S6g.

Set	Condition	Pooled colonoscopy pairs	Within U	Within other group	Gap	Pooled colonoscopy with other	Stratified colonoscopy pairs	C1 / C2 / C3	Holds
U+M19	Dice, P0 (main)	0.820 [0.714, 0.890]	0.859	M19 0.403	−0.039 [−0.165, 0.156]	−0.233 [−0.326, −0.145]	0.625 [0.468, 0.745]	no / yes / yes	no
U+M19	Dice, P1	0.827 [0.719, 0.893]	0.859	M19 0.414	−0.033 [−0.154, 0.161]	−0.231 [−0.325, −0.142]	0.631 [0.470, 0.747]	no / yes / yes	no
U+M19	Dice, P2	0.843 [0.749, 0.903]	0.845	M19 0.425	−0.002 [−0.123, 0.177]	−0.249 [−0.342, −0.160]	0.629 [0.472, 0.741]	no / yes / yes	no
U+M19	NSD 2 px, P0 (main)	0.766 [0.627, 0.855]	0.777	M19 0.577	−0.010 [−0.143, 0.184]	−0.130 [−0.256, −0.015]	0.674 [0.497, 0.776]	no / yes / yes	no
U+M19	NSD 2 px, P1	0.787 [0.650, 0.871]	0.777	M19 0.601	0.010 [−0.128, 0.198]	−0.139 [−0.262, −0.027]	0.686 [0.514, 0.778]	no / yes / yes	no
U+M19	NSD 2 px, P2	0.791 [0.654, 0.872]	0.790	M19 0.618	0.001 [−0.150, 0.209]	−0.127 [−0.255, −0.012]	0.702 [0.523, 0.795]	no / yes / yes	no
U+M19	NSD 1 px	0.769 [0.628, 0.856]	0.769	M19 0.561	0.000 [−0.143, 0.211]	−0.120 [−0.236, −0.014]	0.662 [0.473, 0.769]	no / yes / yes	no
U+M19	NSD 3 px	0.764 [0.619, 0.858]	0.790	M19 0.587	−0.026 [−0.147, 0.165]	−0.135 [−0.259, −0.027]	0.686 [0.514, 0.782]	no / yes / yes	no
U+M19	NSD 5 px	0.791 [0.650, 0.874]	0.818	M19 0.626	−0.027 [−0.147, 0.151]	−0.141 [−0.258, −0.035]	0.719 [0.556, 0.807]	no / yes / yes	no
U+M19	NSD 1% of diagonal	0.768 [0.624, 0.859]	0.796	M19 0.595	−0.028 [−0.149, 0.164]	−0.133 [−0.251, −0.026]	0.693 [0.523, 0.787]	no / yes / yes	no
U+M19	Dice, FUSeg at 256 px	0.820 [0.714, 0.890]	0.859	M19 0.403	−0.039 [−0.165, 0.156]	−0.227 [−0.319, −0.141]	0.625 [0.468, 0.745]	no / yes / yes	no
U+M19	NSD 2 px, FUSeg at 256 px	0.766 [0.627, 0.855]	0.777	M19 0.577	−0.010 [−0.143, 0.184]	−0.125 [−0.248, −0.016]	0.674 [0.497, 0.776]	no / yes / yes	no
U+F	Dice, P0 (main)	0.899 [0.814, 0.935]	0.859	F 0.879	0.020 [−0.107, 0.106]	0.010 [−0.112, 0.135]	0.868 [0.714, 0.927]	no / no / yes	no
U+F	Dice, P1	0.907 [0.827, 0.938]	0.859	F 0.880	0.027 [−0.103, 0.116]	0.040 [−0.089, 0.170]	0.868 [0.710, 0.931]	no / no / yes	no
U+F	Dice, P2	0.903 [0.819, 0.939]	0.845	F 0.885	0.018 [−0.109, 0.129]	0.021 [−0.106, 0.147]	0.862 [0.710, 0.925]	no / no / yes	no
U+F	NSD 2 px, P0 (main)	0.860 [0.754, 0.907]	0.777	F 0.882	−0.022 [−0.134, 0.111]	0.132 [0.009, 0.242]	0.823 [0.666, 0.885]	no / no / yes	no
U+F	NSD 2 px, P1	0.840 [0.727, 0.896]	0.777	F 0.843	−0.003 [−0.135, 0.119]	0.145 [0.032, 0.247]	0.806 [0.644, 0.876]	no / no / yes	no
U+F	NSD 2 px, P2	0.847 [0.746, 0.897]	0.790	F 0.837	0.010 [−0.117, 0.115]	0.170 [0.051, 0.267]	0.811 [0.648, 0.880]	no / no / yes	no
U+F	NSD 1 px	0.853 [0.751, 0.904]	0.769	F 0.871	−0.017 [−0.137, 0.123]	0.139 [0.018, 0.243]	0.813 [0.651, 0.882]	no / no / yes	no
U+F	NSD 3 px	0.861 [0.760, 0.908]	0.790	F 0.887	−0.026 [−0.137, 0.103]	0.130 [0.009, 0.241]	0.832 [0.678, 0.890]	no / no / yes	no
U+F	NSD 5 px	0.867 [0.770, 0.910]	0.818	F 0.897	−0.030 [−0.135, 0.087]	0.142 [0.014, 0.256]	0.853 [0.704, 0.909]	no / no / yes	no
U+F	NSD 1% of diagonal	0.864 [0.763, 0.909]	0.796	F 0.887	−0.023 [−0.135, 0.101]	0.126 [0.010, 0.235]	0.836 [0.682, 0.893]	no / no / yes	no
U+F	Dice, FUSeg at 256 px	0.899 [0.814, 0.935]	0.859	F 0.879	0.020 [−0.107, 0.106]	0.015 [−0.109, 0.142]	0.868 [0.714, 0.927]	no / no / yes	no
U+F	NSD 2 px, FUSeg at 256 px	0.860 [0.754, 0.907]	0.777	F 0.882	−0.022 [−0.134, 0.111]	0.124 [0.008, 0.231]	0.823 [0.666, 0.885]	no / no / yes	no
U+K11	Dice, P0 (main)	0.932 [0.856, 0.962]	0.859	K11 0.713	0.072 [−0.004, 0.174]	0.020 [−0.096, 0.117]	0.804 [0.624, 0.880]	no / no / yes	no
U+K11	Dice, P1	0.933 [0.856, 0.963]	0.859	K11 0.738	0.074 [−0.002, 0.172]	0.022 [−0.095, 0.118]	0.813 [0.640, 0.887]	no / no / yes	no
U+K11	Dice, P2	0.932 [0.863, 0.960]	0.845	K11 0.738	0.087 [0.008, 0.183]	0.022 [−0.094, 0.121]	0.805 [0.630, 0.875]	yes / no / yes	no
U+K11	NSD 2 px, P0 (main)	0.879 [0.753, 0.933]	0.777	K11 0.863	0.016 [−0.098, 0.154]	−0.014 [−0.214, 0.161]	0.809 [0.661, 0.872]	no / no / yes	no
U+K11	NSD 2 px, P1	0.897 [0.798, 0.937]	0.777	K11 0.830	0.067 [−0.058, 0.197]	−0.025 [−0.230, 0.155]	0.797 [0.620, 0.868]	no / no / yes	no
U+K11	NSD 2 px, P2	0.897 [0.783, 0.940]	0.790	K11 0.838	0.059 [−0.096, 0.186]	−0.070 [−0.274, 0.107]	0.808 [0.620, 0.880]	no / no / yes	no
U+K11	NSD 1 px	0.852 [0.699, 0.922]	0.769	K11 0.862	−0.010 [−0.147, 0.150]	−0.025 [−0.235, 0.157]	0.804 [0.641, 0.871]	no / no / yes	no
U+K11	NSD 3 px	0.901 [0.805, 0.937]	0.790	K11 0.835	0.066 [−0.055, 0.178]	0.019 [−0.193, 0.187]	0.807 [0.644, 0.868]	no / no / yes	no
U+K11	NSD 5 px	0.925 [0.858, 0.951]	0.818	K11 0.830	0.095 [−0.019, 0.184]	0.075 [−0.136, 0.238]	0.823 [0.649, 0.884]	no / no / yes	no
U+K11	NSD 1% of diagonal	0.902 [0.807, 0.937]	0.796	K11 0.821	0.081 [−0.049, 0.185]	0.038 [−0.175, 0.208]	0.805 [0.638, 0.867]	no / no / yes	no
U+K11	Dice, FUSeg at 256 px	0.932 [0.856, 0.962]	0.859	K11 0.713	0.072 [−0.004, 0.174]	0.030 [−0.087, 0.125]	0.804 [0.624, 0.880]	no / no / yes	no
U+K11	NSD 2 px, FUSeg at 256 px	0.879 [0.753, 0.933]	0.777	K11 0.863	0.016 [−0.098, 0.154]	0.018 [−0.185, 0.193]	0.809 [0.661, 0.872]	no / no / yes	no

Table S6i. Wound-pair agreement, changes and modality difference under every condition. Single groups (U, F, M19, K11) and pooled sets. Changes are relative to the main condition of the same metric, with paired intervals (Dice, P0 (main) for Dice, P1, Dice, P2, Dice, FUSeg at 256 px; NSD 2 px, P0 (main) for NSD 2 px, P1, NSD 2 px, P2, NSD 1 px, NSD 3 px, NSD 5 px, NSD 1% of diagonal, NSD 2 px, FUSeg at 256 px). Stratified change: pooled sets only. Modality difference over the 55 pairs with its interval and exact p. Exact p: share of the 13,860 distinct relabellings of the 11 modality labels whose modality difference is at least the observed one; (min) marks the smallest attainable p (2/13,860 = 0.00014). Pooled sets U+M19 (37), U+F (32), U+K11 (29): U+M19 and U+F as in the main analysis; U+K11 is U pooled with K11 (Notation), so it differs from U+K. Scores from the rescoring of every method (untrained and from-scratch methods rescored on a local computer, the other trained models in the second GPU session, nnU-Net re-predicted). Conditions: post-processing P0 (main), P1, P2 for Dice and NSD at 2 px; NSD at 1, 3, 5 px and at 1% of the image diagonal (P0); FUSeg replaced by FUSeg at 256 px (P0). For untrained methods P1 equals P0, because their main pipeline already applies closing and opening. Intervals: method bootstrap, B = 2,000, 95% percentile.

Set	Condition	Wound pairs	Change in wound pairs	Change in stratified wound pairs	Modality difference (55)	Exact p
U	Dice, P0 (main)	0.351 [0.122, 0.594]			0.596 [0.349, 0.830]	0.00029
U	Dice, P1	0.351 [0.122, 0.594]	0.000 [0.000, 0.000]		0.596 [0.349, 0.830]	0.00029
U	Dice, P2	0.384 [0.150, 0.621]	0.034 [−0.017, 0.120]		0.633 [0.386, 0.859]	0.00014 (min)
U	NSD 2 px, P0 (main)	0.622 [0.337, 0.834]			0.421 [0.270, 0.543]	0.00014 (min)
U	NSD 2 px, P1	0.622 [0.337, 0.834]	0.000 [0.000, 0.000]		0.421 [0.270, 0.543]	0.00014 (min)
U	NSD 2 px, P2	0.640 [0.366, 0.841]	0.018 [−0.139, 0.232]		0.400 [0.254, 0.530]	0.00014 (min)
U	NSD 1 px	0.606 [0.322, 0.833]	−0.015 [−0.094, 0.036]		0.432 [0.279, 0.558]	0.00014 (min)
U	NSD 3 px	0.632 [0.350, 0.845]	0.010 [−0.048, 0.088]		0.419 [0.274, 0.541]	0.00014 (min)
U	NSD 5 px	0.628 [0.339, 0.841]	0.007 [−0.071, 0.090]		0.412 [0.274, 0.527]	0.00014 (min)
U	NSD 1% of diagonal	0.615 [0.328, 0.827]	−0.007 [−0.085, 0.079]		0.420 [0.278, 0.541]	0.00014 (min)
U	Dice, FUSeg at 256 px	0.404 [0.165, 0.639]	0.053 [−0.024, 0.167]		0.597 [0.357, 0.821]	0.00029
U	NSD 2 px, FUSeg at 256 px	0.626 [0.342, 0.828]	0.004 [−0.036, 0.064]		0.424 [0.281, 0.544]	0.00014 (min)
F	Dice, P0 (main)	0.164 [−0.152, 0.531]			0.467 [0.209, 0.722]	0.0046
F	Dice, P1	0.183 [−0.098, 0.499]	0.019 [−0.149, 0.163]		0.430 [0.220, 0.634]	0.019
F	Dice, P2	0.190 [−0.098, 0.509]	0.026 [−0.147, 0.180]		0.432 [0.234, 0.635]	0.019
F	NSD 2 px, P0 (main)	0.314 [−0.033, 0.626]			0.563 [0.302, 0.822]	0.00014 (min)
F	NSD 2 px, P1	0.146 [−0.172, 0.516]	−0.168 [−0.391, 0.056]		0.454 [0.301, 0.598]	0.0017
F	NSD 2 px, P2	0.168 [−0.171, 0.549]	−0.145 [−0.374, 0.072]		0.469 [0.311, 0.633]	0.00058
F	NSD 1 px	0.345 [−0.009, 0.657]	0.032 [−0.066, 0.119]		0.567 [0.314, 0.817]	0.00014 (min)
F	NSD 3 px	0.267 [−0.078, 0.590]	−0.047 [−0.193, 0.018]		0.538 [0.301, 0.774]	0.00014 (min)
F	NSD 5 px	0.178 [−0.121, 0.483]	−0.136 [−0.352, 0.010]		0.485 [0.265, 0.687]	0.0012
F	NSD 1% of diagonal	0.205 [−0.124, 0.550]	−0.108 [−0.288, 0.006]		0.538 [0.300, 0.768]	0.00014 (min)
F	Dice, FUSeg at 256 px	0.160 [−0.149, 0.538]	−0.004 [−0.102, 0.083]		0.469 [0.220, 0.723]	0.0045
F	NSD 2 px, FUSeg at 256 px	0.176 [−0.134, 0.519]	−0.138 [−0.304, −0.022]		0.550 [0.299, 0.787]	0.00043
M19	Dice, P0 (main)	0.206 [−0.070, 0.491]			0.225 [0.048, 0.457]	0.0061

Set	Condition	Wound pairs	Change in wound pairs	Change in stratified wound pairs	Modality difference (55)	Exact p
M19	Dice, P1	0.256 [−0.021, 0.522]	0.051 [−0.072, 0.165]		0.229 [0.062, 0.433]	0.0084
M19	Dice, P2	0.255 [−0.018, 0.517]	0.049 [−0.066, 0.165]		0.243 [0.069, 0.457]	0.0053
M19	NSD 2 px, P0 (main)	0.479 [0.169, 0.725]			0.360 [0.179, 0.522]	0.00029
M19	NSD 2 px, P1	0.526 [0.245, 0.740]	0.048 [−0.132, 0.225]		0.369 [0.192, 0.543]	0.00029
M19	NSD 2 px, P2	0.522 [0.237, 0.739]	0.044 [−0.150, 0.231]		0.378 [0.195, 0.551]	0.00029
M19	NSD 1 px	0.504 [0.201, 0.750]	0.026 [−0.080, 0.156]		0.309 [0.149, 0.455]	0.0013
M19	NSD 3 px	0.392 [0.080, 0.662]	−0.087 [−0.265, 0.077]		0.383 [0.190, 0.562]	0.00029
M19	NSD 5 px	0.309 [0.001, 0.585]	−0.170 [−0.418, 0.095]		0.453 [0.252, 0.637]	0.00014 (min)
M19	NSD 1% of diagonal	0.434 [0.097, 0.709]	−0.045 [−0.268, 0.197]		0.414 [0.214, 0.599]	0.00014 (min)
M19	Dice, FUSeg at 256 px	0.165 [−0.093, 0.438]	−0.040 [−0.187, 0.067]		0.202 [0.046, 0.393]	0.016
M19	NSD 2 px, FUSeg at 256 px	0.485 [0.184, 0.723]	0.007 [−0.100, 0.119]		0.353 [0.175, 0.504]	0.00043
K11	Dice, P0 (main)	0.258 [−0.053, 0.511]			0.459 [0.134, 0.725]	0.00043
K11	Dice, P1	0.270 [−0.028, 0.507]	0.012 [−0.020, 0.045]		0.486 [0.166, 0.750]	0.00029
K11	Dice, P2	0.267 [−0.036, 0.507]	0.009 [−0.023, 0.040]		0.482 [0.159, 0.750]	0.00029
K11	NSD 2 px, P0 (main)	0.182 [−0.174, 0.592]			0.605 [0.308, 0.837]	0.00043
K11	NSD 2 px, P1	0.114 [−0.199, 0.511]	−0.068 [−0.206, 0.012]		0.595 [0.274, 0.869]	0.00043
K11	NSD 2 px, P2	0.174 [−0.192, 0.592]	−0.008 [−0.159, 0.174]		0.610 [0.250, 0.919]	0.00043
K11	NSD 1 px	0.162 [−0.198, 0.588]	−0.020 [−0.175, 0.191]		0.609 [0.328, 0.852]	0.00043
K11	NSD 3 px	0.214 [−0.125, 0.592]	0.032 [−0.130, 0.198]		0.566 [0.265, 0.790]	0.00029
K11	NSD 5 px	0.230 [−0.119, 0.595]	0.048 [−0.109, 0.185]		0.593 [0.242, 0.863]	0.00029
K11	NSD 1% of diagonal	0.232 [−0.101, 0.595]	0.050 [−0.114, 0.215]		0.567 [0.241, 0.819]	0.00029
K11	Dice, FUSeg at 256 px	0.192 [−0.104, 0.482]	−0.065 [−0.242, 0.000]		0.434 [0.094, 0.718]	0.00043
K11	NSD 2 px, FUSeg at 256 px	0.177 [−0.167, 0.573]	−0.005 [−0.142, 0.097]		0.596 [0.299, 0.823]	0.00043
U+M19	Dice, P0 (main)	0.818 [0.773, 0.866]			1.101 [0.983, 1.197]	0.00014 (min)
U+M19	Dice, P1	0.825 [0.780, 0.871]	0.007 [−0.008, 0.021]	0.026 [−0.032, 0.081]	1.104 [0.992, 1.197]	0.00014 (min)
U+M19	Dice, P2	0.829 [0.783, 0.875]	0.010 [−0.006, 0.028]	0.041 [−0.022, 0.111]	1.132 [1.031, 1.217]	0.00014 (min)
U+M19	NSD 2 px, P0 (main)	0.885 [0.834, 0.928]			1.010 [0.875, 1.123]	0.00014 (min)
U+M19	NSD 2 px, P1	0.888 [0.841, 0.927]	0.004 [−0.021, 0.029]	0.024 [−0.064, 0.118]	1.025 [0.894, 1.137]	0.00014 (min)
U+M19	NSD 2 px, P2	0.894 [0.847, 0.932]	0.009 [−0.024, 0.043]	0.031 [−0.097, 0.169]	1.018 [0.880, 1.134]	0.00014 (min)
U+M19	NSD 1 px	0.879 [0.820, 0.927]	−0.006 [−0.036, 0.018]	0.006 [−0.060, 0.077]	1.003 [0.873, 1.111]	0.00014 (min)
U+M19	NSD 3 px	0.876 [0.825, 0.921]	−0.009 [−0.035, 0.013]	−0.040 [−0.136, 0.044]	1.011 [0.876, 1.129]	0.00014 (min)
U+M19	NSD 5 px	0.861 [0.806, 0.908]	−0.024 [−0.062, 0.012]	−0.084 [−0.212, 0.049]	1.031 [0.901, 1.145]	0.00014 (min)
U+M19	NSD 1% of diagonal	0.879 [0.825, 0.927]	−0.006 [−0.038, 0.027]	−0.026 [−0.148, 0.096]	1.017 [0.884, 1.134]	0.00014 (min)
U+M19	Dice, FUSeg at 256 px	0.819 [0.775, 0.868]	0.001 [−0.021, 0.019]	0.005 [−0.079, 0.079]	1.098 [0.982, 1.191]	0.00014 (min)

Set	Condition	Wound pairs	Change in wound pairs	Change in stratified wound pairs	Modality difference (55)	Exact p
U+M19	NSD 2 px, FUSeg at 256 px	0.886 [0.834, 0.927]	0.001 [−0.014, 0.018]	0.006 [−0.056, 0.069]	1.010 [0.875, 1.123]	0.00014 (min)
U+F	Dice, P0 (main)	0.523 [0.294, 0.694]			0.834 [0.638, 0.999]	0.00014 (min)
U+F	Dice, P1	0.516 [0.286, 0.693]	−0.007 [−0.055, 0.039]	0.008 [−0.069, 0.068]	0.808 [0.618, 0.970]	0.00014 (min)
U+F	Dice, P2	0.538 [0.308, 0.710]	0.015 [−0.034, 0.058]	0.031 [−0.056, 0.105]	0.834 [0.636, 0.997]	0.00014 (min)
U+F	NSD 2 px, P0 (main)	0.555 [0.340, 0.729]			0.713 [0.587, 0.829]	0.00014 (min)
U+F	NSD 2 px, P1	0.501 [0.277, 0.671]	−0.054 [−0.121, 0.007]	−0.073 [−0.181, 0.026]	0.667 [0.549, 0.773]	0.00014 (min)
U+F	NSD 2 px, P2	0.478 [0.268, 0.647]	−0.076 [−0.180, 0.020]	−0.053 [−0.194, 0.101]	0.639 [0.519, 0.752]	0.00014 (min)
U+F	NSD 1 px	0.524 [0.304, 0.701]	−0.031 [−0.075, 0.007]	0.005 [−0.056, 0.059]	0.689 [0.547, 0.812]	0.00014 (min)
U+F	NSD 3 px	0.591 [0.372, 0.753]	0.036 [−0.003, 0.080]	−0.015 [−0.088, 0.037]	0.737 [0.623, 0.841]	0.00014 (min)
U+F	NSD 5 px	0.621 [0.411, 0.775]	0.067 [0.004, 0.141]	−0.055 [−0.156, 0.022]	0.745 [0.634, 0.846]	0.00014 (min)
U+F	NSD 1% of diagonal	0.579 [0.363, 0.742]	0.024 [−0.022, 0.072]	−0.051 [−0.139, 0.014]	0.737 [0.623, 0.840]	0.00014 (min)
U+F	Dice, FUSeg at 256 px	0.539 [0.313, 0.707]	0.015 [−0.009, 0.047]	0.028 [−0.035, 0.100]	0.835 [0.644, 0.996]	0.00014 (min)
U+F	NSD 2 px, FUSeg at 256 px	0.533 [0.320, 0.706]	−0.021 [−0.049, 0.001]	−0.058 [−0.143, −0.002]	0.712 [0.586, 0.828]	0.00014 (min)
U+K11	Dice, P0 (main)	0.200 [0.067, 0.360]			0.667 [0.564, 0.756]	0.00014 (min)
U+K11	Dice, P1	0.200 [0.066, 0.359]	0.000 [−0.011, 0.008]	0.005 [−0.008, 0.018]	0.667 [0.563, 0.756]	0.00014 (min)
U+K11	Dice, P2	0.220 [0.084, 0.374]	0.020 [−0.005, 0.055]	0.024 [−0.011, 0.077]	0.680 [0.576, 0.763]	0.00014 (min)
U+K11	NSD 2 px, P0 (main)	0.554 [0.324, 0.739]			0.751 [0.603, 0.889]	0.00014 (min)
U+K11	NSD 2 px, P1	0.553 [0.324, 0.739]	−0.001 [−0.020, 0.026]	−0.026 [−0.079, 0.003]	0.774 [0.628, 0.917]	0.00014 (min)
U+K11	NSD 2 px, P2	0.586 [0.374, 0.745]	0.032 [−0.086, 0.180]	0.008 [−0.119, 0.148]	0.806 [0.646, 0.964]	0.00014 (min)
U+K11	NSD 1 px	0.566 [0.344, 0.738]	0.012 [−0.044, 0.075]	−0.017 [−0.098, 0.063]	0.748 [0.576, 0.901]	0.00014 (min)
U+K11	NSD 3 px	0.548 [0.321, 0.733]	−0.006 [−0.066, 0.054]	0.018 [−0.049, 0.094]	0.744 [0.617, 0.870]	0.00014 (min)
U+K11	NSD 5 px	0.549 [0.303, 0.735]	−0.005 [−0.112, 0.099]	0.023 [−0.055, 0.089]	0.732 [0.615, 0.858]	0.00014 (min)
U+K11	NSD 1% of diagonal	0.544 [0.304, 0.733]	−0.010 [−0.088, 0.068]	0.015 [−0.064, 0.096]	0.735 [0.609, 0.860]	0.00014 (min)
U+K11	Dice, FUSeg at 256 px	0.226 [0.080, 0.381]	0.026 [−0.016, 0.085]	0.008 [−0.071, 0.088]	0.670 [0.567, 0.755]	0.00014 (min)
U+K11	NSD 2 px, FUSeg at 256 px	0.561 [0.328, 0.747]	0.007 [−0.028, 0.048]	0.001 [−0.052, 0.051]	0.737 [0.592, 0.872]	0.00014 (min)

Table S7a. Selection loss among the 19 modern models (M19, Dice). Rows: the set the model is chosen on (chosen model in brackets); columns: the wound dataset it is deployed on. Mean: over the deployment datasets other than the choice dataset (for the wound rows) or over all 4 (other rows). Mean over the 12 off-diagonal pairs: 0.048 (range 0.000 to 0.119). Selection loss from A to B: best mean Dice on B minus the mean Dice on B of the candidate with the highest mean Dice on A. Random choice: mean over candidates of (best on B minus candidate on B). Synthetic chooser: the candidate with the highest mean Dice on generator B at the given texture amplitude (mean over the 3 synthetic seeds). Point estimates (a loss depends on a maximum over candidates, so no interval is given). Model keys as in Table S3b.

Chosen on (pick)	Deployed on Medetec	Deployed on CO2Wounds-V2	Deployed on FUSeg	Deployed on AZH	Mean
Medetec (manet_r18)	0.000	0.032	0.007	0.110	0.050
CO2Wounds-V2 (segformer_mitb2)	0.039	0.000	0.021	0.000	0.020
FUSeg (manet_r50)	0.113	0.069	0.000	0.119	0.101
AZH (segformer_mitb2)	0.039	0.000	0.021	0.000	0.020
Control (segformer_mitb2)	0.039	0.000	0.021	0.000	0.015
Generator B, amplitude 11 (nnunet_2d)	0.007	0.000	0.033	0.081	0.030
Generator B, amplitude 28 (nnunet_2d)	0.007	0.000	0.033	0.081	0.030
Generator B, amplitude 40 (nnunet_2d)	0.007	0.000	0.033	0.081	0.030
Random choice (expected)	0.061	0.036	0.029	0.126	0.063

Table S7b. Synthetic chooser by candidate set and training seed. Candidate sets as defined in the Notation, with the number of candidates: M19 (19), M (11), M11s (11), P10s (10), N8s (8), M19_s0 (18), M19_s1 (19), M19_s2 (17); in M19_sk each architecture that has training seed k enters as that single model. Pick: the synthetic chooser's model at amplitude 11, 28 and 40, and at amplitude 40 per synthetic seed (2024 / 7 / 101). Loss per wound dataset at amplitude 40 in the order Medetec / CO2Wounds-V2 / FUSeg / AZH, and its mean. Within 0.01 / 0.02: number of the 4 wound datasets where the amplitude-40 pick is within 0.01 or 0.02 Dice of the best. Other wound dataset: mean loss when choosing on one wound dataset and deploying on another (12 ordered pairs). Selection loss from A to B: best mean Dice on B minus the mean Dice on B of the candidate with the highest mean Dice on A. Random choice: mean over candidates of (best on B minus candidate on B). Synthetic chooser: the candidate with the highest mean Dice on generator B at the given texture amplitude (mean over the 3 synthetic seeds). Point estimates (a loss depends on a maximum over candidates, so no interval is given).

Candidate set	n	Pick, amplitude 11	Pick, amplitude 28	Pick, amplitude 40	Pick per synthetic seed (amplitude 40)	Loss per wound dataset (amplitude 40)	Mean loss	Within 0.01 / 0.02	Other wound dataset	Random choice
M	11	nnunet_2d	nnunet_2d	nnunet_2d	nnunet_2d / nnunet_2d / nnunet_2d	0.007 / 0.012 / 0.032 / 0.000	0.013	2 / 3	0.041	0.046
M11s	11	nnunet_2d	nnunet_2d	nnunet_2d	nnunet_2d / nnunet_2d / nnunet_2d	0.007 / 0.000 / 0.026 / 0.000	0.008	3 / 3	0.016	0.044
M19	19	nnunet_2d	nnunet_2d	nnunet_2d	nnunet_2d / nnunet_2d / nnunet_2d	0.007 / 0.000 / 0.033 / 0.081	0.030	2 / 2	0.048	0.063
M19_s0	18	unet_effb3_s0	unet_effb3_s0	unet_effb3_s0	nnunet_2d / unet_effb3_s0 / nnunet_2d	0.090 / 0.041 / 0.035 / 0.101	0.067	0 / 0	0.059	0.060
M19_s1	19	nnunet_2d	nnunet_2d	nnunet_2d	nnunet_2d / nnunet_2d / nnunet_2d	0.027 / 0.033 / 0.033 / 0.109	0.051	0 / 0	0.077	0.085
M19_s2	17	nnunet_2d	nnunet_2d	nnunet_2d	nnunet_2d / nnunet_2d / nnunet_2d	0.005 / 0.000 / 0.024 / 0.066	0.024	2 / 2	0.053	0.059
P10s	10	segformer_mitb0	segformer_mitb0	segformer_mitb0	segformer_mitb0 / unet_mbv2 / segformer_mitb0	0.060 / 0.009 / 0.044 / 0.000	0.028	2 / 2	0.021	0.041
N8s	8	unet_effb3	unet_effb3	unet_effb3	unet_effb3 / unet_effb3 / unet_effb3	0.039 / 0.059 / 0.038 / 0.151	0.072	0 / 0	0.027	0.050

Table S7c. Best model per dataset by training seed. Model with the highest mean Dice on each wound dataset and the control (mean Dice in brackets), among the seed-averaged M19 candidates, the single-seed candidate sets M19_s0, M19_s1, M19_s2, and M. Candidate sets as defined in the Notation, with the number of candidates: M19 (19), M (11), M11s (11), P10s (10), N8s (8), M19_s0 (18), M19_s1 (19), M19_s2 (17); in M19_sk each architecture that has training seed k enters as that single model.

Dataset	M19 (seed averages)	M19_s0	M19_s1	M19_s2	M
Medetec	manet_r18 (0.771)	deeplabv3p_r18 (0.771)	manet_r18_s1 (0.791)	deeplabv3p_r18_s2 (0.769)	deeplabv3p_r18 (0.771)
CO2Wounds-V2	segformer_mitb2 (0.474)	segformer_mitb0 (0.486)	unet_r18_s1 (0.507)	nnunet_2d (0.474)	segformer_mitb0 (0.486)
FUSeg	manet_r50 (0.827)	manet_r18 (0.826)	manet_r50_s1 (0.827)	unetpp_r18_s2 (0.818)	manet_r18 (0.826)
AZH	segformer_mitb2 (0.708)	segformer_mitb2_s0 (0.695)	segformer_mitb2_s1 (0.737)	segformer_mitb2_s2 (0.693)	nnunet_2d (0.627)
Control	segformer_mitb2 (0.673)	segformer_mitb2_s0 (0.671)	unet_swin_t_s1 (0.709)	segformer_mitb2_s2 (0.658)	segformer_mitb0 (0.574)

Table S7d. Misses per architecture. Miss: an image with a non-empty reference mask and Dice below 0.1. Miss rate: misses over non-empty images (AZH 136 images in 113 photographs; FUSeg 108; control 72 in 66 photographs; Medetec 152; CO2Wounds-V2 150). Computed for every individual model; an architecture's rate is the mean over its training seeds (smallest to largest single-seed rate in brackets). A seed-averaged Dice is not used for misses because it is not the output of any model. AZH: second line gives the interval of the seed mean. Intervals: unit bootstrap (resampling units drawn with replacement, the same draws for every model of a dataset, models fixed), B = 2,000, 95% percentile; they cover image sampling only. Resampling units are photographs for Medetec, AZH and the control (Medetec 38 units for 152 images, AZH 118 units for 146 images, Control 71 units for 78 images) and single images otherwise.

Architecture	Set	Seeds	AZH: mean (min to max), interval	FUSeg: mean (min to max)	Control: mean (min to max)	Medetec	CO2Wounds-V2
unet_r18	original	3	32.8% (32.4% to 33.8%) [24.9%, 40.6%]	2.5% (1.9% to 2.8%)	57.9% (56.9% to 58.3%)	1.1%	26.2%
unetpp_r18	original	3	30.1% (29.4% to 31.6%) [22.5%, 37.9%]	1.9% (0.9% to 2.8%)	44.9% (40.3% to 48.6%)	1.8%	29.6%
deeplabv3p_r18	original	3	40.2% (39.7% to 40.4%) [31.7%, 48.2%]	3.4% (0.9% to 4.6%)	63.0% (61.1% to 63.9%)	0.9%	26.0%
fpn_r18	original	3	27.7% (25.7% to 30.9%) [19.8%, 35.6%]	1.9% (1.9% to 1.9%)	44.4% (38.9% to 51.4%)	0.0%	24.0%
pspnet_r18	original	3	28.4% (25.7% to 30.9%) [20.2%, 36.4%]	4.0% (3.7% to 4.6%)	42.6% (36.1% to 50.0%)	6.8%	29.1%
linknet_r18	original	3	30.4% (26.5% to 36.0%) [22.4%, 38.8%]	0.9% (0.0% to 1.9%)	41.2% (30.6% to 51.4%)	0.4%	26.2%
manet_r18	original	2	30.1% (29.4% to 30.9%) [22.5%, 37.8%]	1.9% (1.9% to 1.9%)	47.2% (45.8% to 48.6%)	0.0%	25.3%
segformer_mitb0	original	3	22.3% (18.4% to 24.3%) [15.2%, 30.0%]	4.0% (3.7% to 4.6%)	23.6% (19.4% to 27.8%)	3.3%	27.3%
unet_mbv2	original	3	38.7% (38.2% to 39.7%) [30.0%, 46.8%]	3.1% (2.8% to 3.7%)	57.4% (54.2% to 62.5%)	1.1%	24.0%
unet_effb0	original	3	30.9% (29.4% to 32.4%) [23.3%, 38.6%]	3.4% (2.8% to 3.7%)	40.3% (37.5% to 44.4%)	0.4%	26.9%
unet_r50	new	3	31.6% (30.9% to 32.4%) [23.7%, 39.6%]	2.8% (0.9% to 4.6%)	46.8% (44.4% to 48.6%)	3.9%	30.2%
deeplabv3p_r50	new	3	35.3% (28.7% to 39.7%) [27.0%, 43.8%]	2.5% (1.9% to 2.8%)	53.7% (43.1% to 62.5%)	4.6%	29.1%
upernet_convnext_t	new	3	21.1% (17.6% to 25.0%) [14.7%, 27.9%]	2.8% (1.9% to 3.7%)	31.9% (27.8% to 36.1%)	2.6%	28.4%
unet_swin_t	new	3	21.3% (14.7% to 28.7%) [14.9%, 28.4%]	3.7% (3.7% to 3.7%)	22.2% (15.3% to 26.4%)	4.2%	28.2%
segformer_mitb2	new	3	12.3% (10.3% to 13.2%) [7.4%, 17.3%]	3.1% (2.8% to 3.7%)	16.2% (13.9% to 19.4%)	3.3%	23.3%
unet_effb3	new	3	33.1% (30.1% to 39.0%) [24.8%, 41.4%]	2.8% (1.9% to 3.7%)	45.8% (37.5% to 51.4%)	3.9%	26.2%
fpn_regnety016	new	3	17.2% (12.5% to 19.9%) [11.0%, 23.1%]	1.9% (0.9% to 2.8%)	25.0% (15.3% to 34.7%)	1.1%	28.4%
manet_r50	new	1	32.4% (32.4% to 32.4%) [23.1%, 41.2%]	0.9% (0.9% to 0.9%)	44.4% (44.4% to 44.4%)	3.9%	29.3%
nnunet_2d	nnU-Net	1	1.5% (1.5% to 1.5%) [0.0%, 3.7%]	4.6% (4.6% to 4.6%)	1.4% (1.4% to 1.4%)	0.0%	26.0%

Table S7e. Misses by group and by training seed. Group miss rate: mean over architectures, each architecture weighted equally (rows original 10, new 8, all 18) or mean over the models of one training seed (rows by seed). Minus nnU-Net: paired difference to nnU-Net from the same draws. Ranges: over architecture means and over single models. Intervals: unit bootstrap (resampling units drawn with replacement, the same draws for every model of a dataset, models fixed), B = 2,000, 95% percentile; they cover image sampling only. Resampling units are photographs for Medetec, AZH and the control (Medetec 38 units for 152 images, AZH 118 units for 146 images, Control 71 units for 78 images) and single images otherwise.

Group	Dataset	Mean miss rate [interval]	Minus nnU-Net [interval]	Range of architecture means	Range of single models
original 10	AZH	31.2% [23.9%, 38.3%]	29.7% [22.3%, 36.8%]	22.3% to 40.2%	18.4% to 40.4%
new 8	AZH	25.5% [18.9%, 32.0%]	24.1% [17.4%, 30.3%]	12.3% to 35.3%	10.3% to 39.7%
all 18	AZH	28.7% [21.7%, 35.6%]	27.2% [20.2%, 33.8%]	12.3% to 40.2%	10.3% to 40.4%
nnU-Net	AZH	1.5% [0.0%, 3.7%]			
original 10	FUSeg	2.7% [0.8%, 5.0%]	−1.9% [−4.4%, 0.0%]	0.9% to 4.0%	0.0% to 4.6%
new 8	FUSeg	2.5% [0.6%, 5.2%]	−2.1% [−4.7%, −0.1%]	0.9% to 3.7%	0.9% to 4.6%
all 18	FUSeg	2.6% [0.7%, 5.0%]	−2.0% [−4.5%, −0.1%]	0.9% to 4.0%	0.0% to 4.6%
nnU-Net	FUSeg	4.6% [0.9%, 9.1%]			
original 10	Control	46.2% [37.1%, 55.3%]	44.9% [35.6%, 53.8%]	23.6% to 63.0%	19.4% to 63.9%
new 8	Control	35.8% [27.5%, 44.0%]	34.4% [26.0%, 42.3%]	16.2% to 53.7%	13.9% to 62.5%
all 18	Control	41.6% [33.0%, 50.0%]	40.2% [31.5%, 48.4%]	16.2% to 63.0%	13.9% to 63.9%
nnU-Net	Control	1.4% [0.0%, 4.4%]			
original, seed 0	AZH	31.0% [23.4%, 38.4%]			
original, seed 1	AZH	31.4% [24.2%, 38.6%]			
original, seed 2	AZH	31.2% [23.7%, 38.3%]			
new, seed 0	AZH	24.3% [17.9%, 30.7%]			
new, seed 1	AZH	24.5% [18.0%, 31.1%]			
new, seed 2	AZH	25.9% [19.3%, 32.5%]			
original, seed 0	FUSeg	3.0% [0.9%, 5.6%]			
original, seed 1	FUSeg	2.6% [0.7%, 5.1%]			
original, seed 2	FUSeg	2.6% [0.7%, 4.9%]			
new, seed 0	FUSeg	3.3% [0.8%, 6.5%]			
new, seed 1	FUSeg	2.4% [0.6%, 4.8%]			
new, seed 2	FUSeg	2.4% [0.4%, 5.0%]			
original, seed 0	Control	45.3% [35.9%, 54.5%]			
original, seed 1	Control	45.6% [36.3%, 54.7%]			
original, seed 2	Control	48.0% [38.9%, 57.2%]			
new, seed 0	Control	31.9% [24.3%, 39.8%]			
new, seed 1	Control	34.5% [26.2%, 42.9%]			
new, seed 2	Control	38.5% [29.8%, 47.3%]			

Table S7f. Misses and image padding (all pretrained models). Miss share of an image: share of the individual pretrained models (number in brackets) that miss it. Black share: share of image pixels with every channel at most 8 (padding of the crop). Columns: mean miss share over non-empty images; Spearman correlation of the miss share with the black share and with the lesion fraction; number of images missed by most pretrained models but detected by nnU-Net, and the reverse. Last rows (AZH): mean miss share in images with more vs less padding than the median within each lesion-size tercile (tercile 1 = smallest lesions). Mean black share over all kept images (the basis used in the main text): AZH 0.856 (85.6% of pixels), FUSeg 0.263 (26.3%); over kept images with a non-empty reference mask (the images of this table): AZH 0.850, FUSeg 0.264. Correlational: padding is not manipulated.

Dataset	Models	Mean miss share	Rank correlation with black share	Rank correlation with lesion fraction	Majority miss, nnU-Net detects	nnU-Net misses, majority detects
AZH	original 10, all seeds (29)	0.312	0.74	−0.47	41	0
AZH	new 8, all seeds (22)	0.249	0.68	−0.41	32	0
AZH	all 18, all seeds (51)	0.285	0.73	−0.46	39	0
Control	original 10, all seeds (29)	0.462	0.81	−0.38	33	0
Control	new 8, all seeds (22)	0.350	0.77	−0.43	26	0
Control	all 18, all seeds (51)	0.414	0.81	−0.39	28	0
FUSeg	original 10, all seeds (29)	0.027	0.04	−0.43	0	2
FUSeg	new 8, all seeds (22)	0.027	0.04	−0.39	0	2
FUSeg	all 18, all seeds (51)	0.027	0.08	−0.45	0	2
AZH	all 18, lesion-size tercile 1 (n 45)	0.78 vs 0.23				
AZH	all 18, lesion-size tercile 2 (n 45)	0.38 vs 0.06				
AZH	all 18, lesion-size tercile 3 (n 46)	0.19 vs 0.10				
AZH	new 8, lesion-size tercile 1 (n 45)	0.69 vs 0.16				
AZH	new 8, lesion-size tercile 2 (n 45)	0.32 vs 0.06				
AZH	new 8, lesion-size tercile 3 (n 46)	0.17 vs 0.11				

Table S8a. SAM and MedSAM, mean Dice. SAM and MedSAM (ViT-B) are prompted with boxes. Reference box: one box per connected component of the reference mask (an oracle prompt; these models are therefore never pooled with the other methods). MedSAM official: MedSAM with the preprocessing of its official inference script (second GPU session); MedSAM without that label is the first-session run. nnU-Net box: boxes of the nnU-Net prediction (up to 10 largest components); an image without an nnU-Net component gets no prompt and an empty prediction. Means over the kept images. nnU-Net for comparison, from the first session and from the second-session re-prediction whose masks gave the nnU-Net boxes.

Dataset	n	SAM, reference box	MedSAM, reference box	MedSAM official, reference box	SAM, nnU-Net box	MedSAM official, nnU-Net box	nnU-Net (first session)	nnU-Net (re-prediction)
Medetec	152	0.885	0.376	0.728	0.762	0.589	0.764	0.764
CO2Wounds-V2	150	0.792	0.552	0.659	0.460	0.360	0.474	0.474
FUSeg	112	0.896	0.752	0.728	0.800	0.653	0.794	0.794
AZH	146	0.907	0.580	0.668	0.637	0.518	0.627	0.627
Kvasir	120	0.932	0.792	0.839	0.041	0.042	0.044	0.044
Kvasir-sessile	120	0.885	0.719	0.752	0.013	0.014	0.010	0.010
CVC-ClinicDB	150	0.904	0.736	0.724	0.055	0.037	0.056	0.056
CVC-ColonDB	150	0.887	0.675	0.686	0.003	0.009	0.003	0.003
ETIS-Larib	150	0.896	0.798	0.750	0.021	0.021	0.020	0.020
PH2	150	0.887	0.813	0.561	0.471	0.283	0.440	0.440
TN3K	150	0.764	0.747	0.698	0.131	0.109	0.140	0.140
Control	78	0.875	0.553	0.660	0.532	0.496	0.539	0.539

Table S8b. SAM and MedSAM, mean NSD. SAM and MedSAM (ViT-B) are prompted with boxes. Reference box: one box per connected component of the reference mask (an oracle prompt; these models are therefore never pooled with the other methods). MedSAM official: MedSAM with the preprocessing of its official inference script (second GPU session); MedSAM without that label is the first-session run. nnU-Net box: boxes of the nnU-Net prediction (up to 10 largest components); an image without an nnU-Net component gets no prompt and an empty prediction. Means over the kept images. nnU-Net for comparison, from the first session and from the second-session re-prediction whose masks gave the nnU-Net boxes. NSD at 2 px.

Dataset	n	SAM, reference box	MedSAM, reference box	MedSAM official, reference box	SAM, nnU-Net box	MedSAM official, nnU-Net box	nnU-Net (first session)	nnU-Net (re-prediction)
Medetec	152	0.566	0.096	0.249	0.365	0.141	0.411	0.411
CO2Wounds-V2	150	0.548	0.292	0.357	0.260	0.138	0.257	0.257
FUSeg	112	0.808	0.446	0.374	0.683	0.307	0.666	0.666
AZH	146	0.880	0.298	0.426	0.511	0.257	0.421	0.421
Kvasir	120	0.709	0.388	0.478	0.023	0.020	0.017	0.017
Kvasir-sessile	120	0.653	0.311	0.362	0.011	0.007	0.008	0.008
CVC-ClinicDB	150	0.764	0.357	0.367	0.036	0.013	0.032	0.032
CVC-ColonDB	150	0.684	0.298	0.323	0.001	0.003	0.001	0.001
ETIS-Larib	150	0.832	0.585	0.495	0.007	0.005	0.006	0.006
PH2	150	0.319	0.295	0.147	0.088	0.034	0.100	0.100
TN3K	150	0.415	0.313	0.275	0.071	0.036	0.072	0.072
Control	78	0.858	0.342	0.502	0.402	0.336	0.298	0.298

Table S8c. Paired differences: official preprocessing and nnU-Net-box prompting. Paired differences in mean Dice and mean NSD (2 px), estimate [interval]. nnU-Net-box variants are compared with the nnU-Net re-prediction whose masks gave the boxes. Intervals: unit bootstrap, B = 2,000, 95% percentile. Resampling units are photographs for Medetec, AZH and the control (Medetec 38 units for 152 images, AZH 118 units for 146 images, Control 71 units for 78 images) and single images otherwise. SAM and MedSAM (ViT-B) are prompted with boxes. Reference box: one box per connected component of the reference mask (an oracle prompt; these models are therefore never pooled with the other methods). MedSAM official: MedSAM with the preprocessing of its official inference script (second GPU session); MedSAM without that label is the first-session run. nnU-Net box: boxes of the nnU-Net prediction (up to 10 largest components); an image without an nnU-Net component gets no prompt and an empty prediction. Means over the kept images.

Dataset	Dice: MedSAM official minus MedSAM, reference box	Dice: SAM with nnU-Net box minus nnU-Net	Dice: MedSAM official with nnU-Net box minus nnU-Net	NSD: MedSAM official minus MedSAM, reference box	NSD: SAM with nnU-Net box minus nnU-Net	NSD: MedSAM official with nnU-Net box minus nnU-Net
Medetec	0.352 [0.295, 0.409]	−0.002 [−0.015, 0.012]	−0.175 [−0.210, −0.140]	0.153 [0.116, 0.189]	−0.046 [−0.090, −0.004]	−0.270 [−0.338, −0.203]
CO2Wounds-V2	0.107 [0.062, 0.154]	−0.015 [−0.035, 0.002]	−0.114 [−0.143, −0.086]	0.065 [0.035, 0.096]	0.003 [−0.019, 0.027]	−0.118 [−0.149, −0.089]
FUSeg	−0.024 [−0.051, 0.004]	0.006 [−0.005, 0.017]	−0.141 [−0.169, −0.112]	−0.071 [−0.118, −0.025]	0.018 [−0.009, 0.043]	−0.358 [−0.419, −0.301]
AZH	0.088 [0.052, 0.124]	0.010 [−0.012, 0.030]	−0.109 [−0.148, −0.071]	0.128 [0.091, 0.166]	0.090 [0.057, 0.126]	−0.163 [−0.222, −0.103]
Kvasir	0.047 [0.018, 0.075]	−0.003 [−0.011, 0.003]	−0.002 [−0.014, 0.012]	0.090 [0.045, 0.134]	0.005 [−0.003, 0.015]	0.003 [−0.007, 0.015]
Kvasir-sessile	0.033 [0.004, 0.060]	0.003 [−0.002, 0.012]	0.003 [−0.007, 0.018]	0.051 [0.012, 0.088]	0.003 [−0.003, 0.012]	−0.001 [−0.010, 0.010]
CVC-ClinicDB	−0.012 [−0.045, 0.023]	−0.001 [−0.007, 0.005]	−0.019 [−0.034, −0.007]	0.009 [−0.036, 0.054]	0.004 [−0.005, 0.013]	−0.018 [−0.030, −0.010]
CVC-ColonDB	0.011 [−0.021, 0.041]	−0.001 [−0.002, 0.001]	0.006 [−0.002, 0.017]	0.025 [−0.014, 0.065]	0.000 [−0.001, 0.000]	0.002 [−0.001, 0.005]
ETIS-Larib	−0.048 [−0.071, −0.023]	0.001 [−0.001, 0.003]	0.001 [−0.002, 0.004]	−0.090 [−0.135, −0.046]	0.001 [−0.002, 0.003]	−0.001 [−0.003, 0.001]
PH2	−0.253 [−0.297, −0.210]	0.031 [0.018, 0.043]	−0.157 [−0.188, −0.129]	−0.148 [−0.184, −0.116]	−0.012 [−0.028, 0.004]	−0.066 [−0.095, −0.041]
TN3K	−0.049 [−0.076, −0.020]	−0.009 [−0.021, 0.000]	−0.031 [−0.047, −0.017]	−0.038 [−0.077, 0.001]	−0.001 [−0.012, 0.009]	−0.036 [−0.054, −0.020]
Control	0.108 [0.049, 0.162]	−0.008 [−0.040, 0.026]	−0.043 [−0.104, 0.014]	0.160 [0.098, 0.222]	0.104 [0.046, 0.165]	0.038 [−0.046, 0.123]

Table S8d. Images without an nnU-Net box. An image without an nnU-Net component gets no prompt and an empty prediction, scored 1 against an empty reference and 0 otherwise. The nnU-Net masks were not stored, so the count is exact only for empty references. For non-empty references, nnU-Net Dice 0 with NSD 0 at every tolerance is necessary but not sufficient for an empty mask; candidates where an nnU-Net-box variant scores above 0 had a box and are removed, which gives the range in the last column.

Dataset	n	Empty references	No box, empty reference (exact)	Non-empty candidates	Candidates with a box	No box (range)
Medetec	152	0	0	0	0	0 to 0
CO2Wounds-V2	150	0	0	29	0	0 to 29
FUSeg	112	4	1	5	0	1 to 6
AZH	146	10	0	0	0	0 to 0
Kvasir	120	0	0	87	0	0 to 87
Kvasir-sessile	120	0	0	108	1	0 to 107
CVC-ClinicDB	150	0	0	110	0	0 to 110
CVC-ColonDB	150	0	0	143	0	0 to 143
ETIS-Larib	150	0	0	134	0	0 to 134
PH2	150	0	0	14	0	0 to 14
TN3K	150	0	0	75	1	0 to 74
Control	78	6	0	0	0	0 to 0

Table S9a. Images, exclusions and resampling units. Images in the scored file, images excluded, images kept for every analysis, resampling units and kept images with an empty reference mask. Exclusions: 8 FUSeg validation images are byte-identical to FUSeg training images (7) or to another validation image (1); 4 AZH test images are crops of FUSeg photographs used in this study (all for model training); 72 control images are crops of FUSeg photographs used in this study (70: 51 for model training, 19 for the FUSeg reference) or pixel-identical to an AZH test image (2); the rule and the check are in Table S9e, every excluded file in Table S9f. Resampling units are photographs for Medetec, AZH and the control (Medetec 38 units for 152 images, AZH 118 units for 146 images, Control 71 units for 78 images) and single images otherwise. Synthetic sets: 45 images each, no exclusions.

Dataset	Modality	Source	Images	Excluded	Kept	Units	Empty references
Medetec	wound	Medetec	152	0	152	38	0
CO2Wounds-V2	wound	CO2Wounds	150	0	150	150	0
FUSeg	wound	AZH	120	8	112	112	4
AZH	wound	AZH	150	4	146	118	10
Kvasir	colonoscopy	Kvasir	120	0	120	120	0
Kvasir-sessile	colonoscopy	Kvasir	120	0	120	120	0
CVC-ClinicDB	colonoscopy	CVC	150	0	150	150	0
CVC-ColonDB	colonoscopy	CVC	150	0	150	150	0
ETIS-Larib	colonoscopy	ETIS	150	0	150	150	0
PH2	dermoscopy	PH2	150	0	150	150	0
TN3K	ultrasound	TN3K	150	0	150	150	0
Control	wound	AZH	150	72	78	71	6

Table S9b. nnU-Net: first-session scores against the second-session re-prediction. The same nnU-Net checkpoint predicted again in a second GPU session (half-precision inference). Images: per-image results compared (all images before exclusions). Counts of images whose Dice differs by more than 0.0001 and 0.001; largest absolute per-image difference in Dice and NSD (2 px); mean Dice over the kept images in each session and the difference (second minus first). Direction of the 107 larger differences: 63 higher, 44 lower in the second session. nnU-Net's rank among the 11 modern models of M is the same in both sessions on every dataset, for Dice and NSD. The main analyses use the first-session scores.

Dataset	Images	Dice differs > 0.0001	Dice differs > 0.001	Largest Dice difference	Largest NSD difference	Mean Dice, first session	Mean Dice, second session	Difference of means
Medetec	152	14	0	0.0004	0.0042	0.76394	0.76394	7.0e−06
CO2Wounds-V2	150	12	1	0.0034	0.0049	0.47417	0.47420	2.6e−05
FUSeg	120	6	0	0.0004	0.0016	0.79428	0.79428	−1.2e−06
AZH	150	20	0	0.0006	0.0044	0.62726	0.62727	1.3e−05
Kvasir	120	7	0	0.0005	0.0002	0.04443	0.04443	2.5e−06
Kvasir-sessile	120	1	0	0.0005	0.0000	0.01023	0.01022	−4.0e−06
CVC-ClinicDB	150	3	0	0.0003	0.0000	0.05635	0.05635	1.7e−06
CVC-ColonDB	150	1	0	0.0006	0.0000	0.00329	0.00329	4.0e−06
ETIS-Larib	150	0	0	0.0000	0.0000	0.01977	0.01977	3.4e−07
PH2	150	9	0	0.0002	0.0007	0.43992	0.43992	−6.3e−07
TN3K	150	8	1	0.0012	0.0007	0.13986	0.13985	−1.2e−05
Control	150	26	0	0.0004	0.0054	0.53925	0.53926	8.2e−06
All	**1712**	**107**	**2**	**0.0034**	**0.0054**			

Table S9c. Trainings that diverged. Second-session trainings: 44; saved: 41; every saved model had 0 non-finite training steps. MA-Net trainings with the cumulative number of optimiser steps with a non-finite loss at epochs 10, 20, 30 and 40, the logged mean loss at epoch 40, and the outcome. A model with non-finite weights after training was not saved and is not imputed: manet_r18 is averaged over seeds 0 and 1, manet_r50 is a single model (seed 1).

Model	Non-finite steps, epoch 10	Epoch 20	Epoch 30	Epoch 40	Mean loss, epoch 40	Outcome
manet_r18_in_s1	0	0	0	0	0.112	saved
manet_r18_in_s2	0	1	4	4	0.140	not saved (36 non-finite tensors)
manet_r50_s0	0	2	134	367	not finite	not saved (397 non-finite tensors)
manet_r50_s1	0	0	0	0	0.103	saved
manet_r50_s2	41	247	497	747	not finite	not saved (397 non-finite tensors)

Table S9d. Wound dataset excluded before scoring. A fifth wound dataset was prepared and checked automatically before any model was run on it. Masks whose size differed from their image by more than the stated share indicate an unreliable image-mask pairing, so the dataset was excluded as a whole; no score exists for it.

Dataset	Image-mask pairs checked	Masks not matching their image size	Size difference	Decision
WoundSeg	2686	322	more than 5%	excluded before any scoring

Table S9e. Overlap of AZH test and control images with FUSeg photographs. Rule, applied to every evaluation set before any analysis: an image is removed if it is a copy or a crop of a photograph that another part of the study uses, that is a FUSeg training image used to train a model (training images 0 to 392) or to tune a threshold (images 400 to 549), an image of the FUSeg reference (the first 120 FUSeg validation images, without those excluded as duplicates), or, for the control, an image of the AZH test set. Crops of FUSeg photographs that nothing in the study uses are kept: they are not leakage and are not shared between evaluation sets. Check: scale-invariant feature transform (SIFT) keypoints matched with a ratio test at 0.75 and a similarity transform fitted by random sample consensus (RANSAC) with a 3 px tolerance; an image matches a photograph with at least 25 inliers. Every AZH test and control image was compared with all 810 FUSeg training and 200 validation photographs; where several photographs tie at the best inlier count, the one used in this study is counted (training or validation split refers to that photograph). AZH test against control: pixel identity, then the same feature check. The check found again all 47 control crops recorded before it was run. Counts of images; the files are listed in Table S9f.

Quantity	AZH test	Control
Images checked	150	150
Matched to a FUSeg photograph	**61**	**71**
of which the photograph is in the FUSeg training split	49	52
of which the photograph is in the FUSeg validation split	12	19
of which the photograph is used in this study: model training	4	51
of which the photograph is used in this study: threshold tuning	0	0
of which the photograph is in the FUSeg reference (first 120 validation images, duplicates excluded)	0	19
of which the photograph is not used in this study (image kept)	57	1
Excluded as crops of FUSeg photographs used in this study	4	70
Images with a pixel-identical image in the other set	4	4
control image already excluded as a crop of a FUSeg photograph	n/a	2
control image excluded as a copy of an AZH test image	n/a	2
Crops of an AZH test image (not pixel-identical)	n/a	0
Excluded in total	**4**	**72**
Kept	**146**	**78**
Largest inlier count among images without a match (threshold 25)	23	n/a

Table S9f. Every excluded image. File name in the dataset as distributed, the reason, and for crops of FUSeg photographs how the study uses that photograph. FUSeg file names refer to the FUSeg training and validation folders. SIFT inliers: the number of geometrically consistent keypoint matches (Table S9e); an equally good match is a second photograph with the same inlier count.

Dataset	File	Reason	Photograph used for
FUSeg	0096.png	byte-identical to FUSeg training image 0983.png	n/a
FUSeg	0201.png	byte-identical to FUSeg training image 0803.png	n/a
FUSeg	0256.png	byte-identical to FUSeg training image 0997.png	n/a
FUSeg	0284.png	byte-identical to FUSeg training image 0121.png	n/a
FUSeg	0339.png	byte-identical to FUSeg validation image 0303.png	n/a
FUSeg	0356.png	byte-identical to FUSeg training image 0636.png	n/a
FUSeg	0485.png	byte-identical to FUSeg training image 0980.png	n/a
FUSeg	0545.png	byte-identical to FUSeg training image 0569.png	n/a
AZH	c8cf2da63be1a27497baa852387fe31f_0.png	crop of FUSeg training image 0132.png (74 SIFT inliers; equally good match: training image 0825.png)	model training
AZH	c8cf2da63be1a27497baa852387fe31f_1.png	crop of FUSeg training image 0132.png (39 SIFT inliers; equally good match: training image 0825.png)	model training
AZH	c95105b1727344ae36f8dca2a1ba4543_2.png	crop of FUSeg training image 0426.png (27 SIFT inliers; equally good match: training image 0826.png)	model training
AZH	d950fc3ee06a1970c0361244bc6d039a_0.png	crop of FUSeg training image 0399.png (25 SIFT inliers; equally good match: training image 0892.png)	model training
Control	002985bcbd572241a950bf39b4188d70_0.png	crop of FUSeg validation image 0002.png (45 SIFT inliers)	FUSeg reference
Control	00437234527092743faeb39bc59bd7ee_0.png	crop of FUSeg validation image 0003.png (30 SIFT inliers)	FUSeg reference
Control	00be2ad4736f6657b767683186615f3f_0.png	crop of FUSeg validation image 0004.png (31 SIFT inliers)	FUSeg reference
Control	00d87e85e220404257d75a9c6d574e0b_0.png	crop of FUSeg validation image 0006.png (27 SIFT inliers)	FUSeg reference
Control	01289b5d0d3f47b0de56c003ec81614a_0.png	crop of FUSeg validation image 0007.png (75 SIFT inliers)	FUSeg reference
Control	0272003ec882522e7bec012aac93cc26_1.png	crop of FUSeg training image 0012.png (29 SIFT inliers)	model training
Control	0301eae58a63514483f1f05436437ac2_0.png	crop of FUSeg training image 0014.png (96 SIFT inliers)	model training
Control	0301eae58a63514483f1f05436437ac2_1.png	crop of FUSeg training image 0014.png (72 SIFT inliers)	model training
Control	035b9aff921b413432a9b2b6d68e4c50_1.png	crop of FUSeg training image 0016.png (35 SIFT inliers)	model training
Control	035b9aff921b413432a9b2b6d68e4c50_2.png	crop of FUSeg training image 0016.png (44 SIFT inliers)	model training
Control	03722f11a500f35511add8b2ecf54e61_0.png	crop of FUSeg training image 0017.png (31 SIFT inliers)	model training
Control	03a29338d44c41efd1856034029cb9cb_0.png	crop of FUSeg training image 0018.png (28 SIFT inliers)	model training
Control	03b7f4b22ab4357bbf58da1540daf836_0.png	crop of FUSeg training image 0019.png (31 SIFT inliers)	model training
Control	04475f8d2c89fbf998cf4713d234ce50_0.png	crop of FUSeg training image 0020.png (41 SIFT inliers)	model training
Control	045069bee69b9a6b457a69c564589895_0.png	crop of FUSeg training image 0021.png (36 SIFT inliers)	model training
Control	04ff12bacbf32cde2df83fc1199518d8_1.png	crop of FUSeg training image 0026.png (51 SIFT inliers)	model training
Control	054008cac63fbda02ca31e4c9cd5ee46_0.png	crop of FUSeg training image 0028.png (25 SIFT inliers)	model training
Control	05bf122b0e8bf667fc2aeedd8b70d9f3_0.png	crop of FUSeg validation image 0030.png (26 SIFT inliers)	FUSeg reference
Control	06d5643601a23d986ce7a24ca4bb2152_0.png	crop of FUSeg training image 0032.png (40 SIFT inliers)	model training
Control	07a31c50fc96dcf841177d2c832767b9_0.png	crop of FUSeg training image 0035.png (26 SIFT inliers)	model training
Control	07cd49d4077e7392d36501f291706ddb_1.png	crop of FUSeg training image 0036.png (62 SIFT inliers)	model training
Control	08b680ecad190d66a8295c5255550bbf_0.png	crop of FUSeg training image 0037.png (51 SIFT inliers)	model training
Control	09174d2f9d9ac1b948ba8fb889081e88_0.png	crop of FUSeg training image 0039.png (33 SIFT inliers)	model training
Control	095919973cde59672f21c88850a4aa09_0.png	crop of FUSeg training image 0041.png (25 SIFT inliers)	model training
Control	0a5a3ada2bb75c4af2daf8e4726bf099_1.png	crop of FUSeg validation image 0046.png (25 SIFT inliers)	FUSeg reference
Control	0a5a3ada2bb75c4af2daf8e4726bf099_2.png	crop of FUSeg validation image 0046.png (36 SIFT inliers)	FUSeg reference
Control	0abb9eba5adac0db89364735c88860e4_0.png	crop of FUSeg validation image 0047.png (41 SIFT inliers)	FUSeg reference

Dataset	File	Reason	Photograph used for
Control	0ac40b1b34bdfebeab8fdad265e76554_0.png	crop of FUSeg training image 0048.png (27 SIFT inliers)	model training
Control	0b12ef4301fbef4071bc8a7b0beeff49_0.png	crop of FUSeg training image 0050.png (32 SIFT inliers)	model training
Control	0b1e2ebd480a39038df6a4a8b273ded2_0.png	crop of FUSeg training image 0051.png (28 SIFT inliers)	model training
Control	0bc5e8bfd852b40e2a2af5cd933e00bb_0.png	crop of FUSeg training image 0056.png (58 SIFT inliers)	model training
Control	0c2ef774c755a57b93845eed12e8029f_0.png	crop of FUSeg validation image 0057.png (33 SIFT inliers)	FUSeg reference
Control	0c79e72868d6bbc3673fd091109ff6ff_0.png	crop of FUSeg training image 0059.png (30 SIFT inliers)	model training
Control	0dac4ad9e961452a81183370848d3ec3_0.png	crop of FUSeg training image 0063.png (42 SIFT inliers)	model training
Control	0ddc7e1cf8508f1fcae87bd41106d8c6_0.png	crop of FUSeg training image 0064.png (30 SIFT inliers)	model training
Control	0e3efb99f7529906a52c26558d37af9c_0.png	crop of FUSeg training image 0070.png (30 SIFT inliers)	model training
Control	0e9585008d43fdf08d840064a49674f3_0.png	crop of FUSeg training image 0071.png (25 SIFT inliers)	model training
Control	0f06905655751ee938d3c71e516f3084_0.png	crop of FUSeg training image 0072.png (26 SIFT inliers)	model training
Control	0f42ca66d0a67e868683aab958c70b70_0.png	crop of FUSeg training image 0073.png (35 SIFT inliers)	model training
Control	0f9e59822589ead1b752065cf09d4b87_0.png	crop of FUSeg training image 0074.png (40 SIFT inliers)	model training
Control	100e9e226ad344e054a1a36191308da1_0.png	crop of FUSeg training image 0076.png (33 SIFT inliers)	model training
Control	11c4368cd336af0da7ff5681c6146ccf_0.png	crop of FUSeg training image 0081.png (35 SIFT inliers)	model training
Control	11ebfc1753d3ef581684f936c934b55e_0.png	crop of FUSeg validation image 0082.png (28 SIFT inliers)	FUSeg reference
Control	12f816e49ae64a52d95a484a559ac2d4_0.png	crop of FUSeg validation image 0085.png (25 SIFT inliers)	FUSeg reference
Control	1576e366442683aac204404e081eb938_1.png	crop of FUSeg training image 0088.png (75 SIFT inliers; equally good match: training image 0151.png)	model training
Control	1576e366442683aac204404e081eb938_2.png	crop of FUSeg training image 0088.png (66 SIFT inliers; equally good match: training image 0151.png)	model training
Control	15ee4679bf92307b89e2d818e8fbccd7_0.png	crop of FUSeg validation image 0089.png (37 SIFT inliers)	FUSeg reference
Control	160fcc9963c057d0a4754c8a39f87f1c_0.png	crop of FUSeg training image 0090.png (25 SIFT inliers)	model training
Control	16180f16e73ffb069b756f56859aec57_0.png	crop of FUSeg training image 0091.png (40 SIFT inliers)	model training
Control	16489f399948a5dfc46c056f1d26da40_0.png	crop of FUSeg training image 0093.png (46 SIFT inliers)	model training
Control	16bbfa585c420f96ef4cd0812e9c163d_0.png	crop of FUSeg training image 0095.png (35 SIFT inliers)	model training
Control	18d60d8c820b5487f59dce1237f31f47_0.png	crop of FUSeg training image 0099.png (26 SIFT inliers)	model training
Control	19d43ee6c12f70035f87d57ff561a41b_1.png	crop of FUSeg training image 0104.png (49 SIFT inliers)	model training
Control	19ff49b10de40e6a431a38b19a212ec0_0.png	crop of FUSeg validation image 0107.png (29 SIFT inliers)	FUSeg reference
Control	1a4b3982e17b4f019481ebcba701fc24_0.png	crop of FUSeg validation image 0108.png (36 SIFT inliers)	FUSeg reference
Control	1a969119fbcdf79012ff301231d59e95_0.png	crop of FUSeg training image 0109.png (30 SIFT inliers)	model training
Control	1ad70e82589acbd9638b6e465366a7fe_0.png	pixel-identical to AZH test image d89317533d1ca29abe05734f949b0934_0.png	n/a
Control	1ad97384d93aa85123fe2173047c3dce_0.png	crop of FUSeg training image 0112.png (33 SIFT inliers)	model training
Control	1b43b69901255b1f2fc7aedcfefa7912_0.png	crop of FUSeg training image 0116.png (26 SIFT inliers)	model training
Control	1b68ab7a408b0c74157999bd480962a6_0.png	crop of FUSeg training image 0117.png (26 SIFT inliers)	model training
Control	1c1864fe796dc3d59999950a2c697756_1.png	crop of FUSeg training image 0119.png (72 SIFT inliers)	model training
Control	1c58fc115fbca1fff4e83c069493625b_0.png	crop of FUSeg validation image 0124.png (51 SIFT inliers)	FUSeg reference
Control	1c700b5e74caa409d88eb73957906b05_0.png	crop of FUSeg training image 0125.png (37 SIFT inliers)	model training
Control	1c700b5e74caa409d88eb73957906b05_1.png	crop of FUSeg training image 0125.png (28 SIFT inliers)	model training
Control	1c8e887ad6ec44f3c6cc91e88dd59f56_0.png	crop of FUSeg validation image 0128.png (31 SIFT inliers)	FUSeg reference
Control	1d557cd47276255d999c71bc49ccf493_0.png	crop of FUSeg training image 0132.png (74 SIFT inliers; equally good match: training image 0825.png)	model training
Control	1d557cd47276255d999c71bc49ccf493_1.png	crop of FUSeg training image 0132.png (39 SIFT inliers; equally good match: training image 0825.png)	model training
Control	1d557cd47276255d999c71bc49ccf493_2.png	pixel-identical to AZH test image c8cf2da63be1a27497baa852387fe31f_2.png	n/a
Control	1d87e42d6146a984a037cf13a150eec0_0.png	crop of FUSeg validation image 0133.png (26 SIFT inliers)	FUSeg reference
Control	1dbb00d92741863fcc6451afd48daf31_0.png	crop of FUSeg validation image 0136.png (29 SIFT inliers)	FUSeg reference

Dataset	File	Reason	Photograph used for
Control	1e172e50acdceb692355ba080e2146cc_0.png	crop of FUSeg training image 0140.png (26 SIFT inliers)	model training
Control	1e172e50acdceb692355ba080e2146cc_1.png	crop of FUSeg training image 0140.png (42 SIFT inliers)	model training

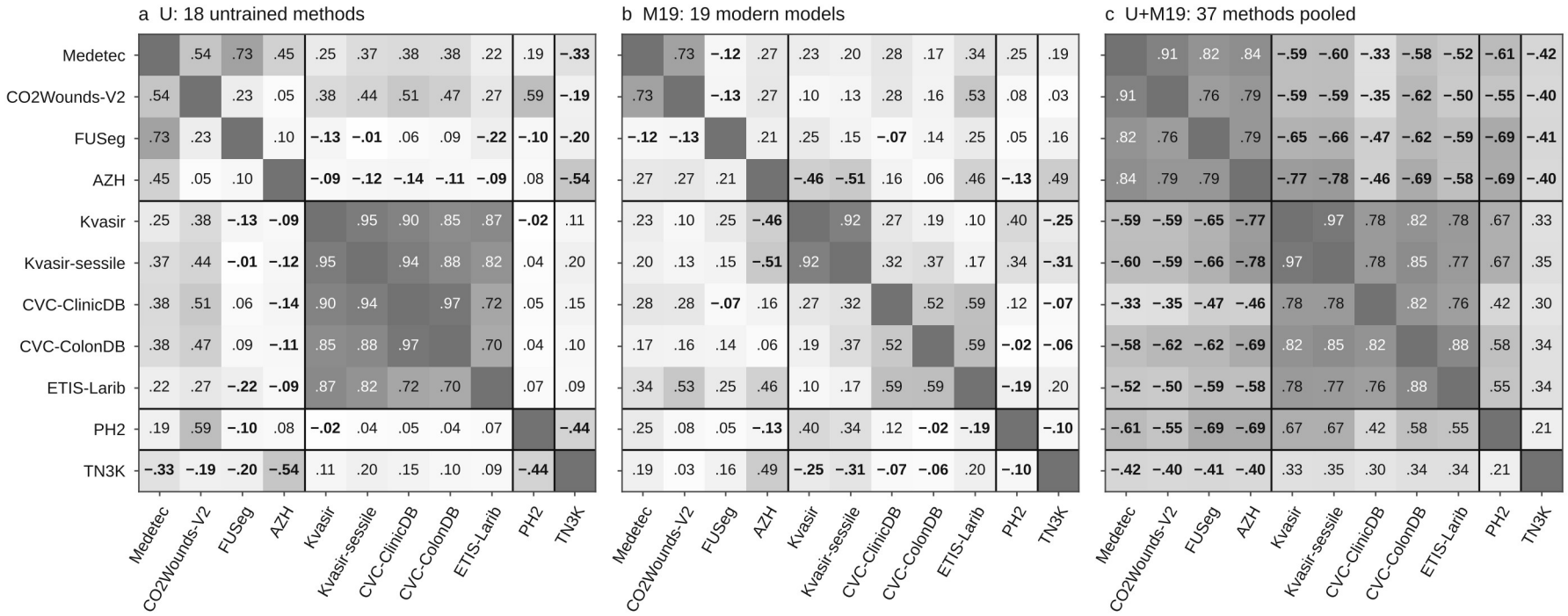


Fig. S1. Agreement matrices (Dice). Each cell is the Spearman agreement of per-method mean Dice between the row dataset and the column dataset over the methods of one set: (a) U, 18 untrained methods; (b) M19, 19 modern models (seed averages and nnU-Net); (c) U+M19, the 37 methods pooled. Values to 2 decimals with the leading zero omitted (for example .54 is 0.54); negative values in bold; shading by absolute value (darker is stronger); diagonal not shown. Lines separate the wound, colonoscopy, dermoscopy and ultrasound datasets. Unrounded values for these and all other method sets are in Online Resource 2 (sheet Pair agreement).